Internal Use Only

MOBILE PHONE
SERVICE MANUAL

CAUTION
BEFORE SERVICING THE UNIT, READ THE “SAFETY PRECAUTIONS” IN THIS MANUAL

MODEL : LG-X220ds

Date: Apr, 2016 / Issue 1.0

Table Of Contents

1.1 Purpose

This manual provides the information necessary to repair, calibration, description and download the features of this model.

1.2 Regulatory Information

A. Security

This material is prohibited to share and release to unauthorized person, in accordance with the regulations, LG Electronics, Civil / criminal responsibility in accordance with the relevant provisions violate.

B. Precautions for repair

- In case of Disassembly or Assembly to repair product, be careful of a product failure caused by RF signals and Static electricity.
- When using Magnetic tool for the Phone's SVC repair, you should check affect the Electric parts according to effect of Magnet.
- When fastening the screw, be careful not to damage the head of screw and even product.

C. Attention

Boards, which contain Electrostatic Sensitive Device (ESD), are indicated by the  **sign.**

Following information is ESD handling:

- Service personal should ground themselves by using a wrist, strap when exchange system board.
- When repair are made to a system board, they should spread the floor with anti-static mat which is also grounded.
- Use a suitable, grounded soldering iron.
- Keep sensitive parts in these protective packages until these are used.
- When returning system board or parts like EEPROM to the Factory, use the protective package as described.

2.1 Band Specification

Support Band	TX Freq (MHz)	RX Freq (MHz)
WCDMA(FDD1)	1920 – 1980	2110 – 2170
WCDMA(FDD8)	880 – 915	925 – 960
EGSM	880 – 915	925 – 960
GSM850	824 – 849	869 – 894
DCS1800	1710 – 1785	1805 – 1880
PCS1900	1850 – 1910	1930 – 1990

2.2 HW Features

List	Type / Spec.	
1. Phone Type	Bar Type	
2. Size	141.5 x 71.6 x 8.9 mm	
3. Weight	128g (with Battery)	
4. Battery	1,900mAh(Typ) (Li-Ion), 1,820mAh(Min.)	
5.Chipset	MT6580A/WM	
6. Memory	8GB(EMMC) + 1GB(LPDDR3)External Memory(SD Card) : Up to 32GB	
7. LCD	Size	5 inch
	Display Type	Active matrix TFT, Transmissive Type
	Color	16.7M colors
	Resolution	854(H) x 480(W) 196ppi
8. Touch	Type	5 inch Capacitive type
9. Main Camera (5M)	Type	CMOS image sensor
	Resolution	2560(H) X 1920 (V) pixels.
	Image Scaling Down	5M(2560x1920), 4M(2560x1440), 3M(2048x1536), 2M(1600x1200), 1.3M(1280x960), 1M(1280x720)
	Format	Image : JPG, Video : MP4

2.2 HW Features

10.Front Camera(2M)	Type	CMOS image sensor
	Resolution	2,560(H) x 1,920(V)
	Format	Image : JPG, Video : 3GP
11. Audio	Receiver/Speaker 2in1	18 X 10 X 3.1T Speaker
	Format	MP3, AAC, MIDI, EAAC+, OGG, AMR, AAC+, WAV, FLAC
12. Bluetooth	Standard	Bluetooth 4.1
	Effective Distance	10M
	Distance	0 m ~ 10 m (depend on environment)
13. WLAN	Standard	IEEE 802.11 b/g/n
	Throughput	Max 40Mbps (SDIO Driver performance)
	Depend on environment	0 ~ 50m (depend on environment)
14. GPS	type	A-GPS
15. FM	type	FM Radio, 3.5pi Ear-jack

2.3 RSSI Display

Antenna BAR	Specification		Unit
	WCDMA(RSCP)	GSM(RSSI)	
5□4	-85 ±3dBm	-91 ±3dBm	dBm
4□3	-92 ±3dBm	-96 ±3dBm	
3□2	-98 ±3dBm	-98 ±3dBm	
2□1	-102 ±3dBm	-102 ±3dBm	
1□0	-106 ±3dBm	-104 ±3dBm	

2.4 Current consumption

Item	Specification	
	WCDMA	GSM
1. Current(Sleep & Idle AVG)	Under 6mA(@DRX7)	Under 6mA(@PG 5)
2. Talking Current	≤260mA @10dBm ≤270mA @12dBm	≤340mA(@PCL5,GSM900/GSM850) ≤280mA(@PCL0,DCS1800/PCS1900)

2.5 Battery bar

Battery Bar	Specification	
BAR 7 (Full)	91% over	remain%
BAR 6	90%-76%	
BAR 5	75%-61%	
BAR 4	60%-50%	
BAR 3	49%-36%	
BAR 2	35%-16%	
BAR 1	15%-5%	
BAR 0	4%-0%	
Low Battery Pop-up	15%	

2.6 SW Specification

Item	Feature	Comment
RSSI	0 ~ 5 Levels	
Battery Charging	0 ~ 100%	
Key Volume	No	No key volume setting
Audio Volume	0 ~ 15 Level	
Time / Date Display	Yes	
Multi-Language	Yes	depending on build language
Quick Access Mode	Phone / Contact / Browser / Messaging	
PC Sync	No	
Speed Dial	Yes	Voice mail center -> 1 key
Profile	No	not same with feature phone setting
CLIP / CLIR	Yes	
Phone Book	Name / Phonetic / Nickname / Photo / Phone / E mail / Postal addresses / Birthday / Company / N otes / IM / Relationship	There is no limitation on the number of it ems. It depends on available memory amount.
Last Dial Number	Yes	
Last Received Number	Yes	
Last Missed Number	Yes	
Search by Number/Name	Yes	
Group	Yes	There is no limitation on the number of it ems. It depends on available memory amount.
Fixed Dial Number	Yes	
Service Dial Number	No	
Own Number	No	My Profile (add/edit/delete are supporte d)

2.6 SW Specification

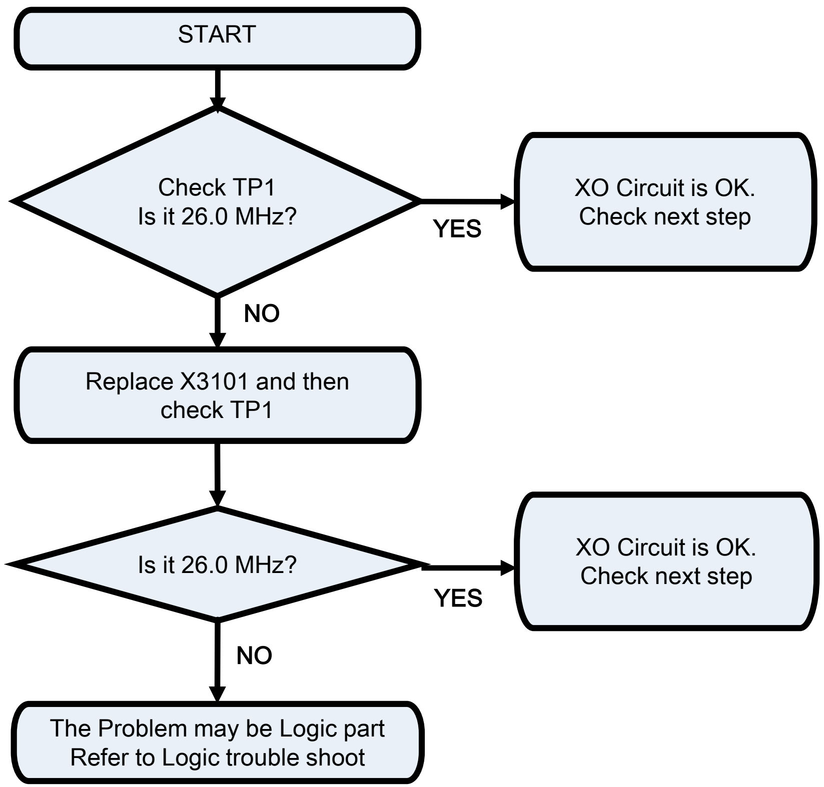
Voice Memo	Yes	Support voice recorder
Call Reminder	No	
Network Selection	Automatic	
Mute	Yes	
Call Divert	Yes	
Call Barring	Yes	
Call Charge (AoC)	No	
Call Duration	Yes	
SMS (EMS)	There is no limitation on the number of items. It depends on available memory amount.	EMS does not support.
SMS Over GPRS	Yes	
EMS Melody / Picture Send / Receive / Save	No	
MMS MPEG4 Send / Receive / Save	Yes	Ø**Send / Receive** : Yes Ø**Save** : depends on content type Support video content type list 1.video/mp4 2.video/h263, h264 3.video/3gpp2 video/3gpp
Long Message	MAX 450 characters	SMS
Cell Broadcast	Yes	
Download	Over the Web	
Game	No	
Calendar	Yes	
Memo	No	
World Clock	Yes	
Knock on	Yes	The Knock ON feature allows you to double-tap the screen to easily turn it on or off.

2.6 SW Specification

Unit Convert	No	
Stop Watch	Yes	
Wall Paper	Yes	
WAP Browser	No	Support only web browser based on webkit. WAP stack and wml are not supported.
Download Melody / Wallpaper	Yes	Over web browser
SIM Lock	No	
SIM Toolkit	Yes	
MMS	Yes	Google MMS Client
EONS	Yes	
CPHS	Yes	V4.2
ENS	No	
Camera	Yes	5M FF / Digital Zoom : x4
JAVA	No	
Voice Dial	Yes	US English only
IrDa	No	IrRC
Bluetooth	Yes	Ver. 4.1
FM radio	Yes	
GPRS	Yes	Class 12
EDGE	Yes	Class 12(Rx only)
Hold / Retrieve	Yes	
Conference Call	Yes	Max. 6
DTMF	Yes	
Memo pad	No	
TTY	No	
AMR	Yes	
SyncML	Yes	
IM	Yes	Google Hangout
Email	Yes	
Gesture shot	Yes	Taking the photos with a gesture, There are two methods for using the Gesture shot feature.

The out put frequency(26.0MHz) of XO(X3101) is used as the reference one of MT6580 internal VCO

Checking Flow



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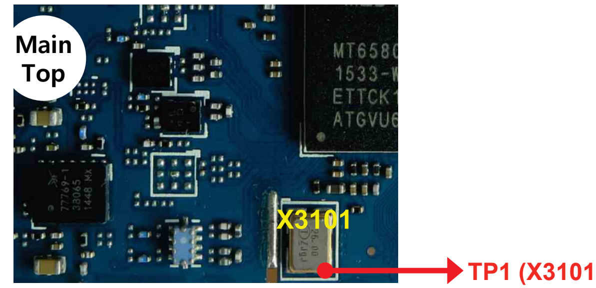
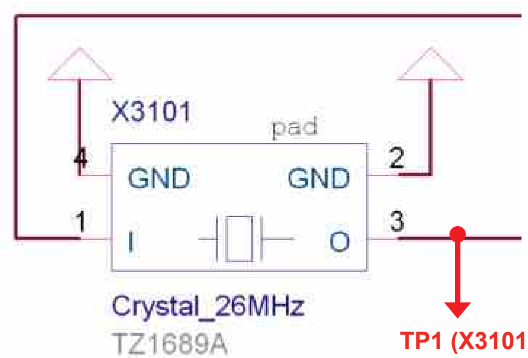
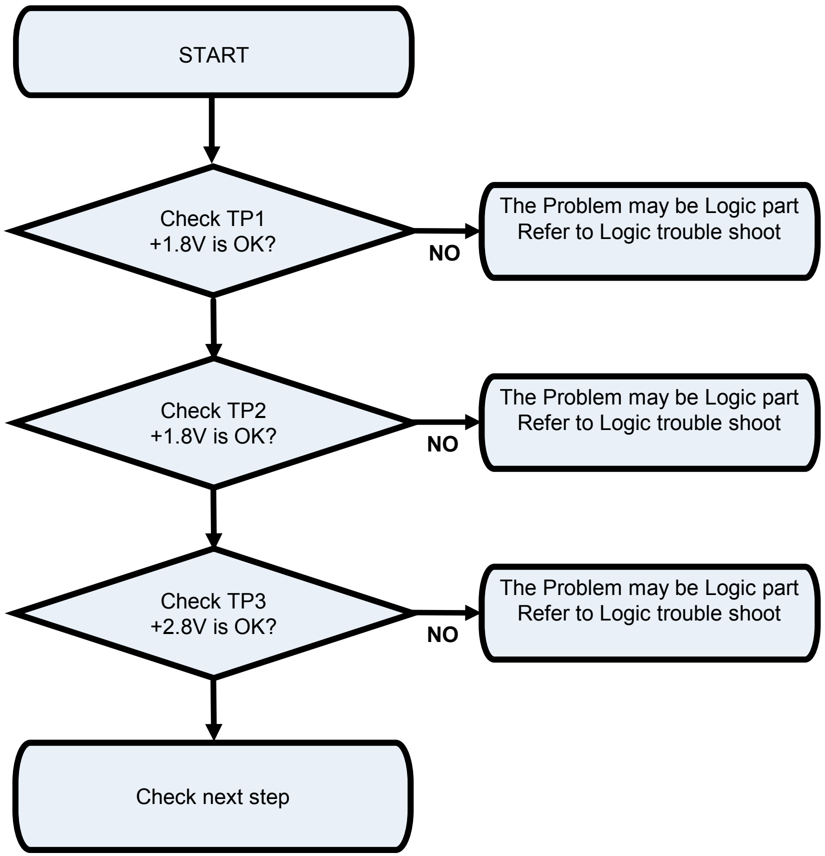


Circuit Diagram



The MT6580 operating voltages used three voltage sources 1.8V and 1.8V and 2.8V

Checking Flow



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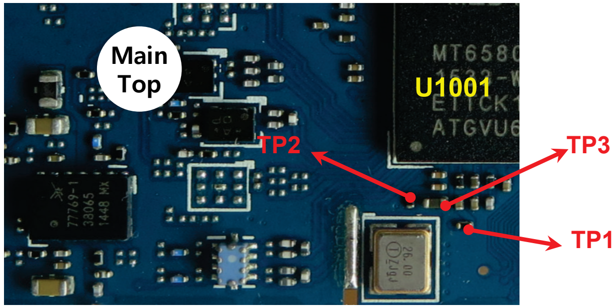
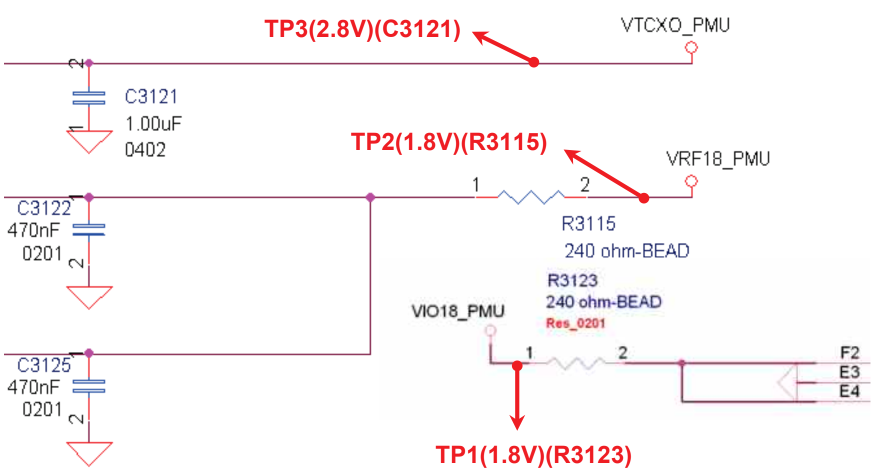


Circuit Diagram



3.3.1 . FEM Mode Control Logic Table

FEM Mode Control Logic Table

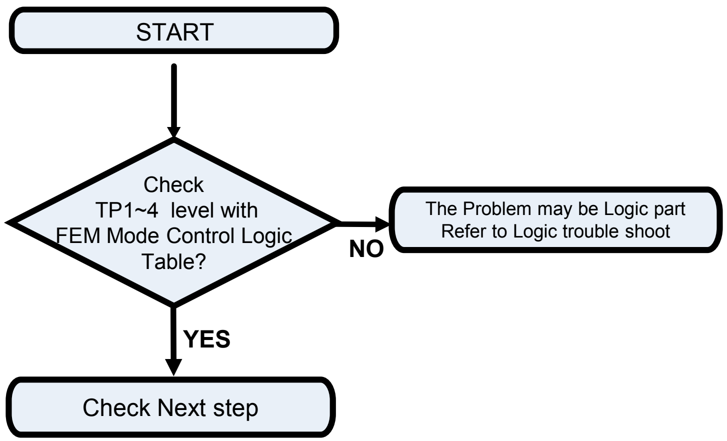
TABLE 3. SKY77909-11 MODE CONTROL LOGIC

Mode	Input Control Bits			
	TXEN	MODE	BS1	BS2
STANDBY	0	0	0	0
LB_GMSK_TX	1	0	0	1
HB_GMSK_TX	1	0	1	1
TRX1	0	1	0	0
TRX2	0	1	1	0
TRX3	0	1	0	1
TRX4	0	1	1	1
TRx5	0	0	1	0
TRx6	0	0	0	1
Forward Isolation	0	0	X	1

X = Don't care

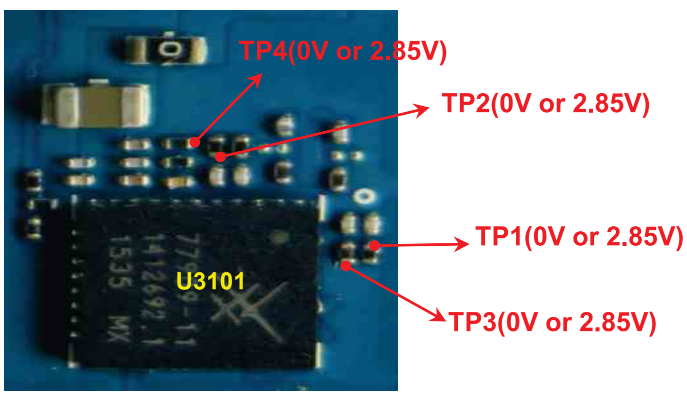
3.3.2 . All Band FEM Block (GSM 850/900/1800/1900, WCDMA B2/5)

Checking Flow

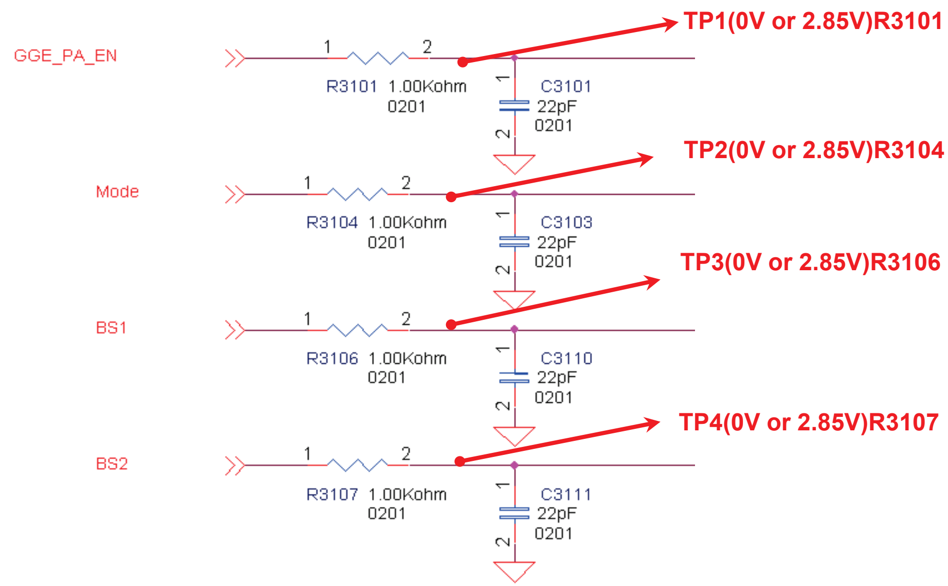


Image

Main
Bot



Circuit Diagram



GSM RF Part support GSM850/900/1800/1900 with ASM, PAM, Transceiver component

Checking Flow

START

↓

Check XO Block
Refer 3.1

↓

Check Transceiver DC Power supply Block
Refer 3.2

↓

Check FEM Block
Refer 3.3

↓

Check T/RX Block

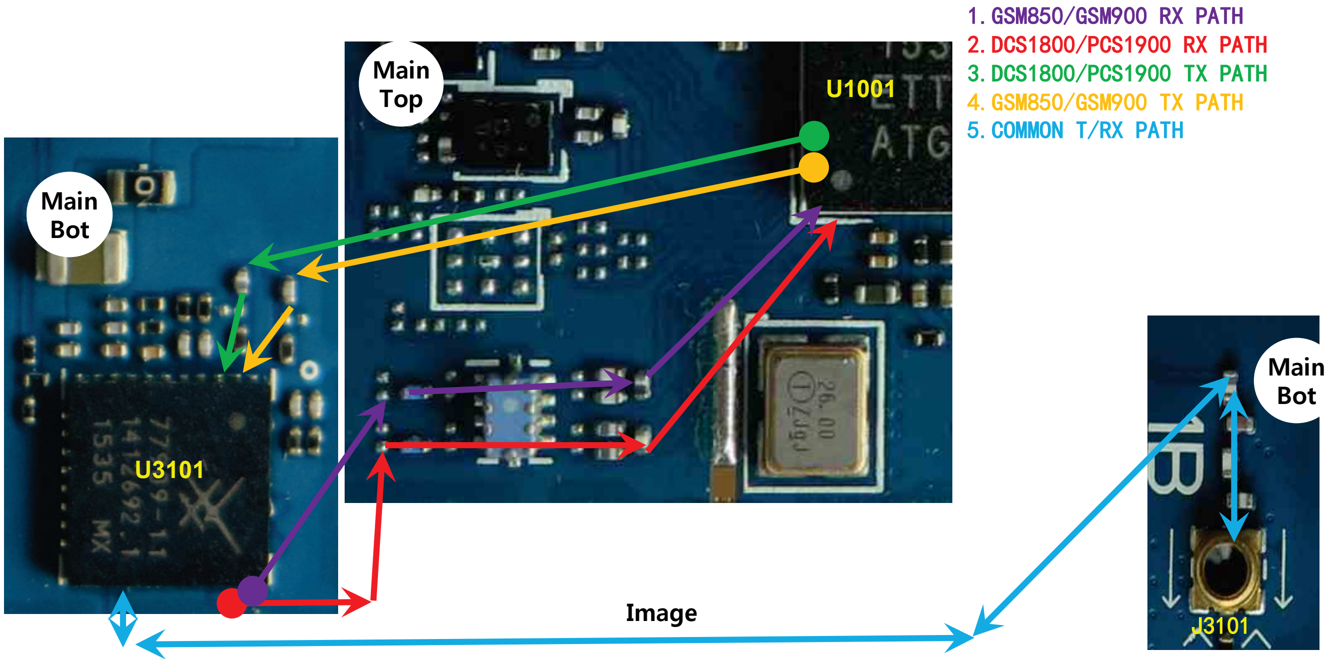
↓

Re-Download & Cal

↓

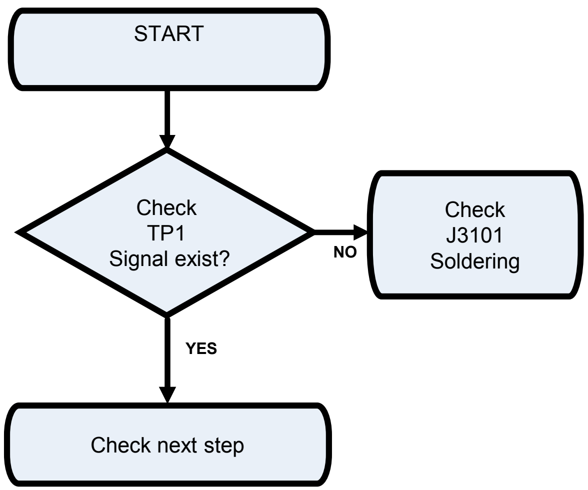
Change the Board

3.4.1 GSM850/900/1800/1900 Rx



3.4.1.1 Checking RF signal path (SW)

Checking Flow



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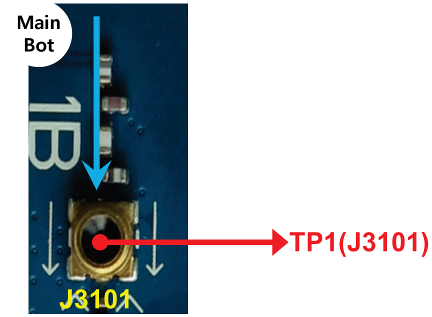
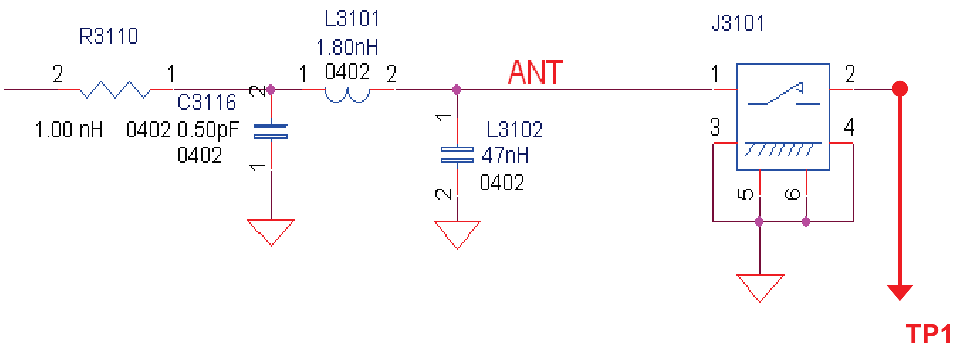
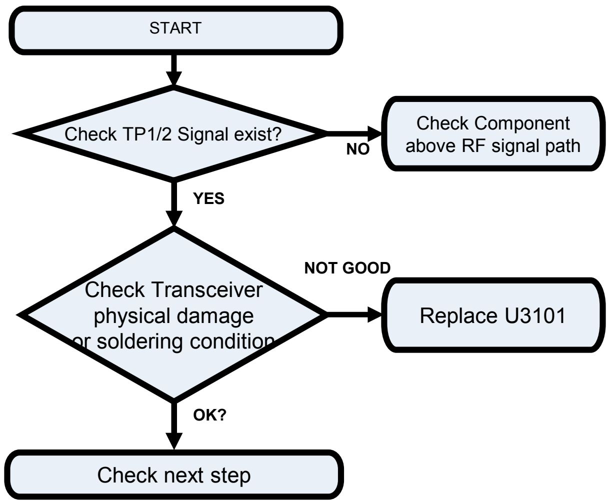


Circuit Diagram



3.4.1.2 Checking RF Signal RX path(GSM850/GSM900/DCS1800/PCS1900)

Checking Flow



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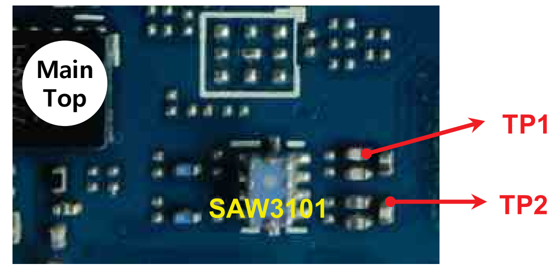
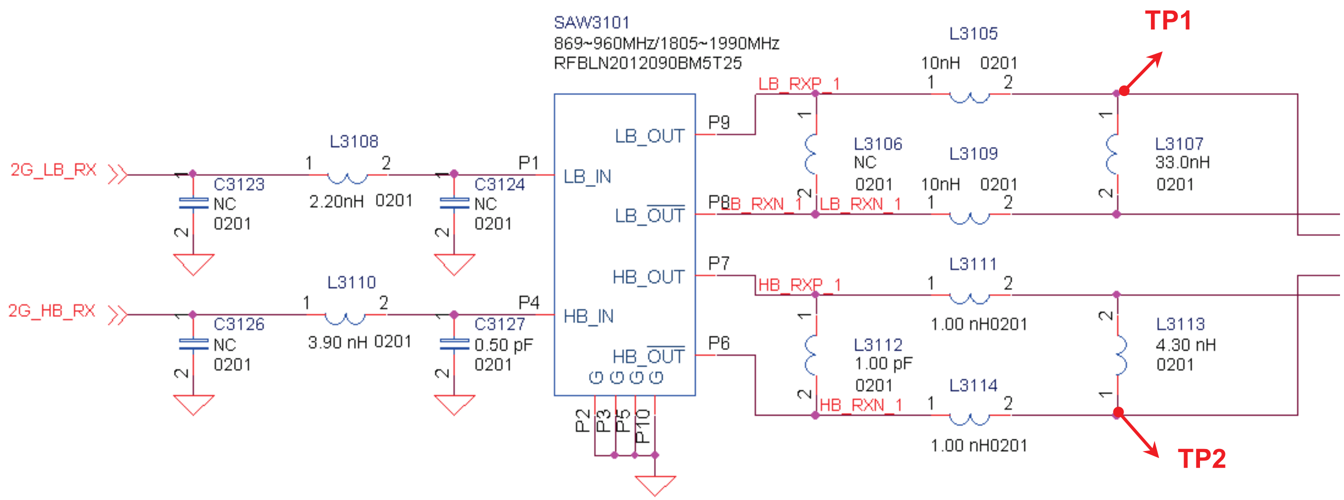
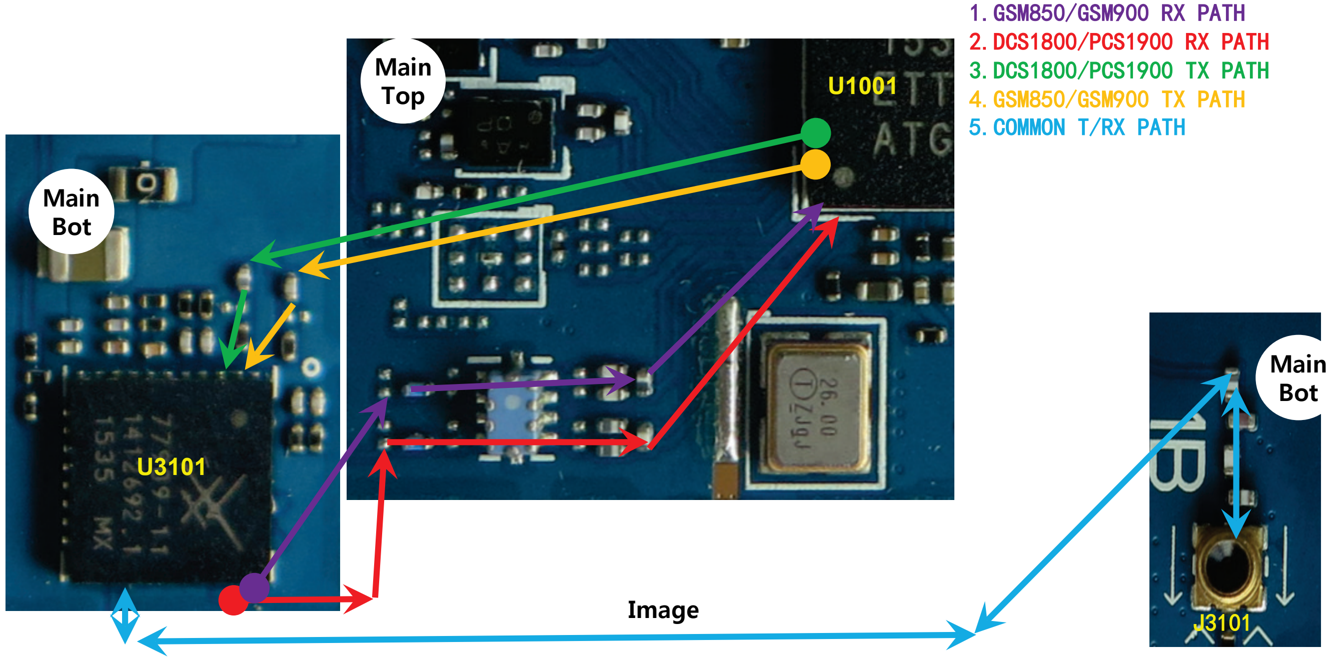


Circuit Diagram

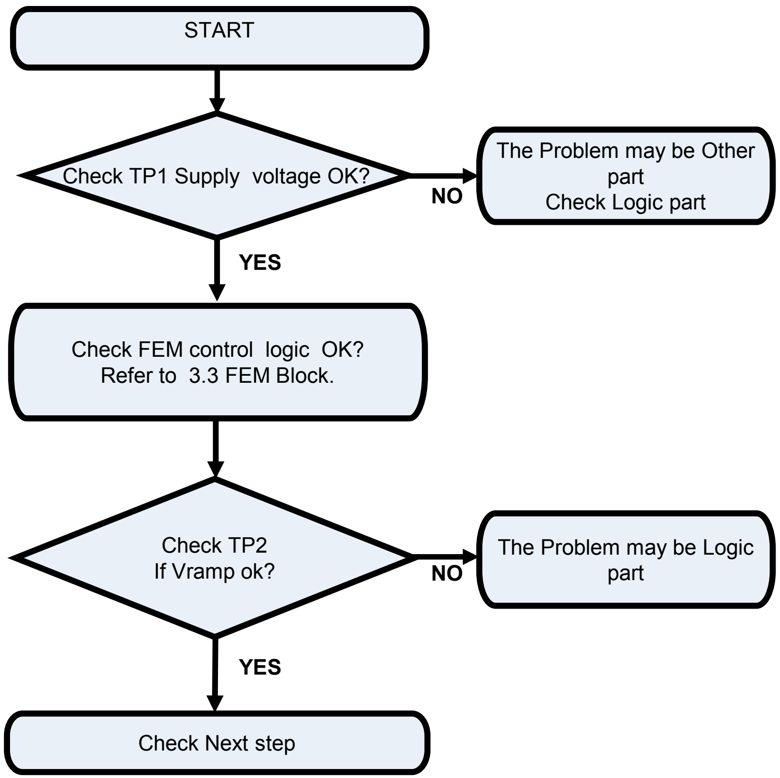


3.4.2.1 GSM850/900/1800/1900 Tx



3.4.2.2 Checking GSM PAM DC Power Circuit

Checking Flow



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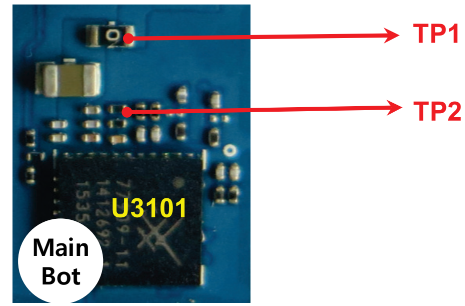
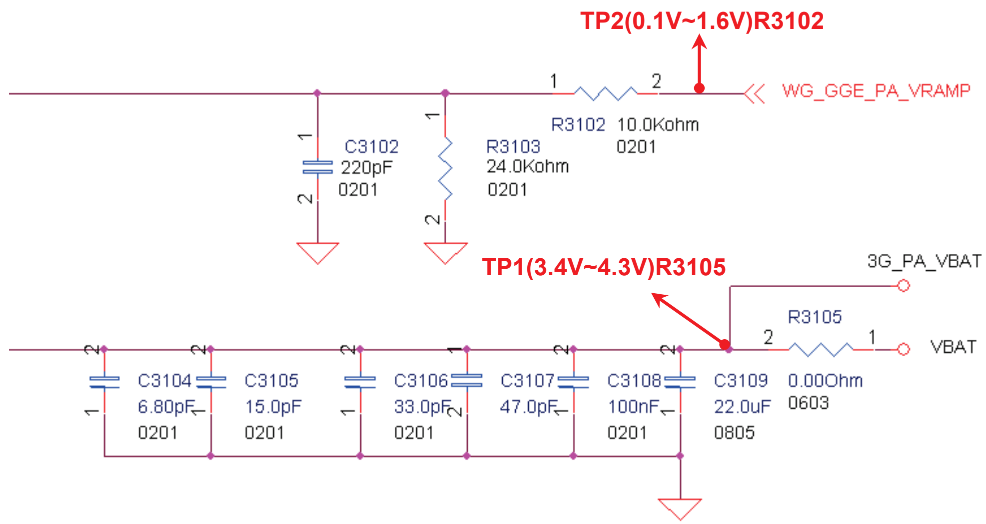
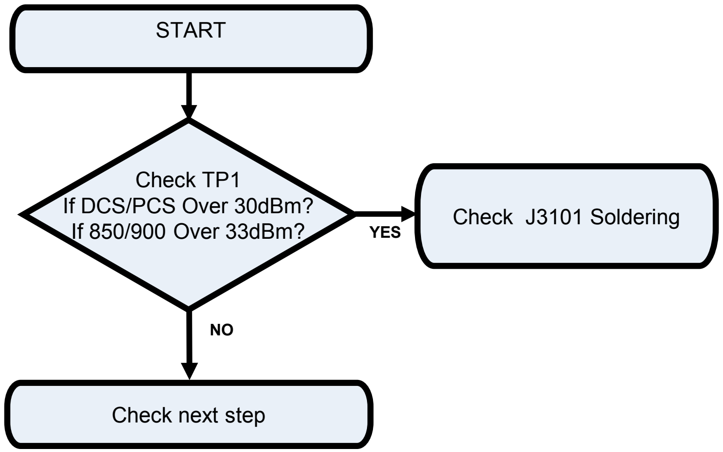


Circuit Diagram



3.4.2.3 Checking RF signal path(SW)

Checking Flow



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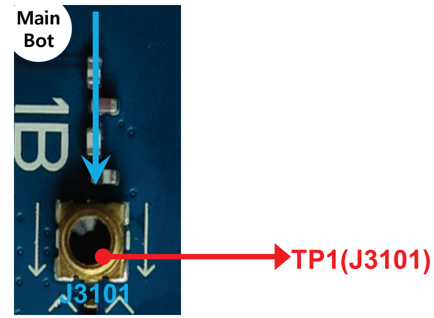
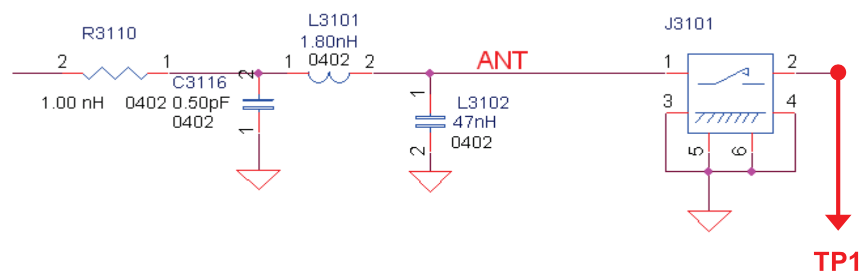
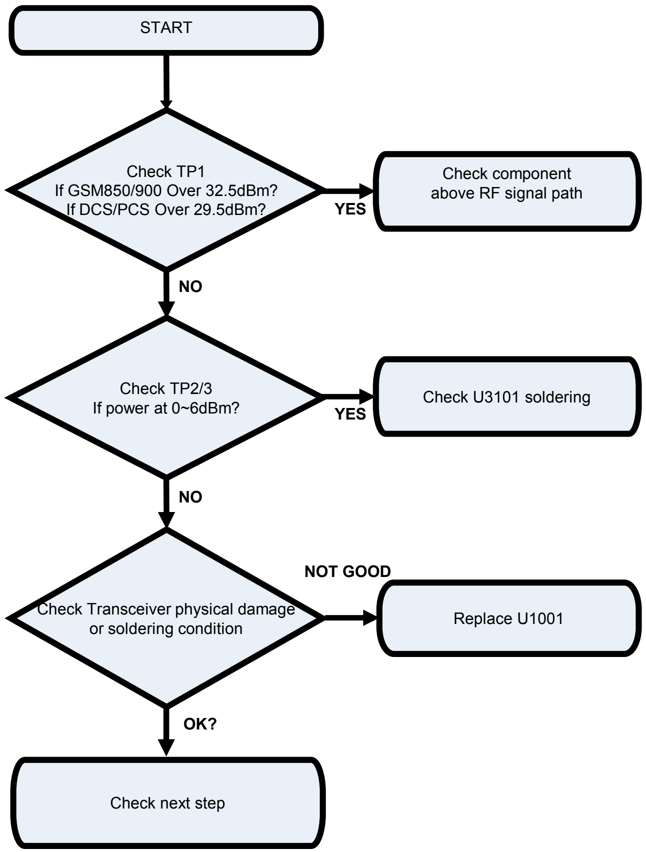


Circuit Diagram



3.4.2.4 Checking RF signal path

Checking Flow



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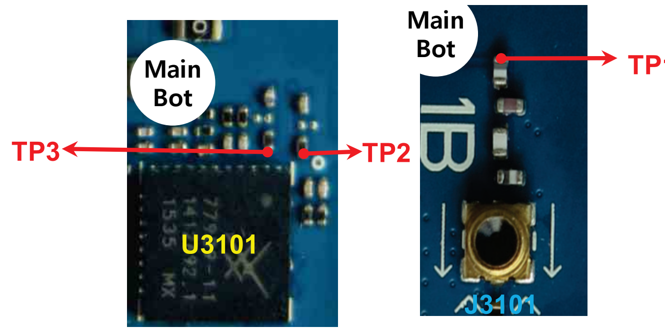
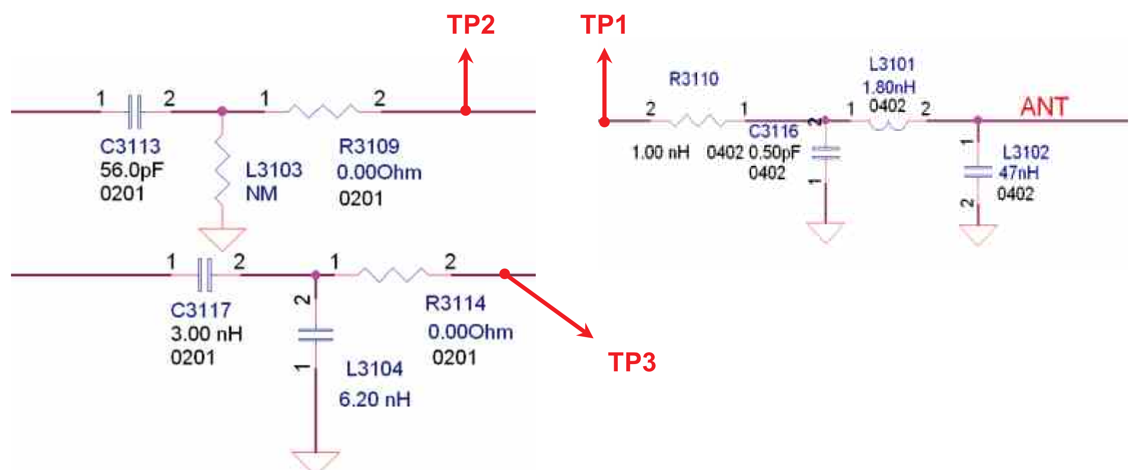
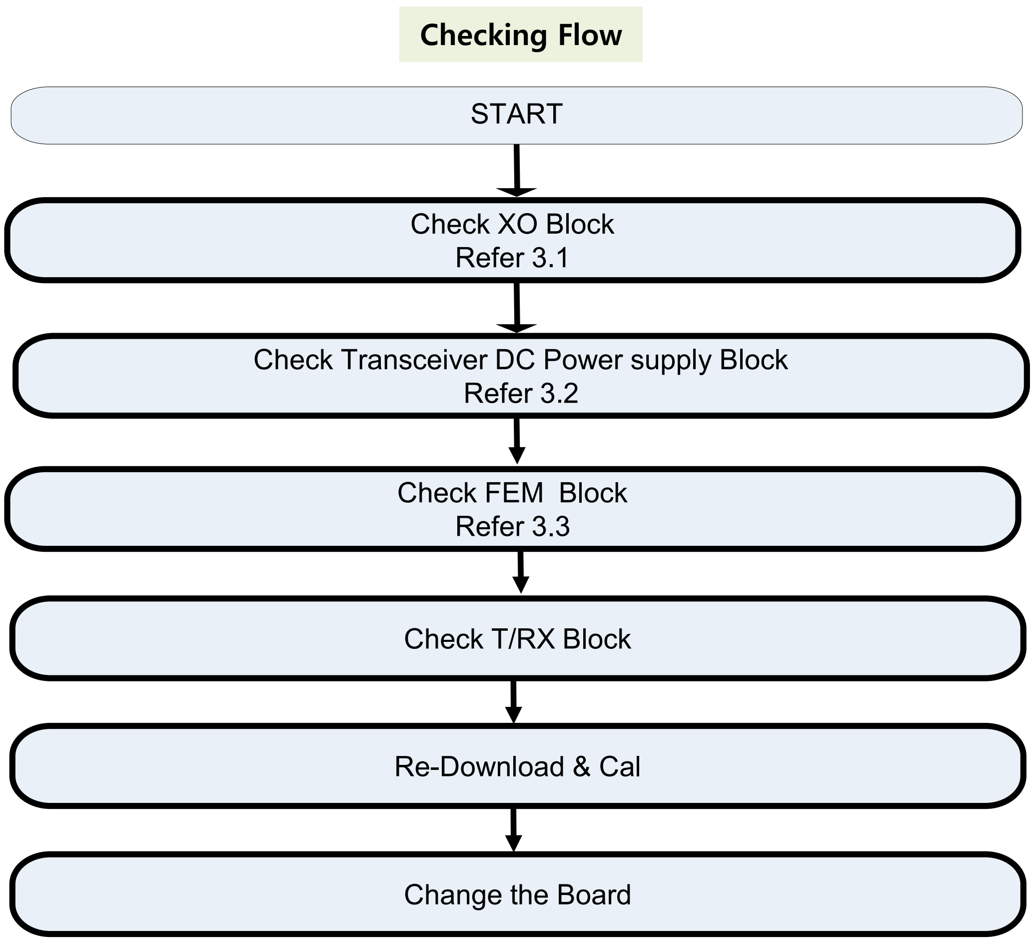


Circuit Diagram



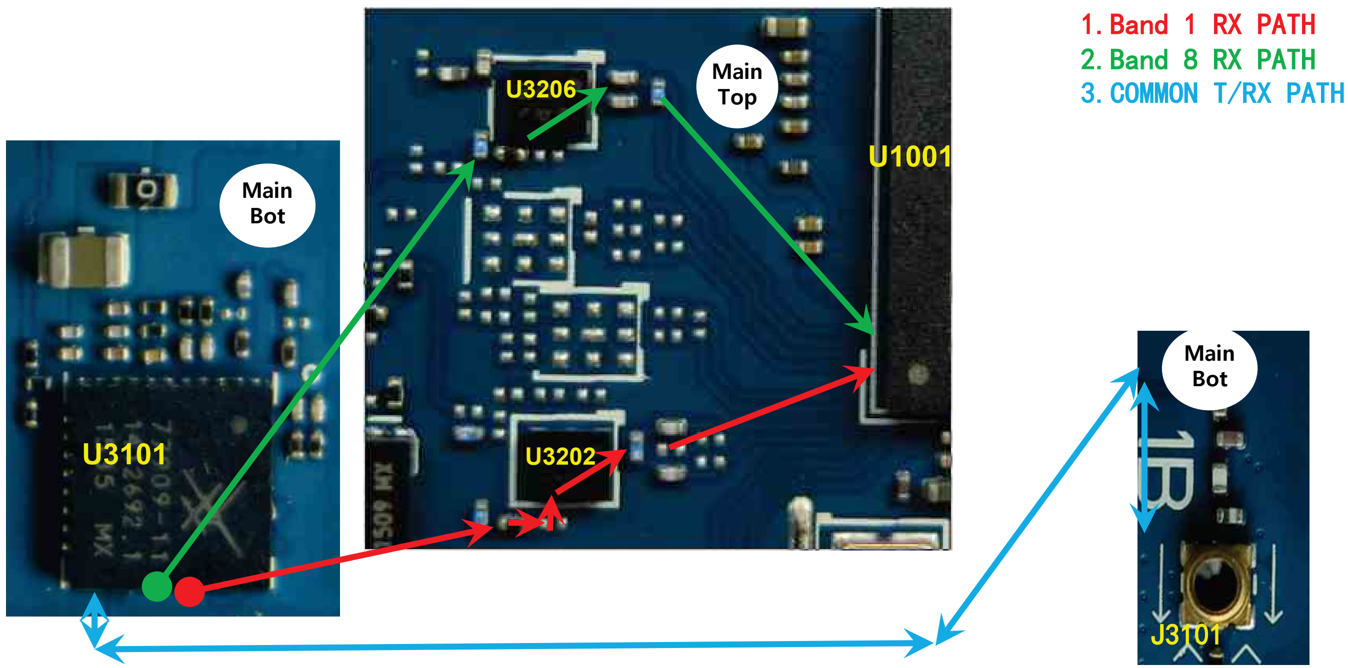
WCDMA RF Part support WCDMA B1/8 with FEM, PAM, Transceiver component



3.5.1 WCDMA RF Part RX PATH (B1/B8)

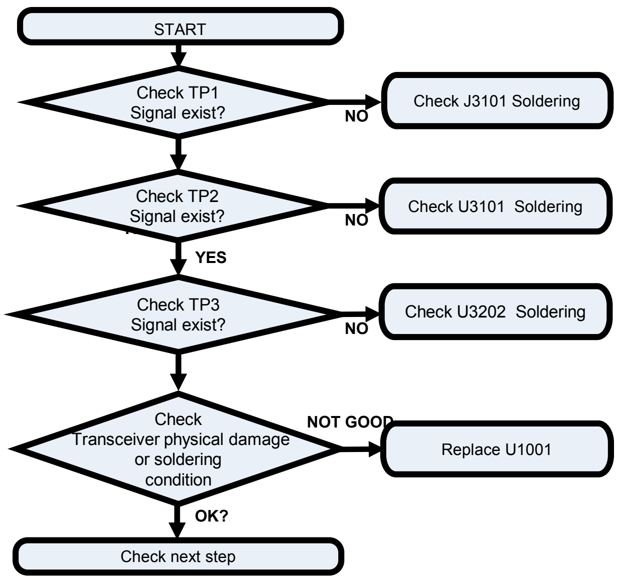
1) WCDMA RX Path

Image



3.5.1.1 Checking RF signal path(Band 1)

Checking Flow



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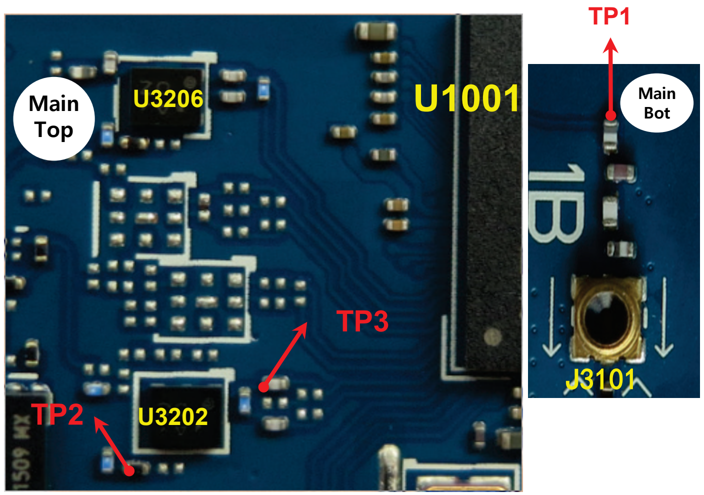
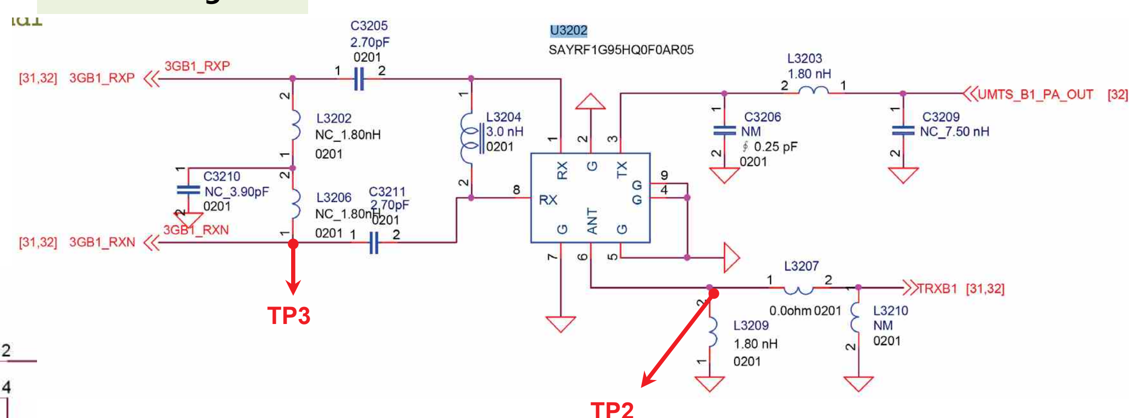
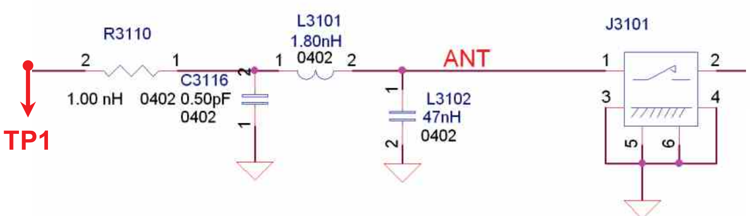


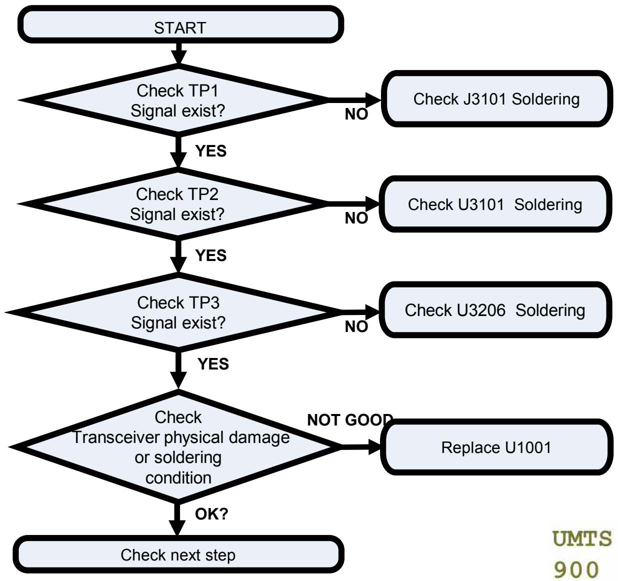
Circuit Diagram





3.5.1.2 Checking RF signal path(Band 8)

Checking Flow



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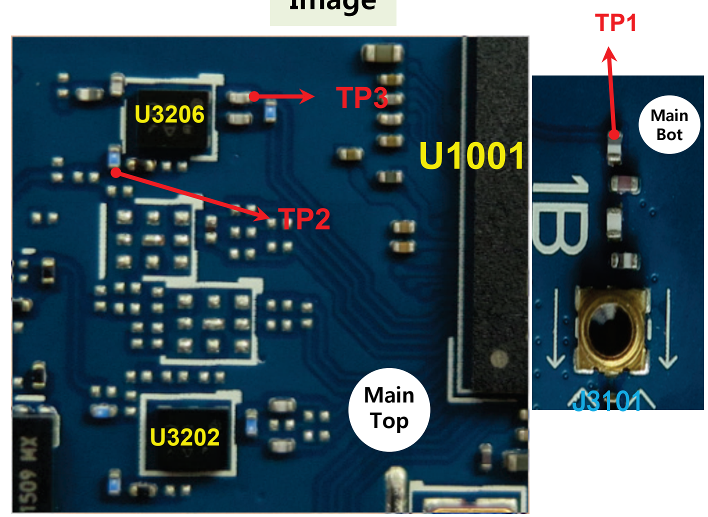
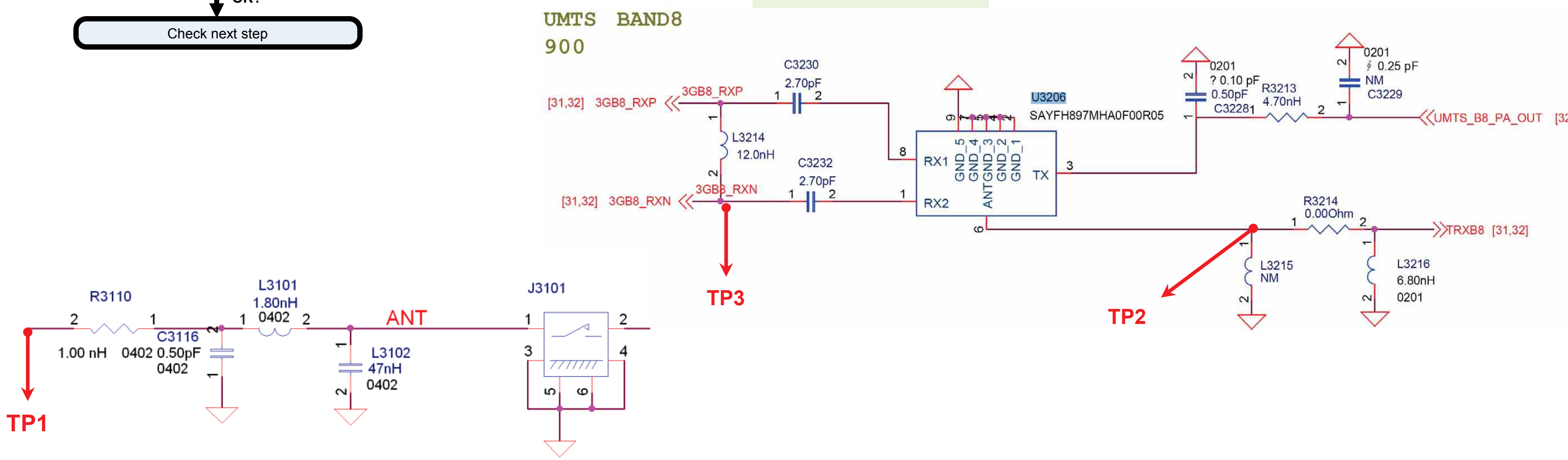


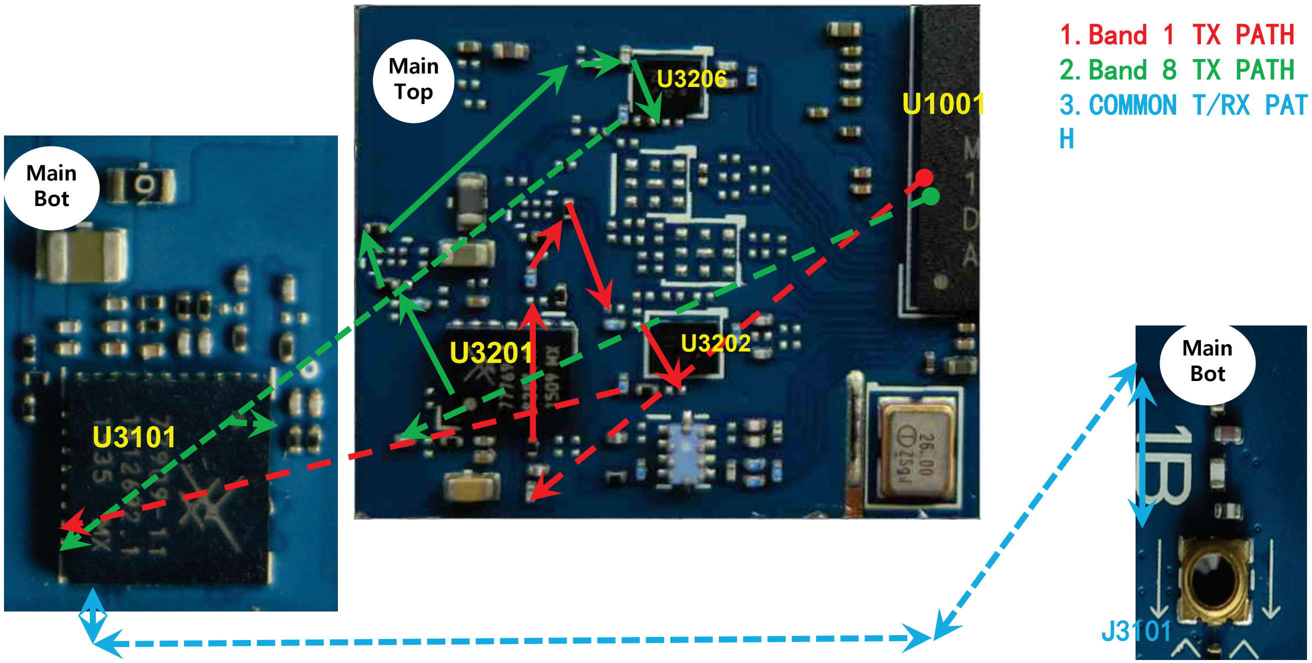
Circuit Diagram



3.5.2 WCDMA RF Part TX PATH (B1/B8)

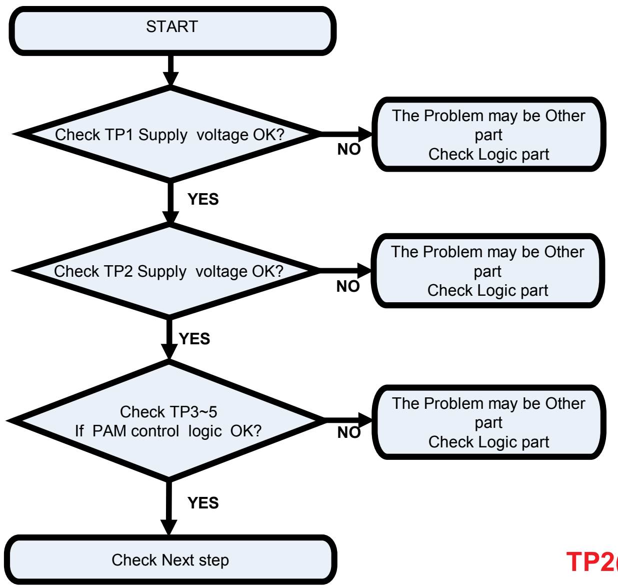
2) WCDMA TX Path

Image



3.5.2.1 Checking WCDMA PAM DC Power & Control Circuit

Checking Flow



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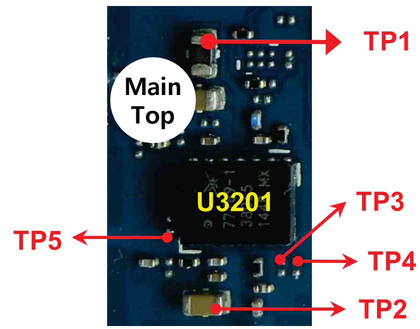
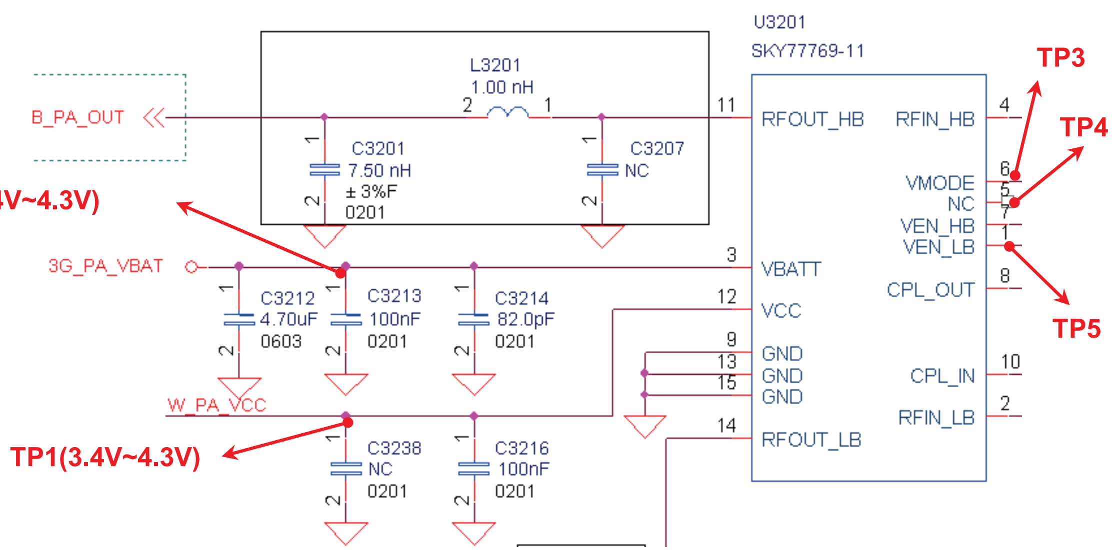


Circuit Diagram



3.5.2.1 Checking WCDMA PAM DC Power & Control Circuit

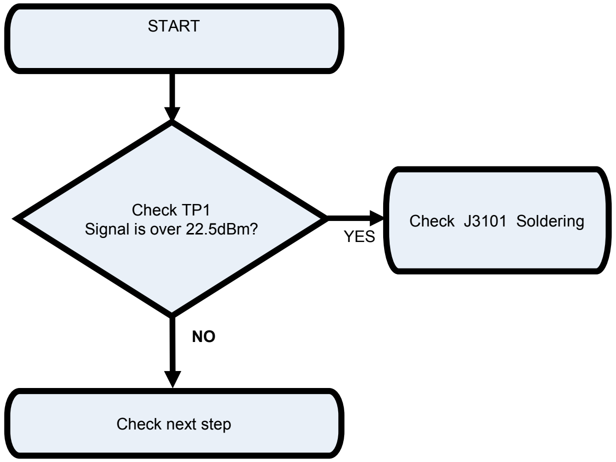
U3201 PAM CONTROL LOGIC

TABLE 3. SKY77769 MODES OF OPERATION

Power Setting		Band	VENHB	VENLB	VMODE	VBATT
Power Down Mode			Low	Low	Low	On
Standby Mode			Low	Low	X	On
Low Power Mode	($P_{OUT} \leq$ 17 dBm)	I, II, IV	High	Low	High	On
High Power Mode	(P_{OUT} = 17 dBm to P_{MAX})	I, II, IV	High	Low	Low	On
Low Power Mode	($P_{OUT} \leq$ 17 dBm)	V, VIII	Low	High	High	On
High Power Mode	(P_{OUT} = 17 dBm to P_{MAX})	V, VIII	Low	High	Low	On

3.5.2.2 Checking RF signal path(SW)

Checking Flow



Image

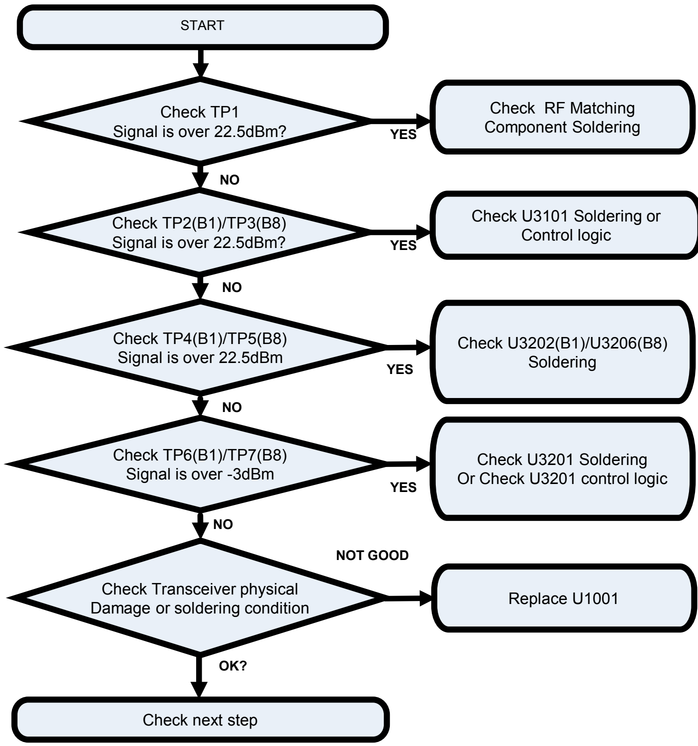
Refer to 3.5.1.1

Circuit Diagram

Refer to 3.5.1.1

3.5.2.3 Checking RF signal path

Checking Flow



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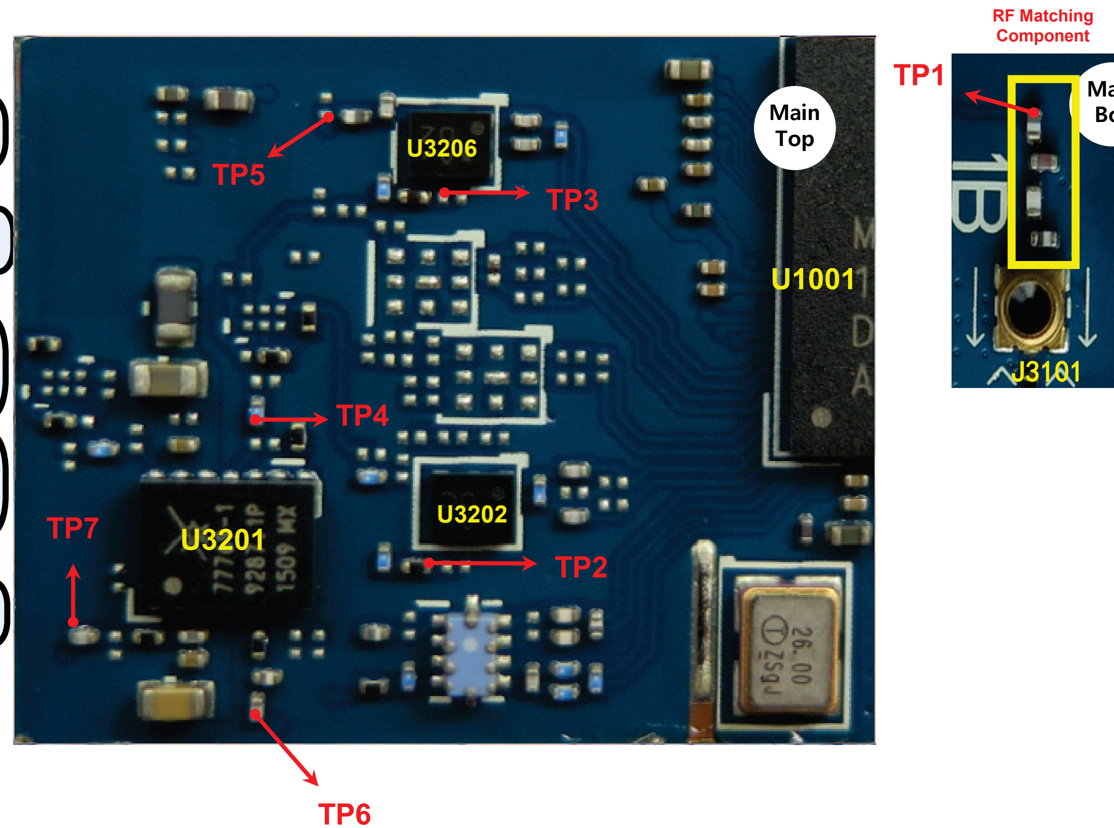
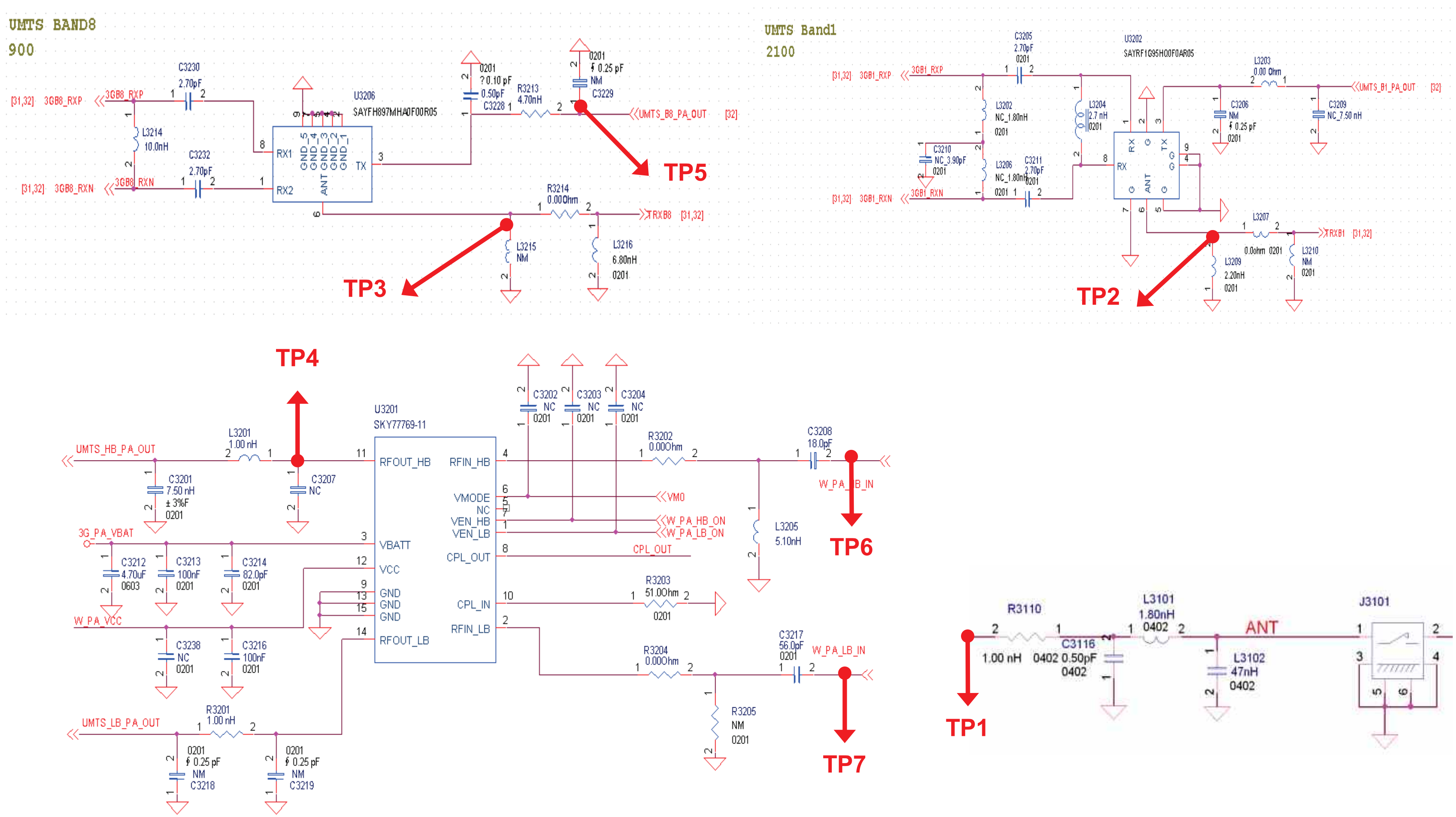


3.5.2.3 Checking RF signal path

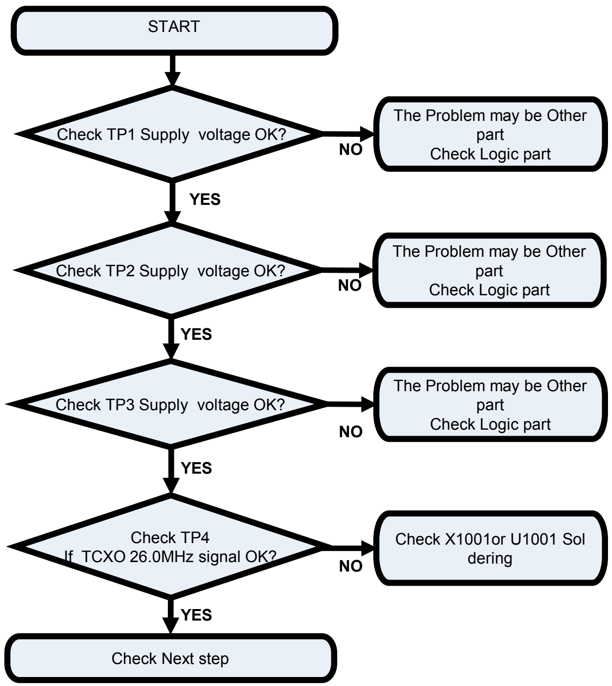
Circuit Diagram



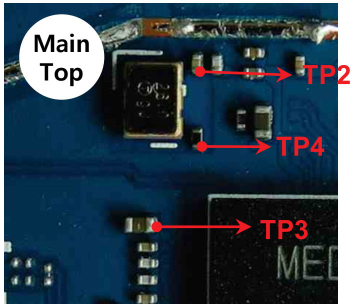
LGE Internal Use Only

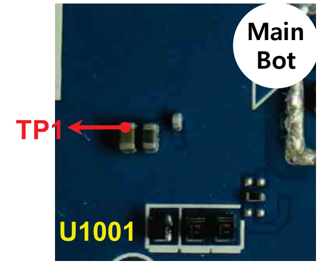
3.6.1. Checking Connectivity DC Power & TCXO Circuit

Checking Flow

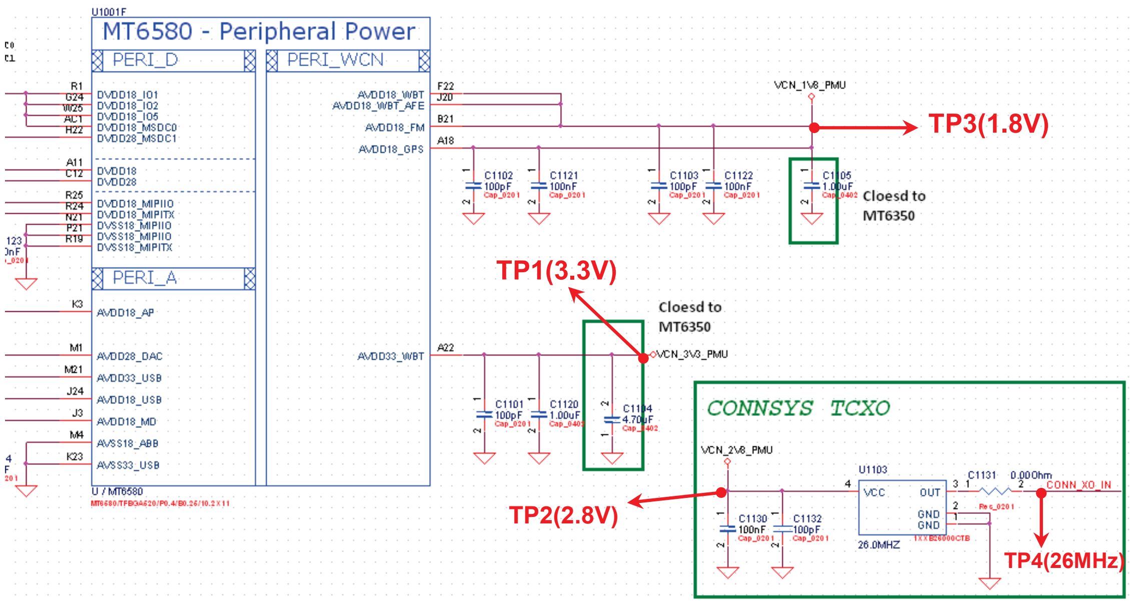


Image



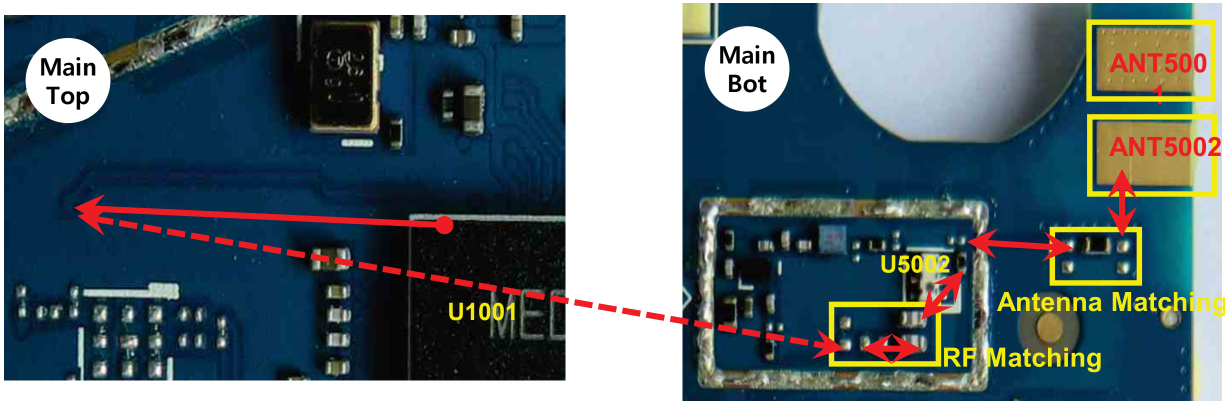


Circuit Diagram



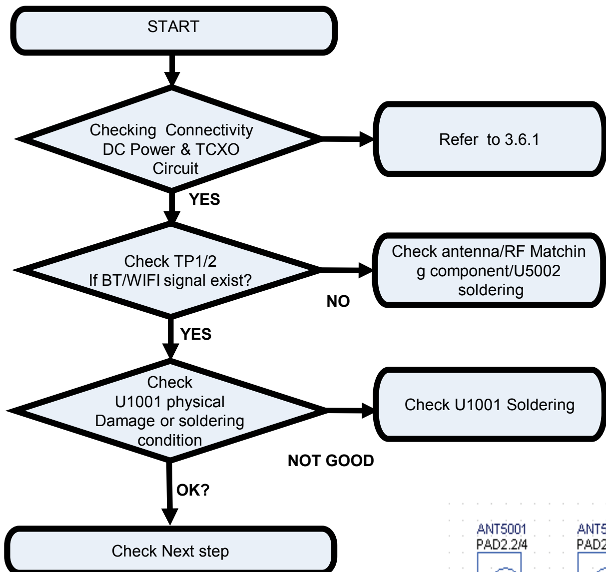
3.6. 2. 1. BT&WIFI Trouble Shooting

BT/WIFI Signal Path

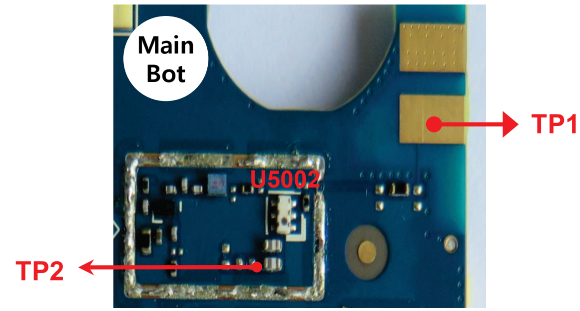


3.6. 2. 2. BT&WIFI Trouble Shooting

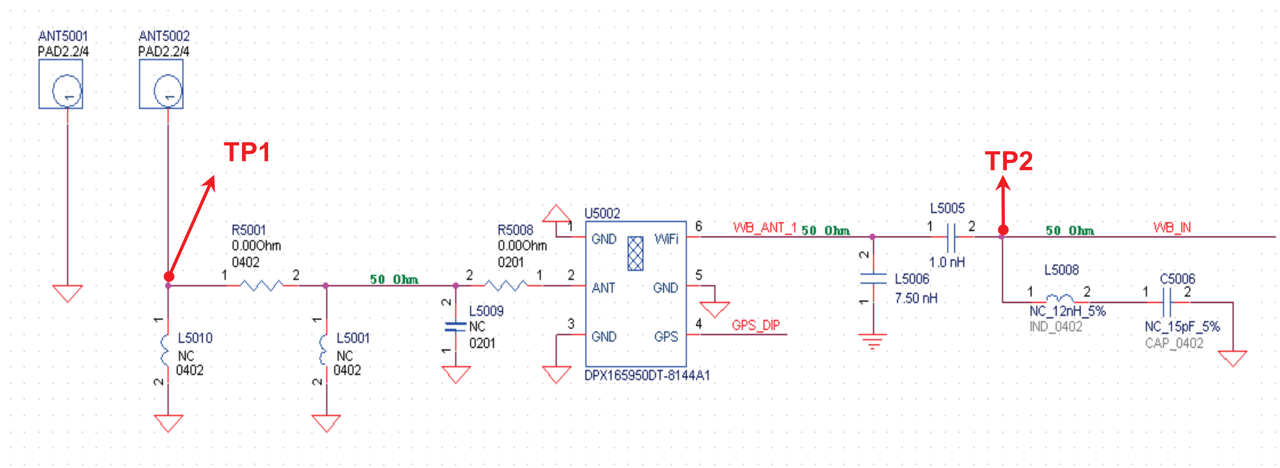
Checking Flow



Image

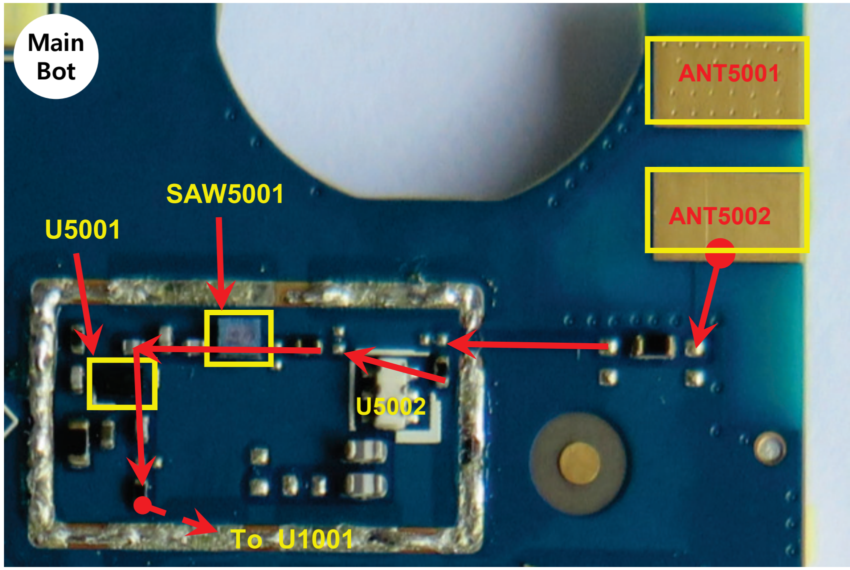


Circuit Diagram



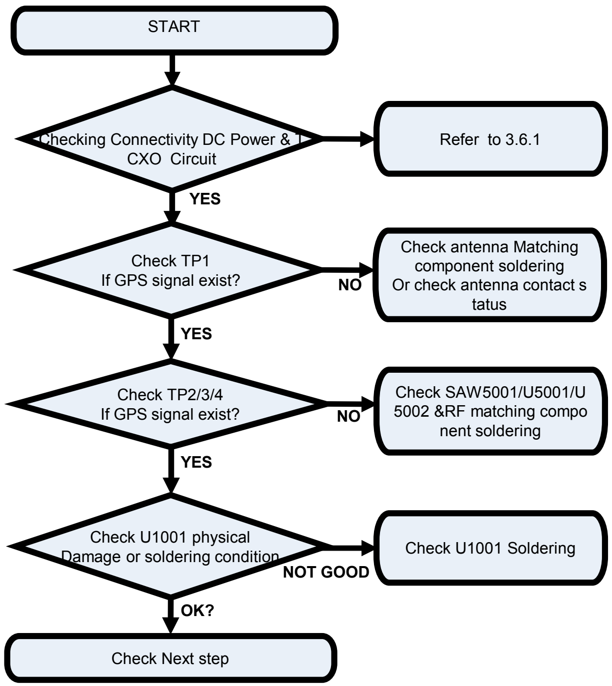
3.6. 3. 1. GPS Trouble Shooting

GPS Signal Path

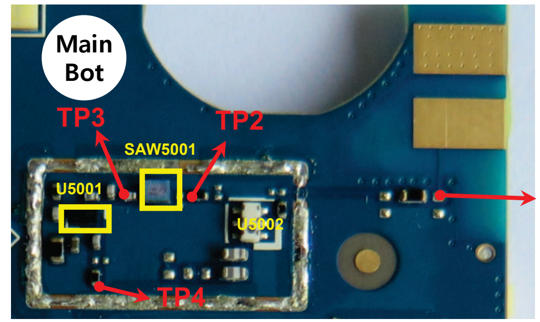


3.6. 3. 2. GPS Trouble Shooting

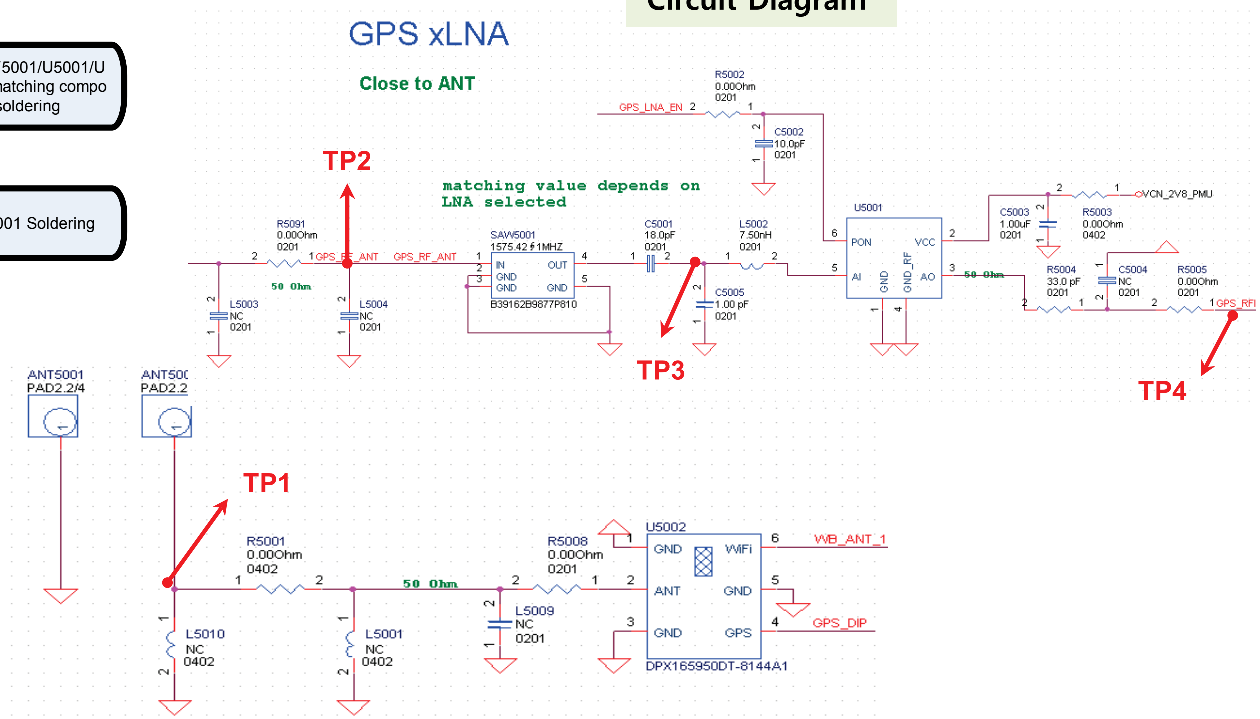
Checking Flow



Image

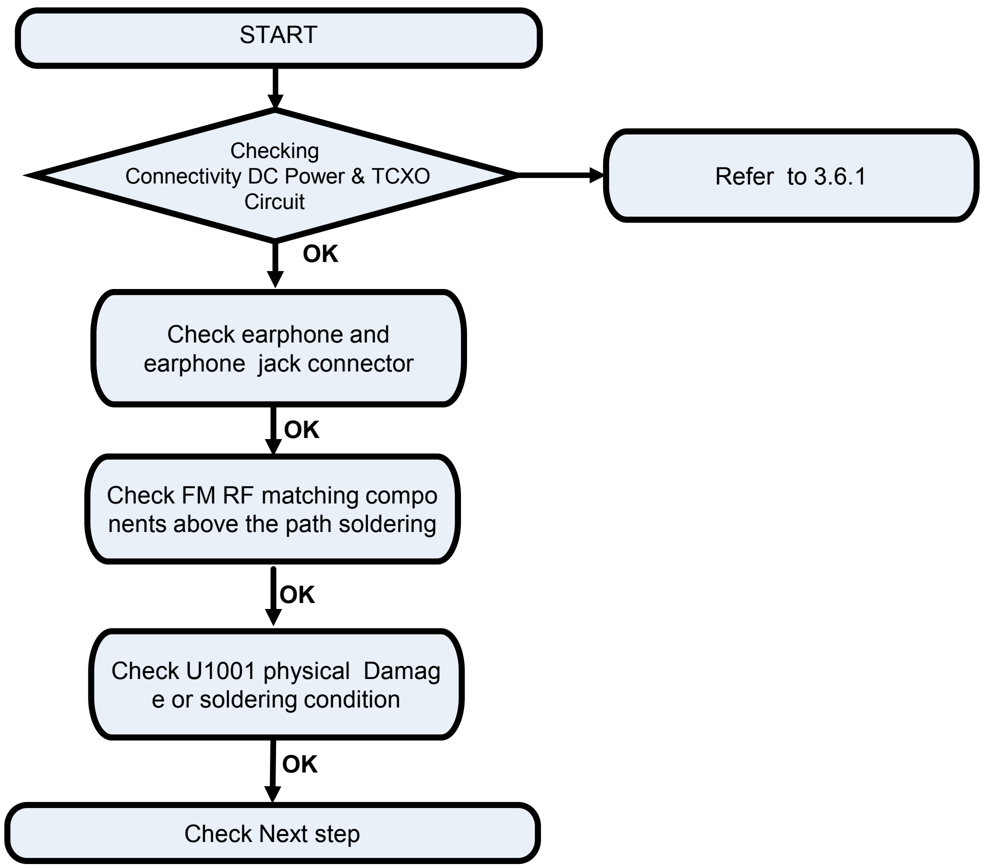


Circuit Diagram

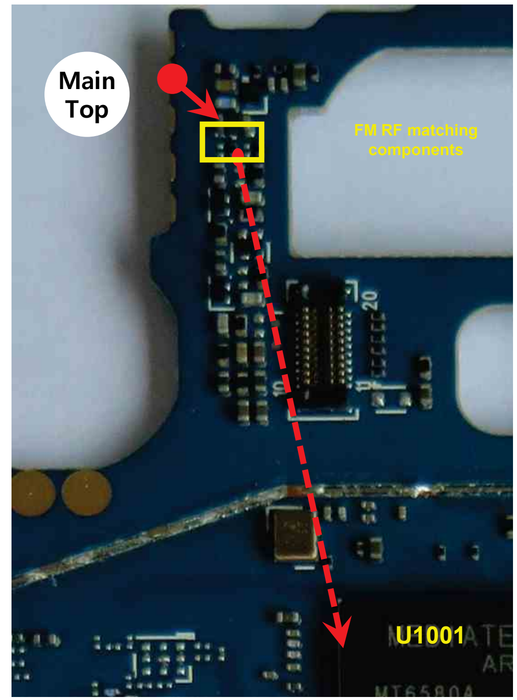


3.6. 4. 1. FM Trouble Shooting

Checking Flow

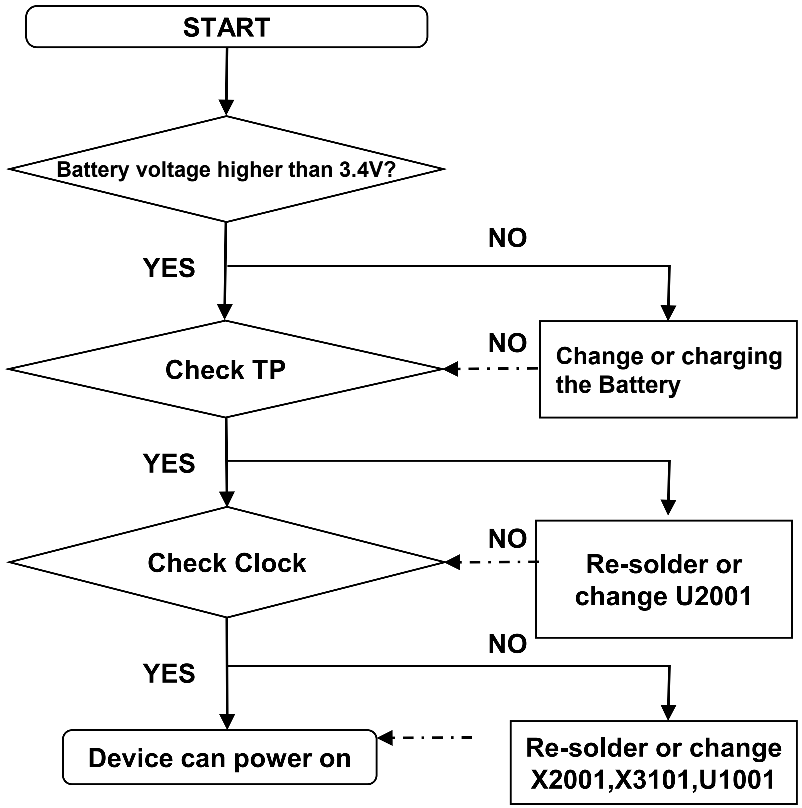


FM Signal Path

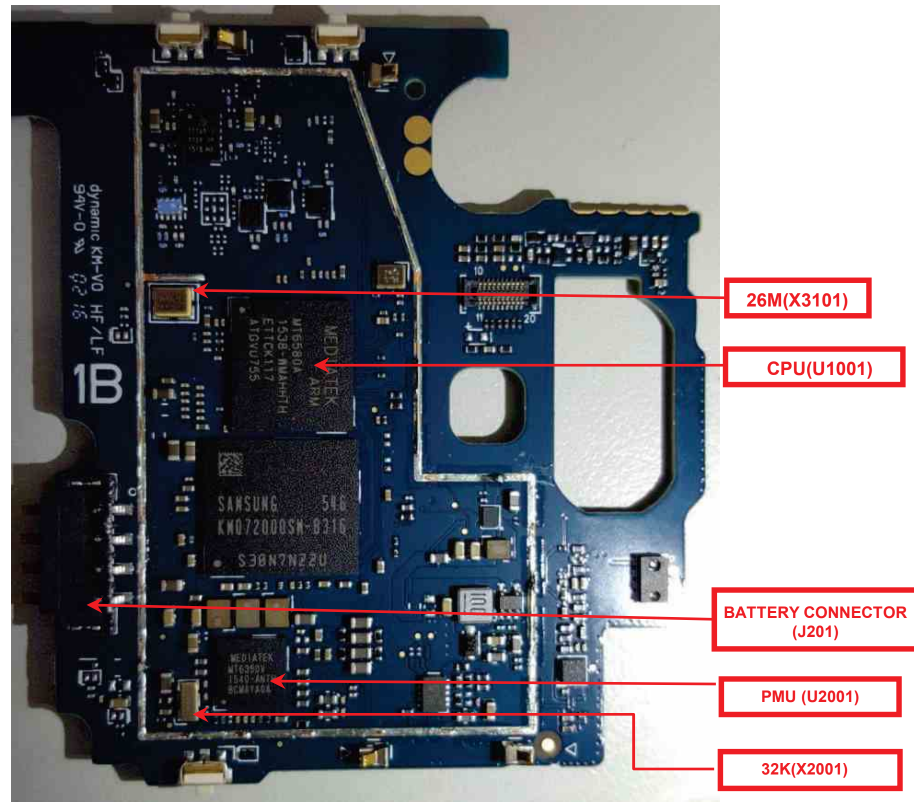


Checking Power signal(Battery connector, PMU, Clock)

Checking Flow



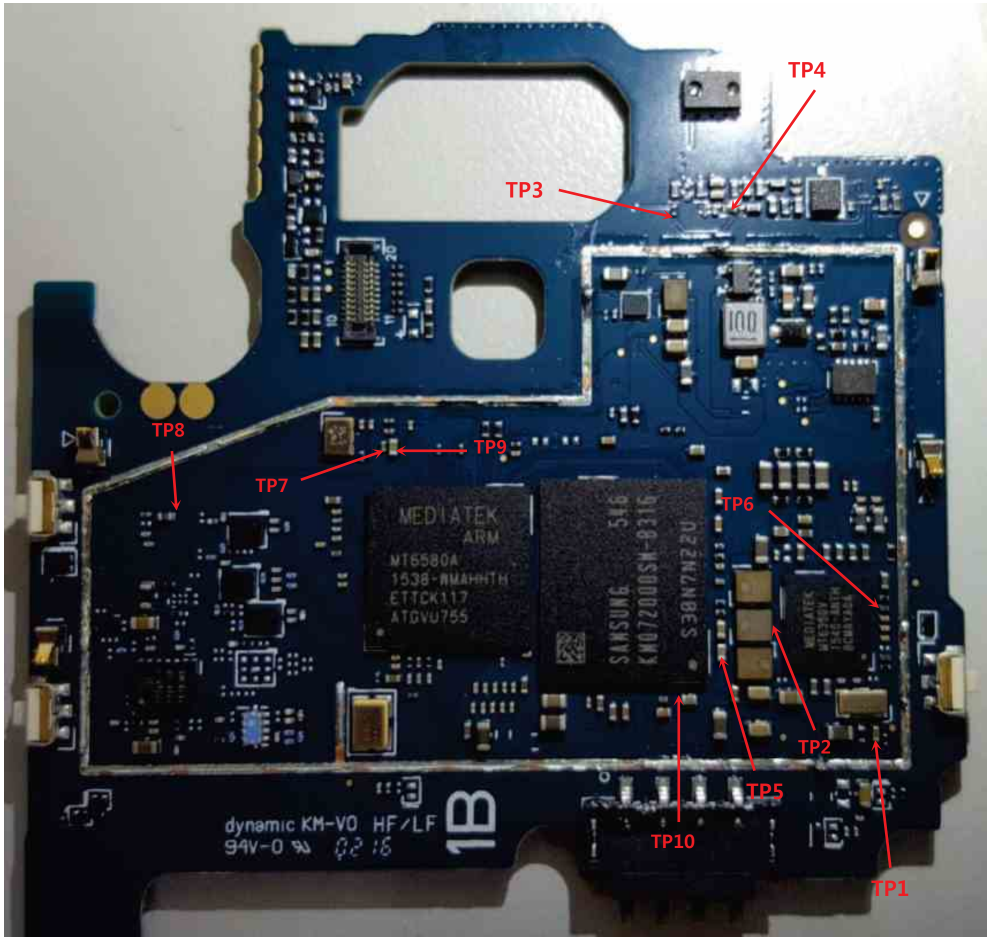
Image



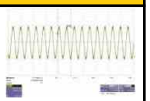
Checking Power signal(Battery connector, PMU, Clock)

Check Output Voltage of U301 (PMU)

Power on voltage check list				
Power domain	**Configurable Voltage**	**Signal Name**	**Measurment Location**	**TP**
VRTC	2.8 V	VRTC	C2022	TP1
VPROC	0.75 ~ 1.3V (25mV/step)	VPROC_PMU	L2002	TP2
VIO18	1.8 V	VIO18 _PMU	R803	TP3
VIO28	2.8 V	VIO28_PMU	R802	TP4
VM	1.2 V	VM_PMU	C4006	TP5
VA	2.8 V	VA_PMU	C2003	TP6
VUSB	3.3V	VUSB_PMU	C1112	TP7
VTCXO	2.8V	VTCXO_PMU_RF	B3201	TP8
VMC	1.8 V/3.3 V	VMC_PMU	C1107	TP9
VEMC_3V3	3.3 V	VEMC_3V3	C4111	TP10



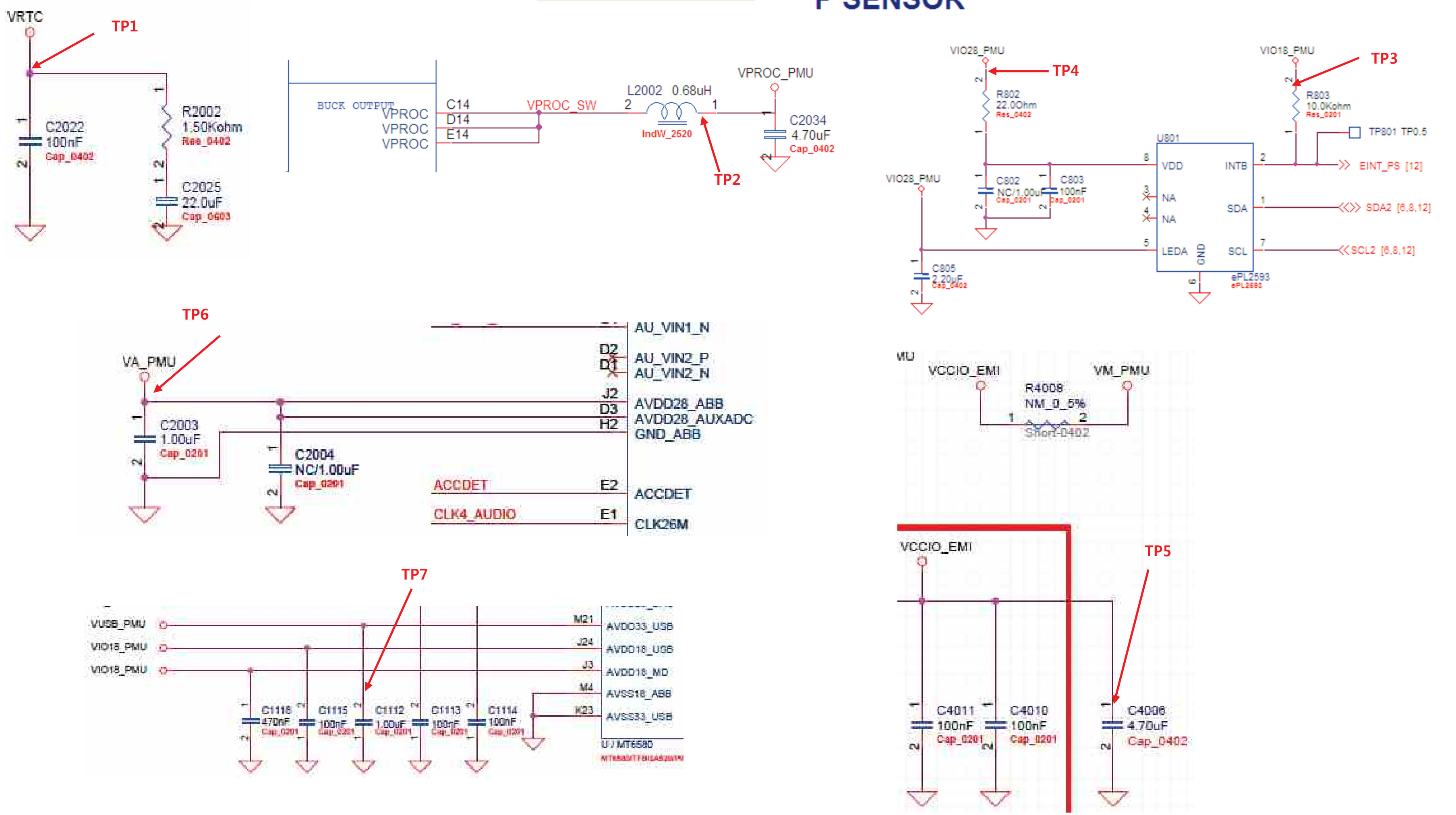
Oscillate frequency measurement

Signal name	Description	Frequency MHz	Measurment Location	Check Clock Result
X2001 32.768 KHz crystal		32.768kHz	C2031	
CLK32K_CK	32.768KHz for MT 6580	32.768kHz	X	
X3101 26MHz Crystal		26MHz	X	

Checking Power signal(Battery connector, PMU, Clock)

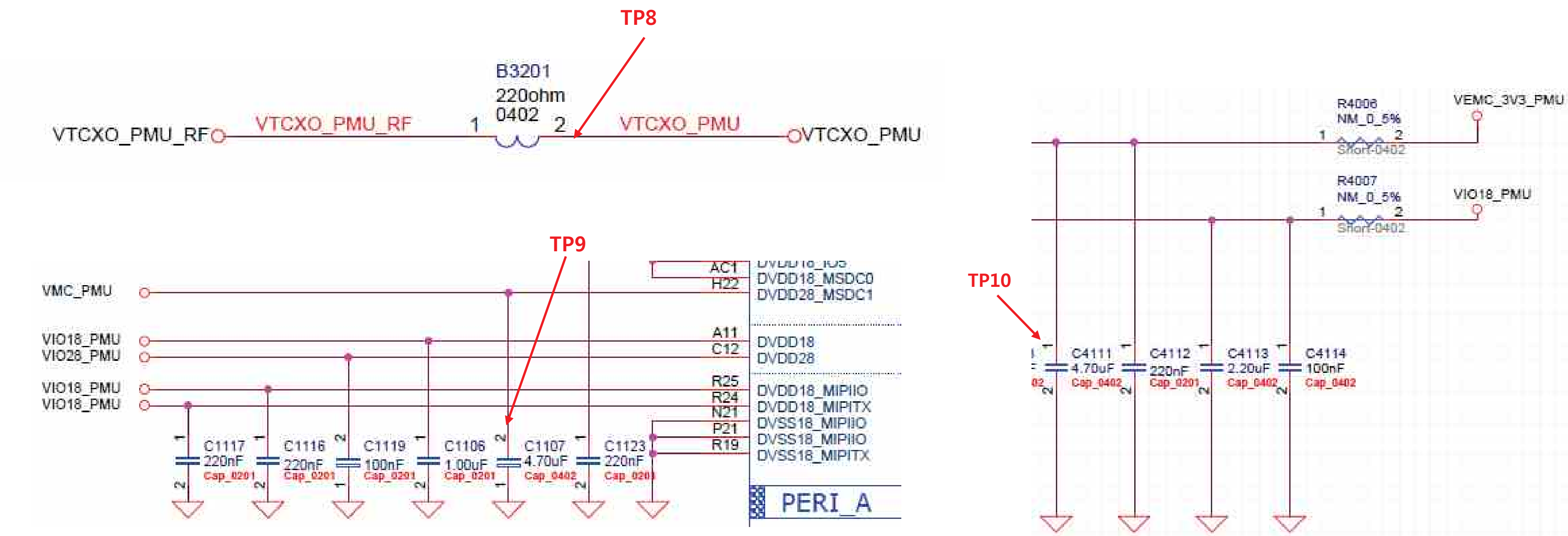
Circuit Diagram

P SENSOR



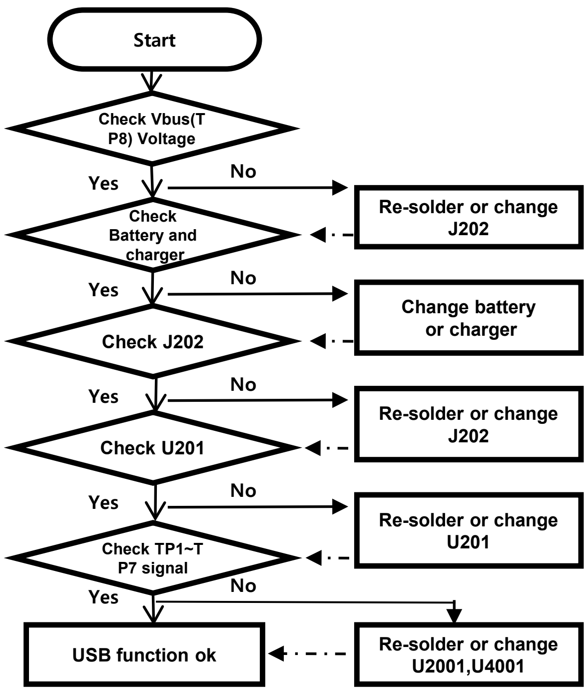
Checking Power signal(Battery connector, PMU, Clock)

Circuit Diagram

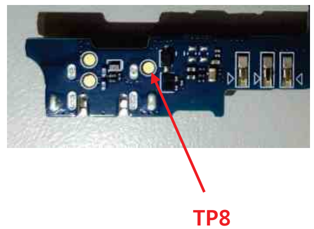


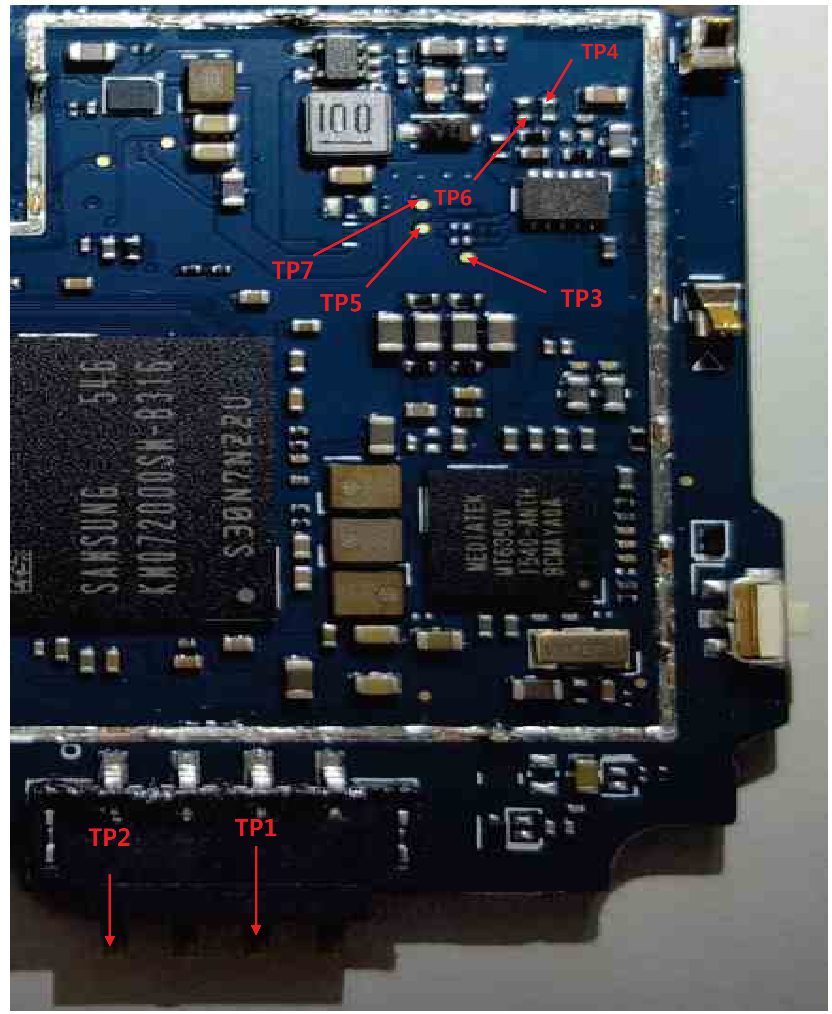
The charging controlled by BB chip MT6580 U1001),PMU chip MT6350(U2001),charge IC RT9536(U201).

Checking Flow



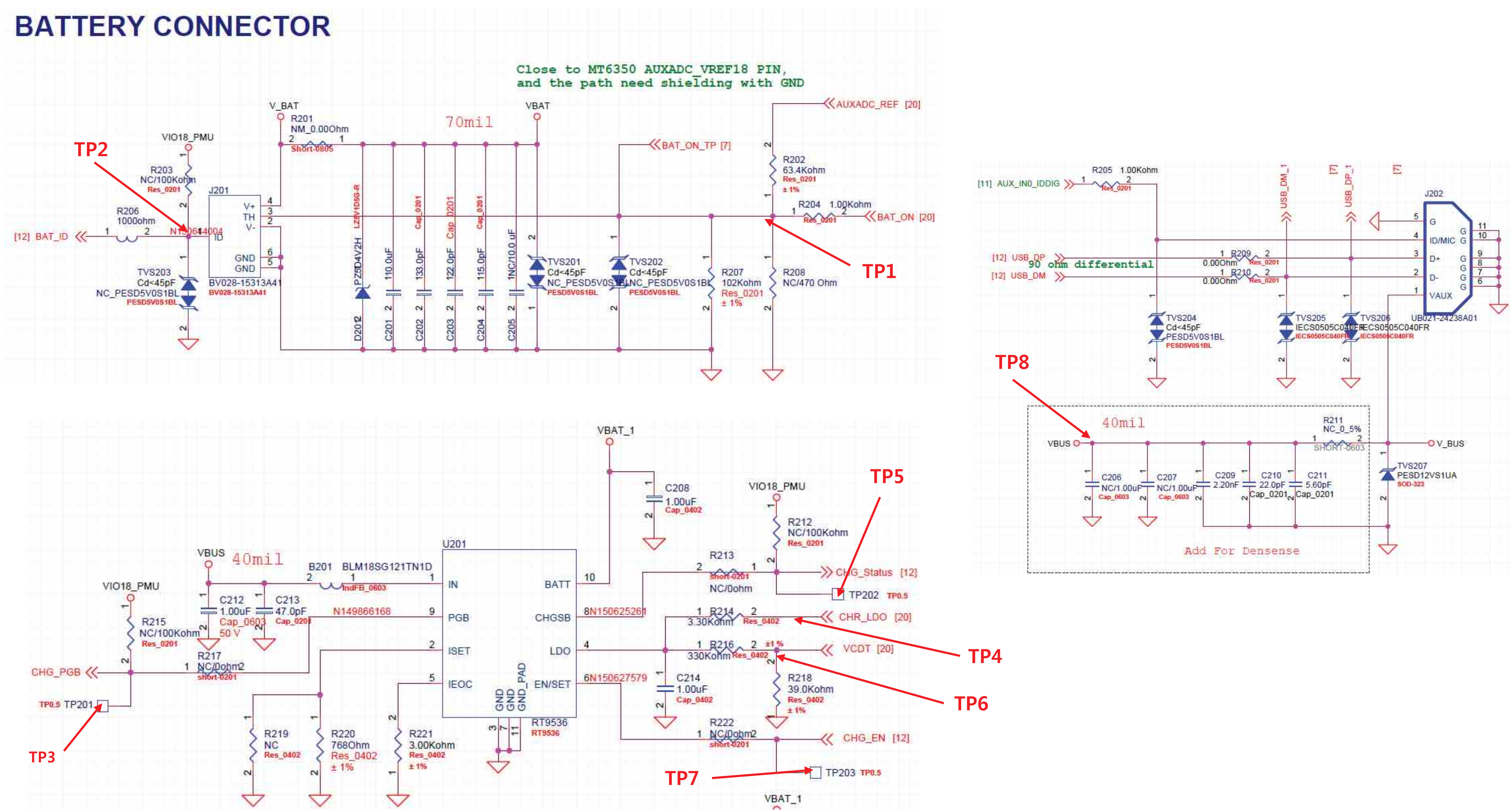
Image





The charging controlled by BB chip MT6580 U1001),PMU chip MT6350(U2001),charge IC RT9536(U201).

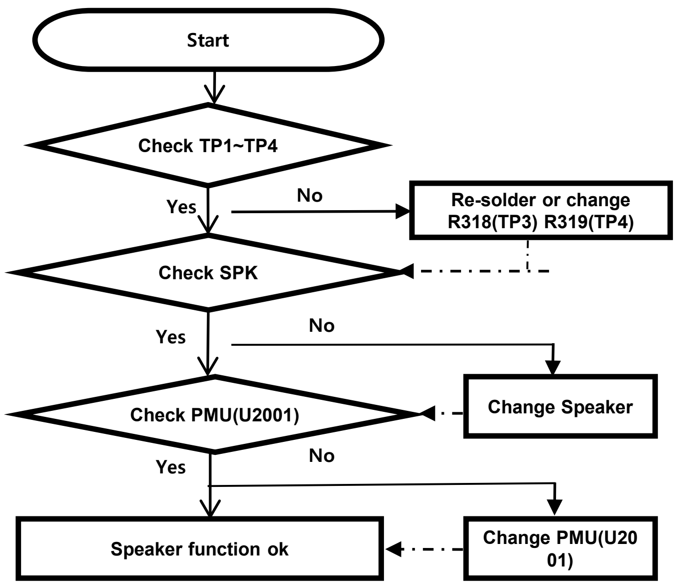
Circuit Diagram



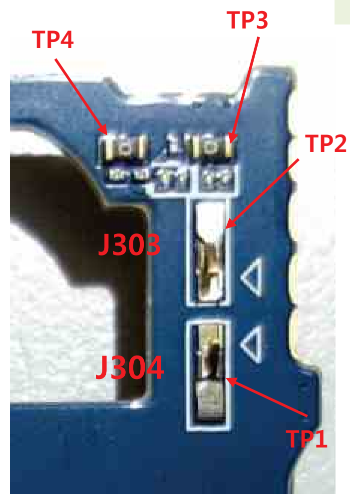
3.9.1 Audio_2in1 speaker

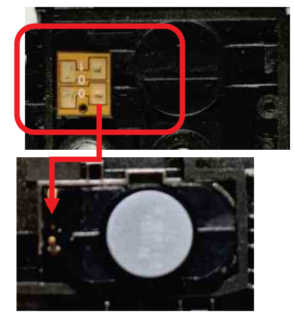
The 2in1 Speaker control signals are generated by PMU chip MT6350(U2001), the PMU chip and the speaker are to be checked out.

Checking Flow

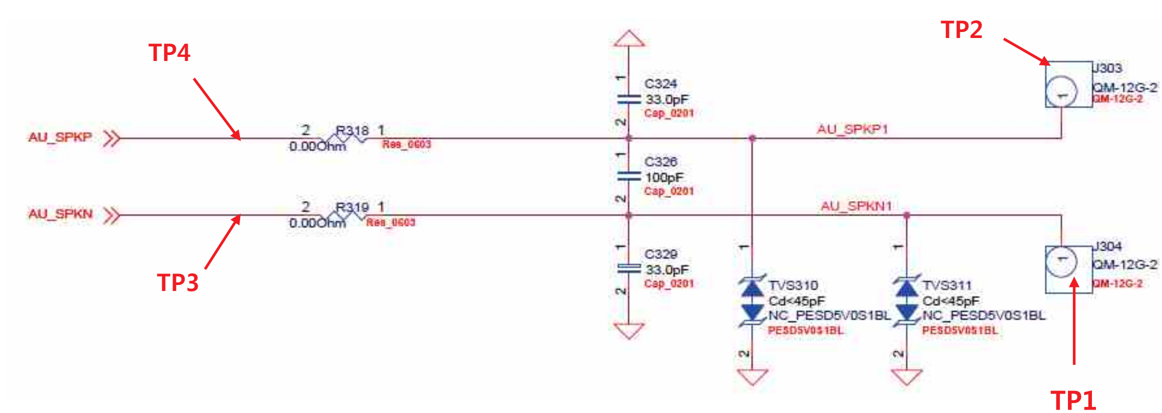


Image





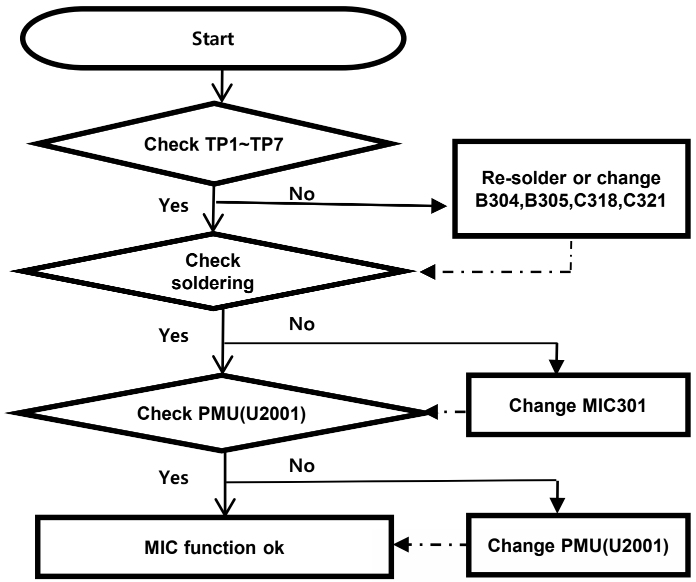
Circuit Diagram



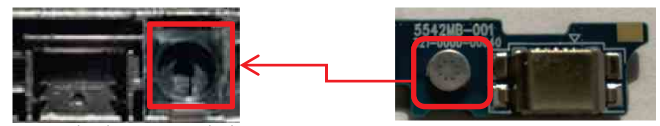
3.9.2 Audio_Main MIC

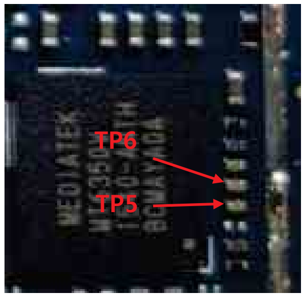
The MIC control signals are generated by PMU chip MT6350(U2001), the PMU chip and the MIC are to be checked out.

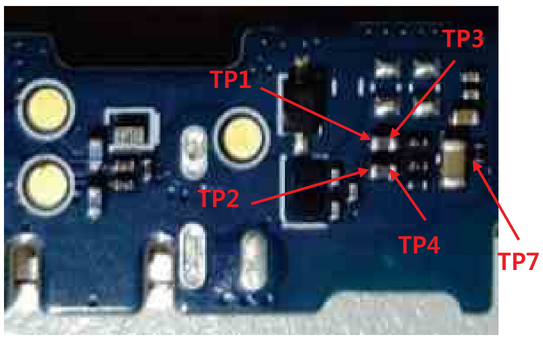
Checking Flow



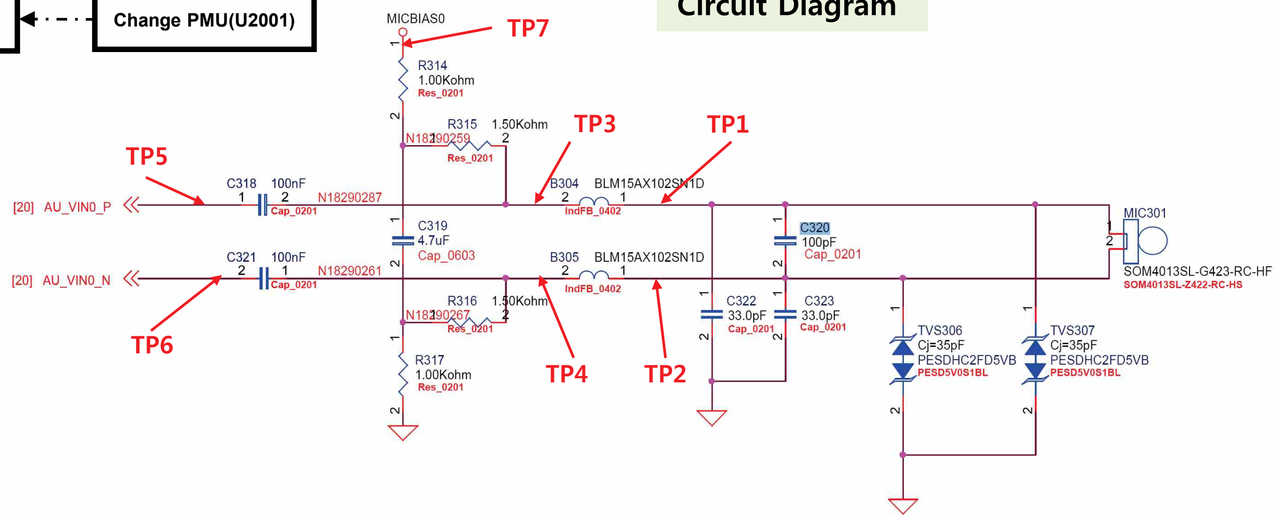
Image


Check Main Mic hole.





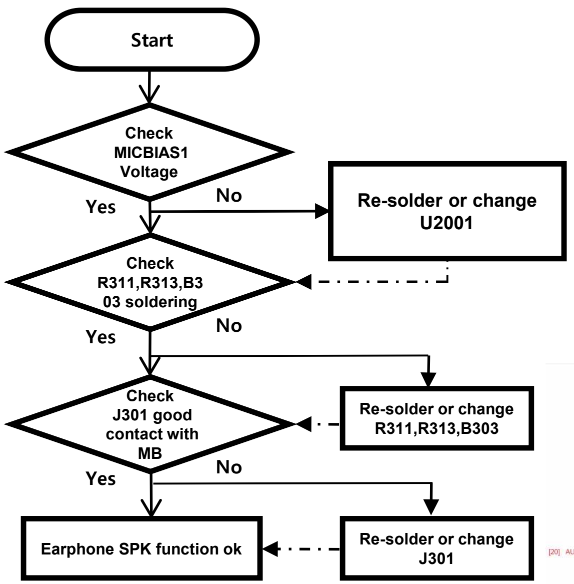
Circuit Diagram



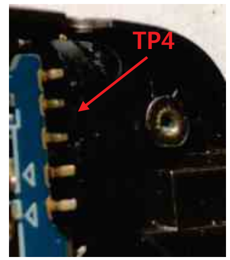
3.9.3 Audio_Ear MIC

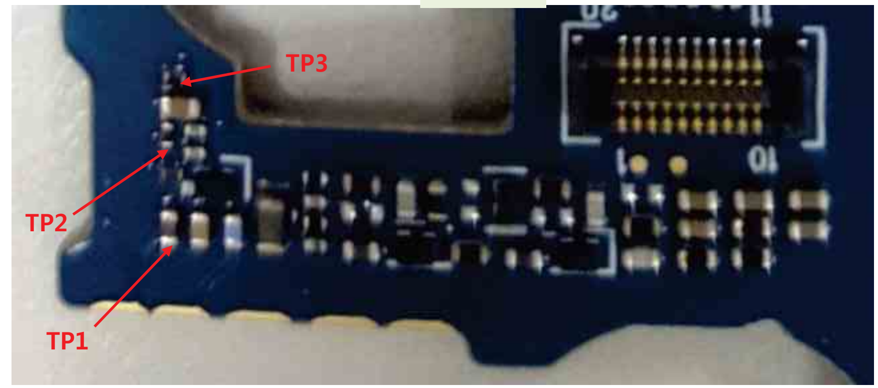
Disable detecting headset insert or No sound from Earphone, Check the Ear Mic and PMU(U2001).

Checking Flow

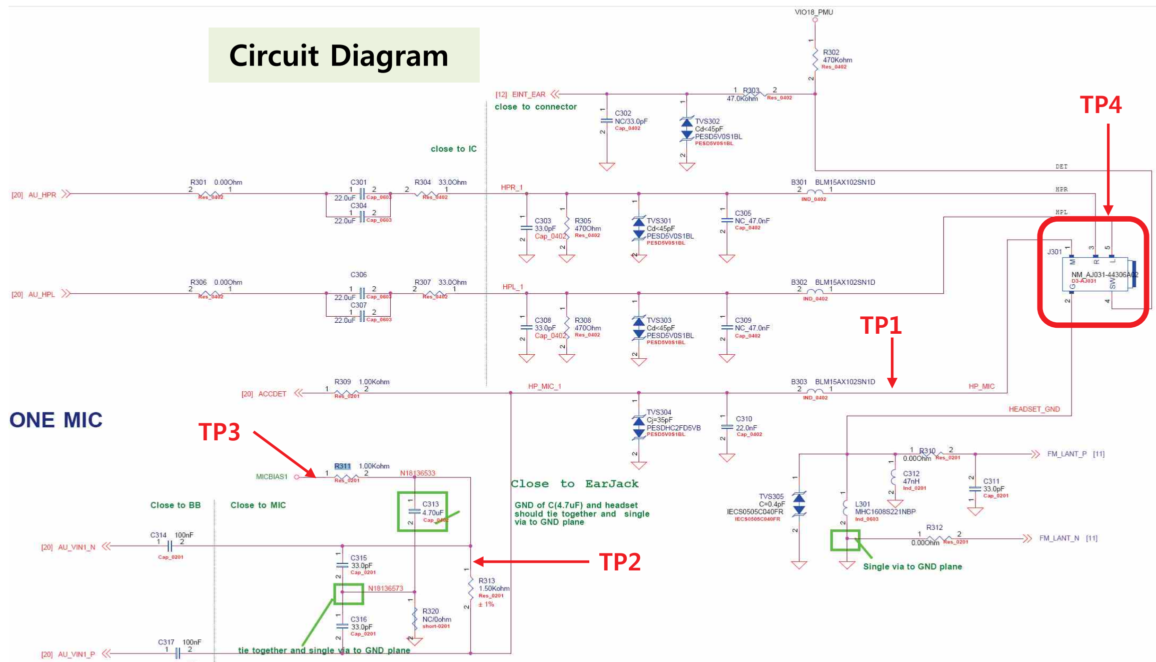


Image



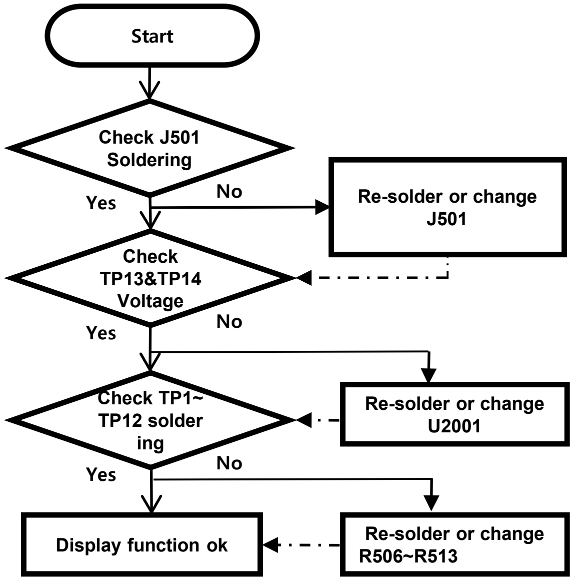


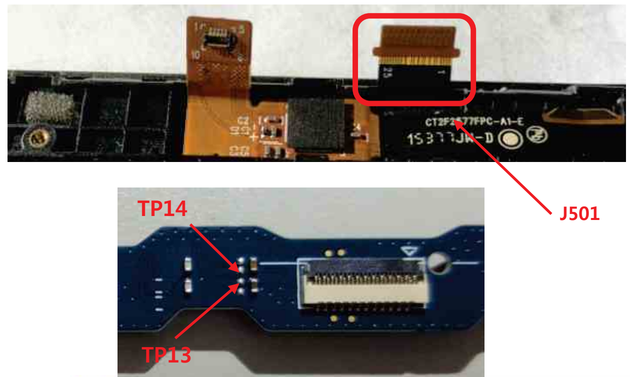
Circuit Diagram



The LCD control signals are generated by MT6580. Its interface is MIPI having four data lanes and one clock lane.

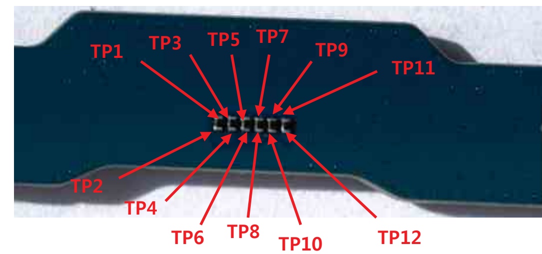
Checking Flow



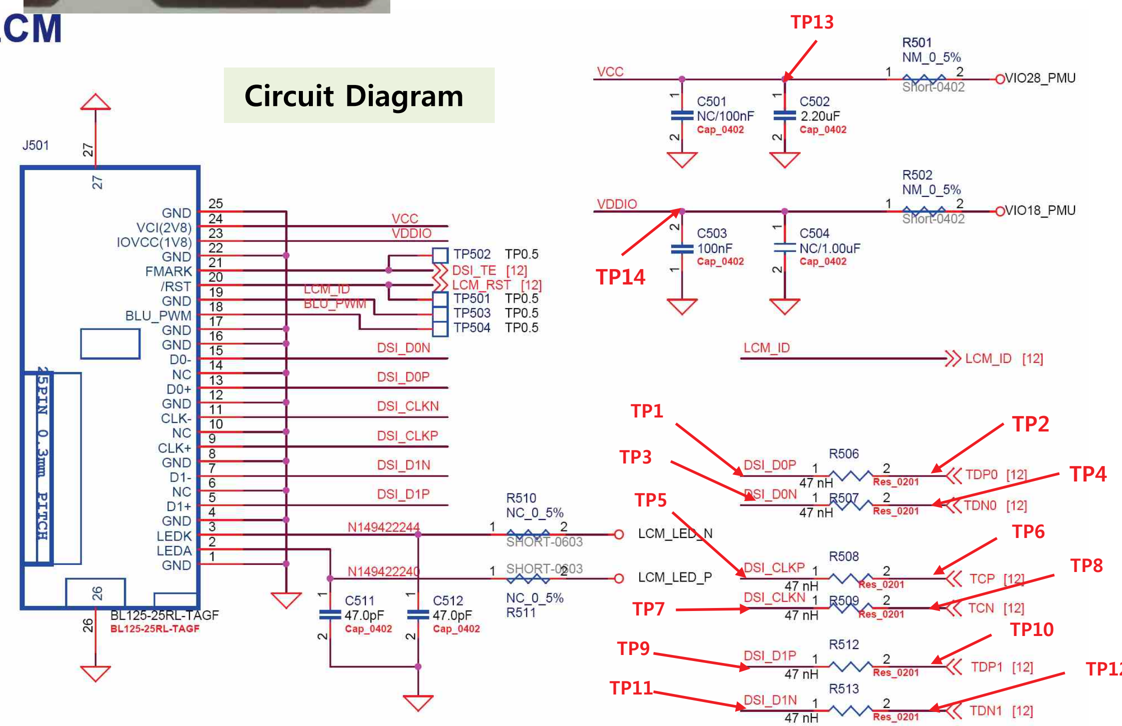


LCM

Image

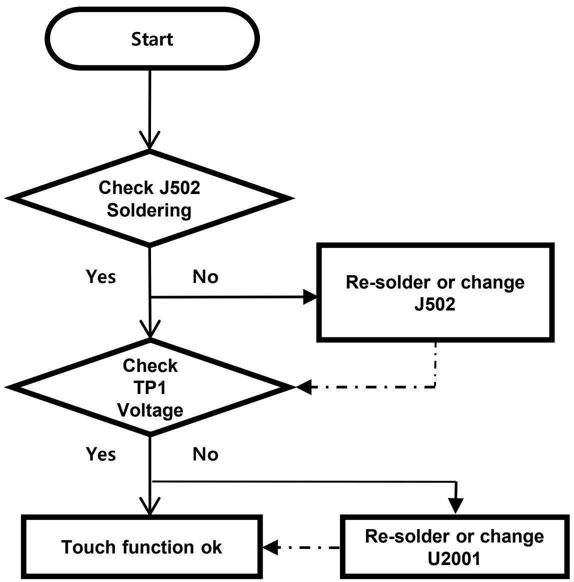


Circuit Diagram

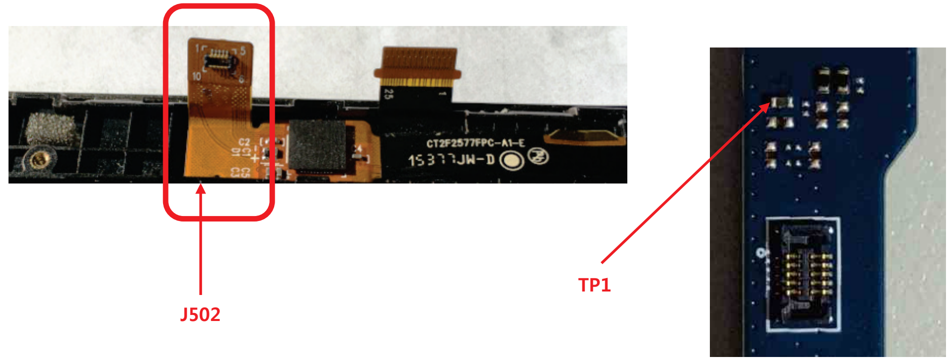


The Touch control signals are generated by MT6580 and MT6350. It is assembled with LCD.

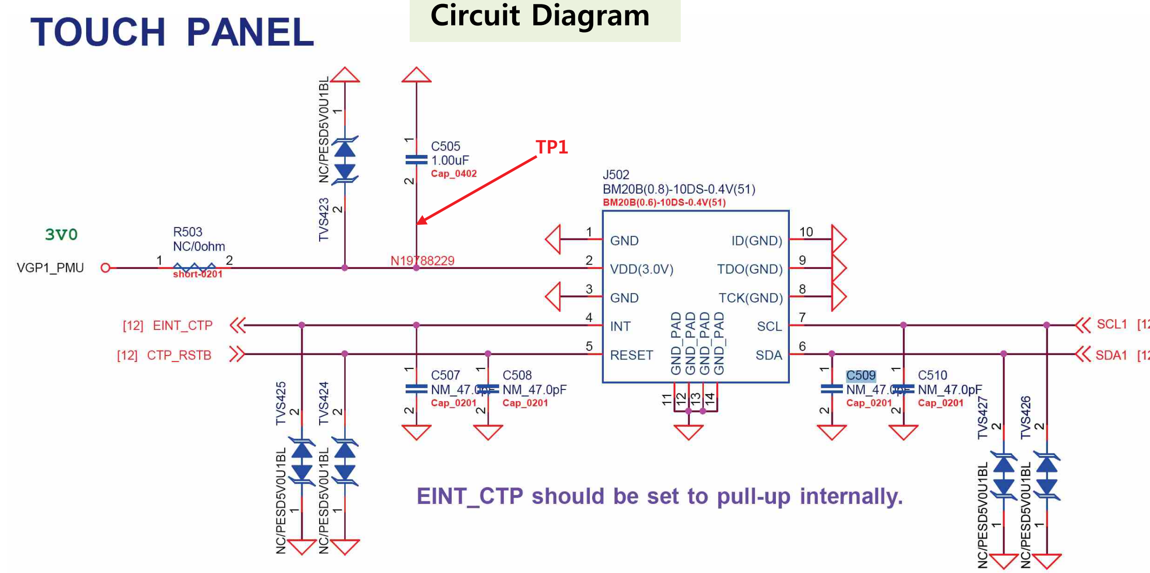
Checking Flow



Image



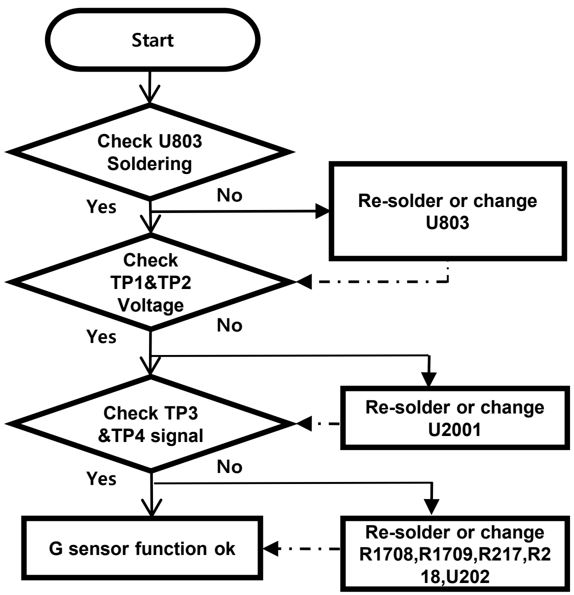
Circuit Diagram



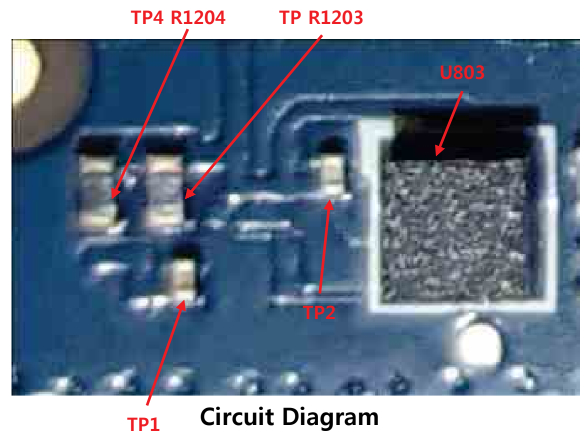
3.12.1 Checking G sensor Block

The G sensor is calibrated by using SW algorithm.

Checking Flow

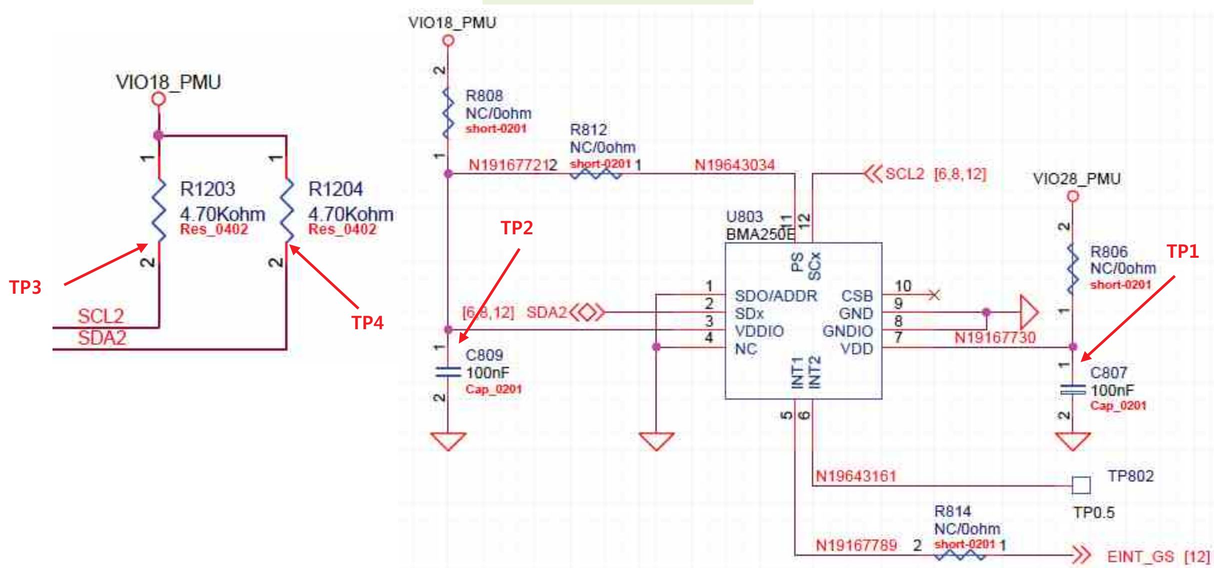


Image



Circuit Diagram
: Next Page

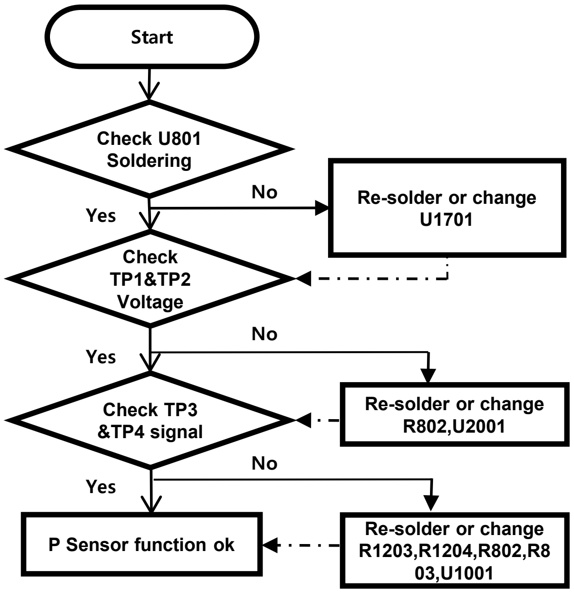
Circuit Diagram



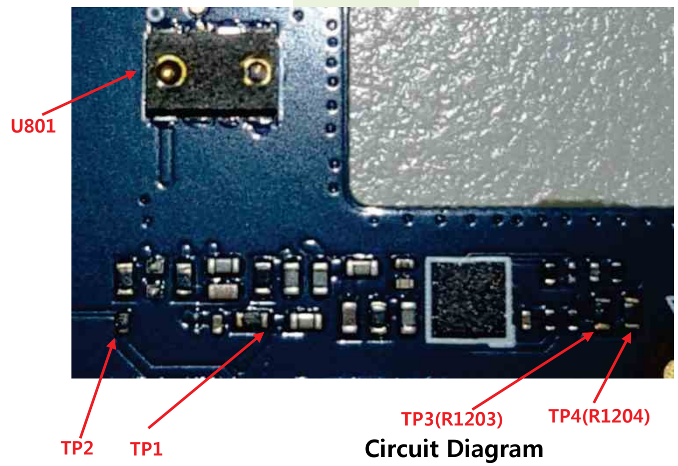
3.12.2 Checking proximity sensor Block

Proximity Sensor is worked as below: Send Key click, Phone number click, Call connected, Object moved at the sensor, Control the screen's on/off operation automatically.

Checking Flow

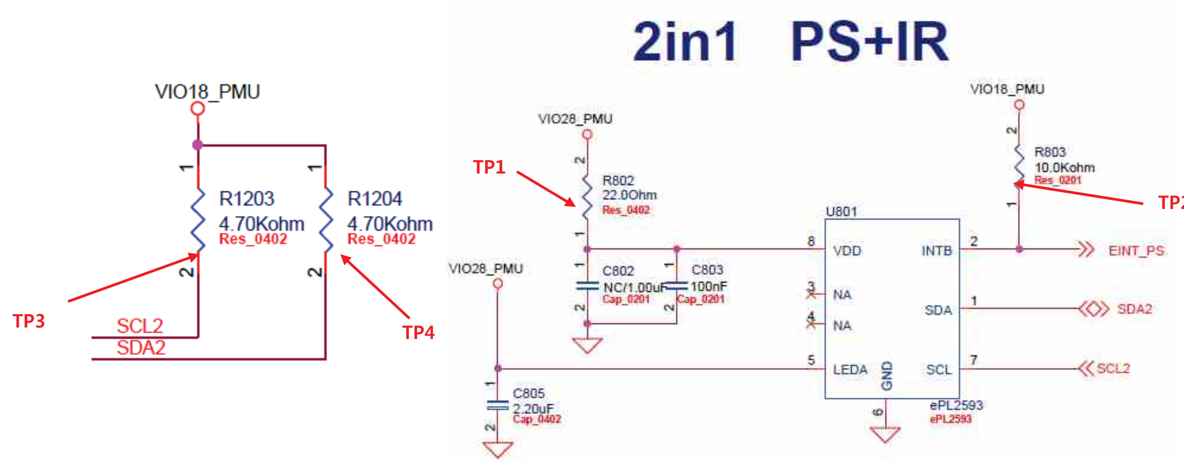


Image



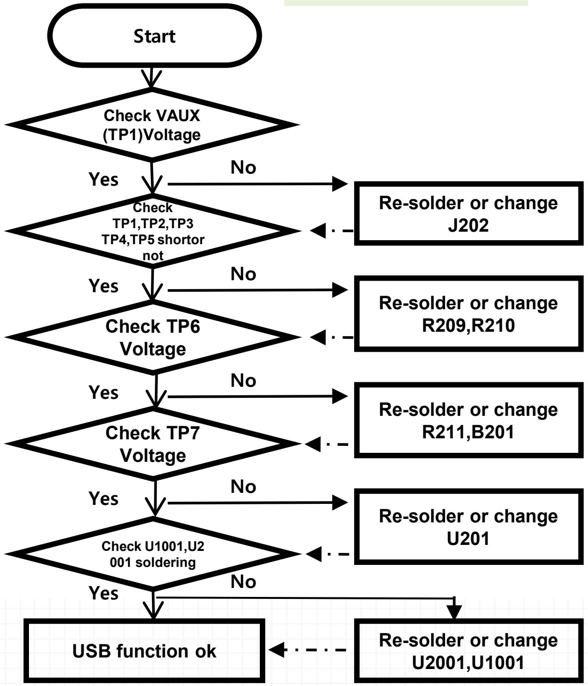
Circuit Diagram
: Next Page

Circuit Diagram

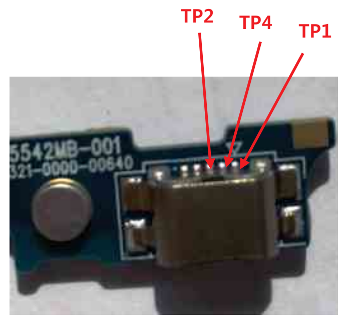


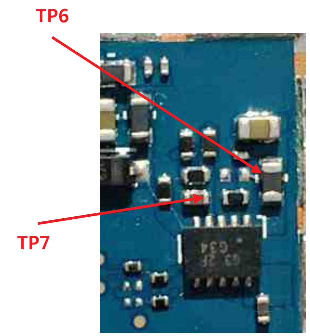
I/O connector is used as the USB port.

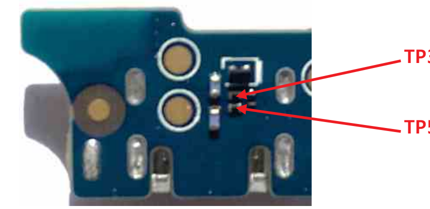
Checking Flow



Image

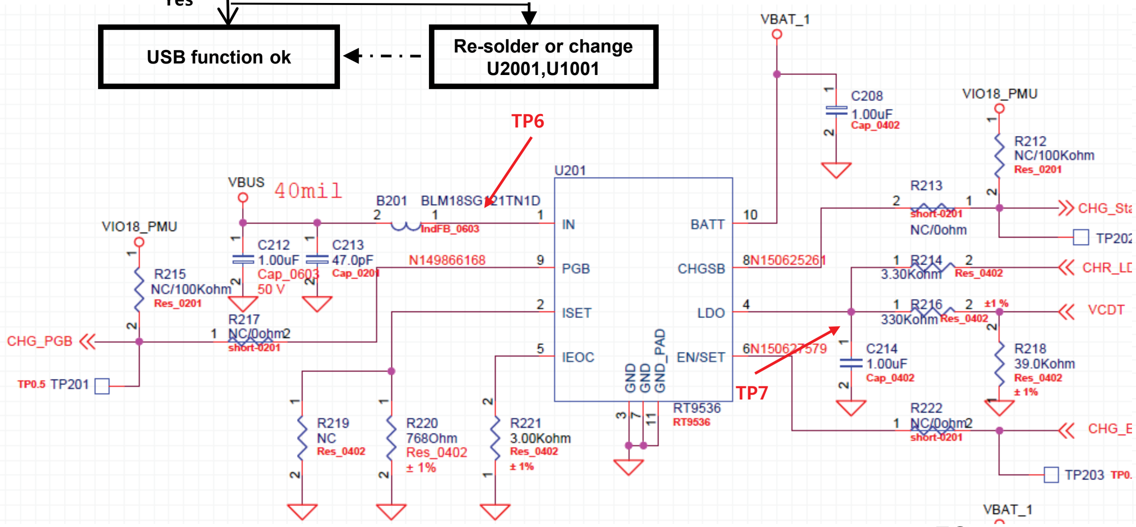


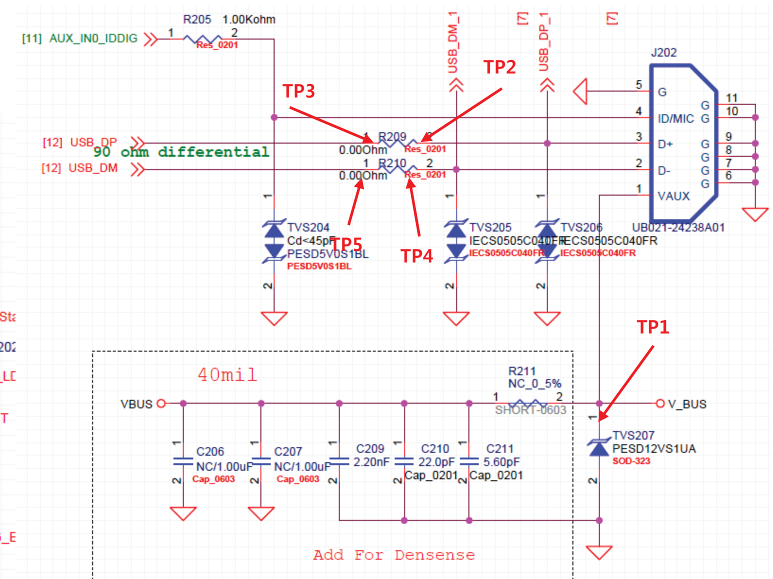




Circuit Diagram
: Next Page

Circuit Diagram

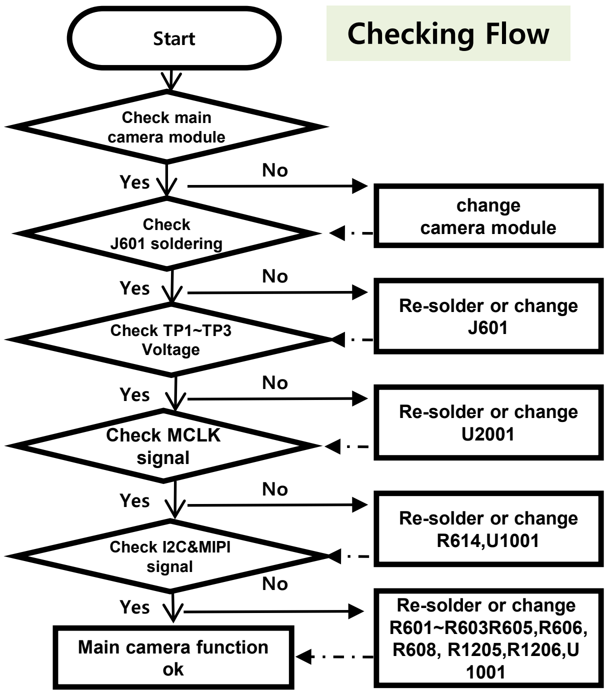




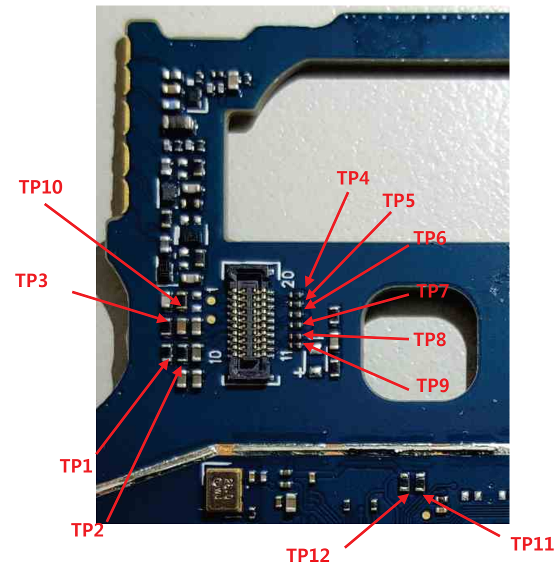
3.14.1 Checking Main camera Block

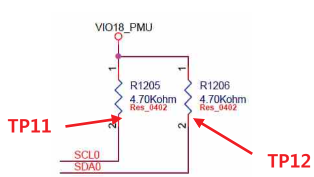
5M FF camera control signals are generated by MT6580(U1001). And powered by PMU(U2001).

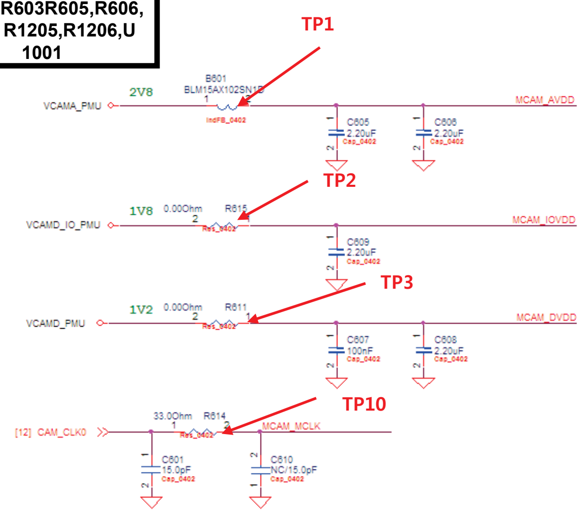
Checking Flow

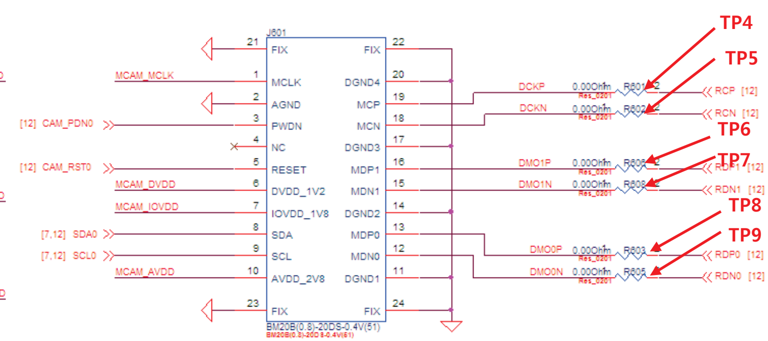


Image





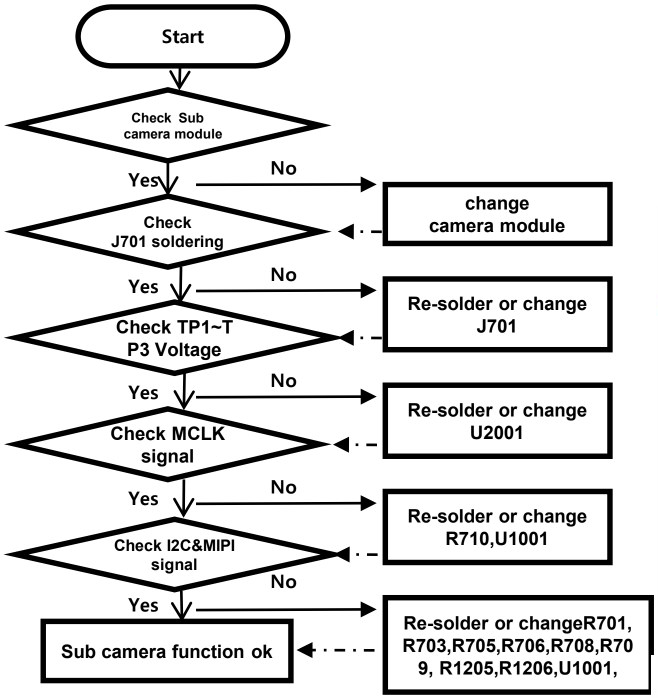




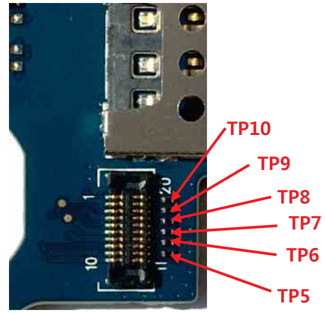
3.14.2 Checking sub camera Block

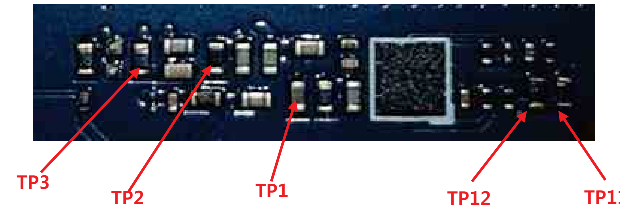
Sub camera control signals are generated by MT6580(U1001). And powered by PMU(U2001).

Checking Flow

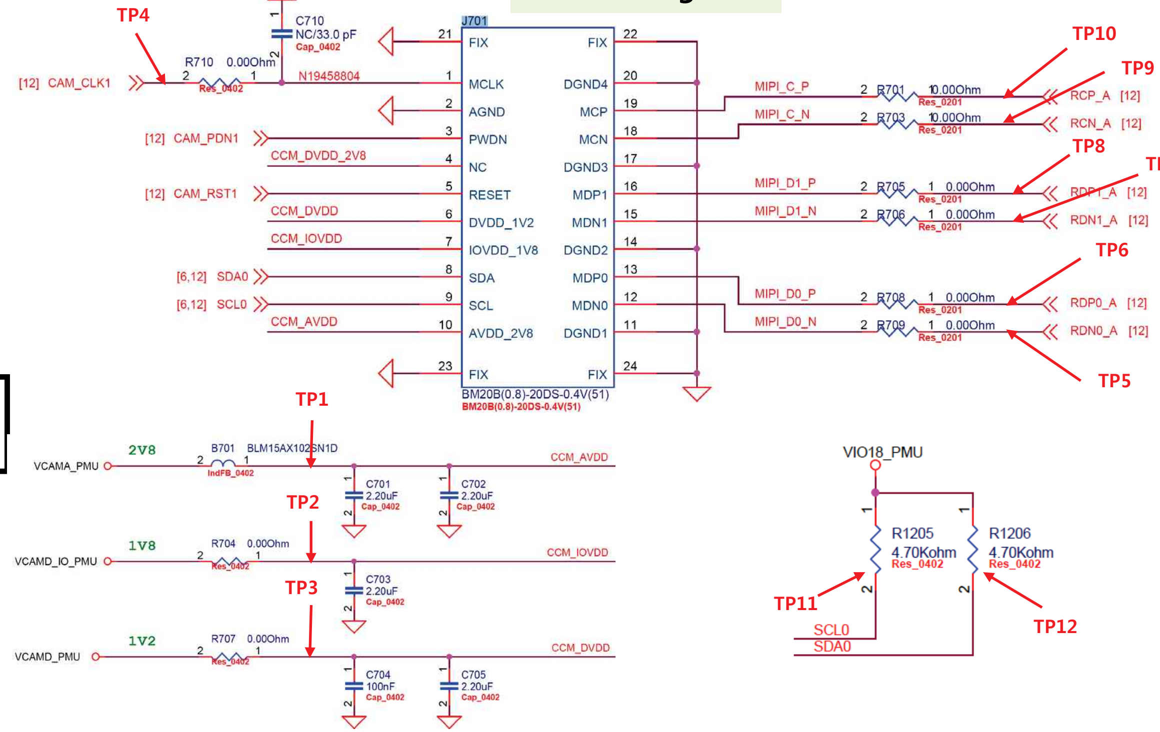


Image

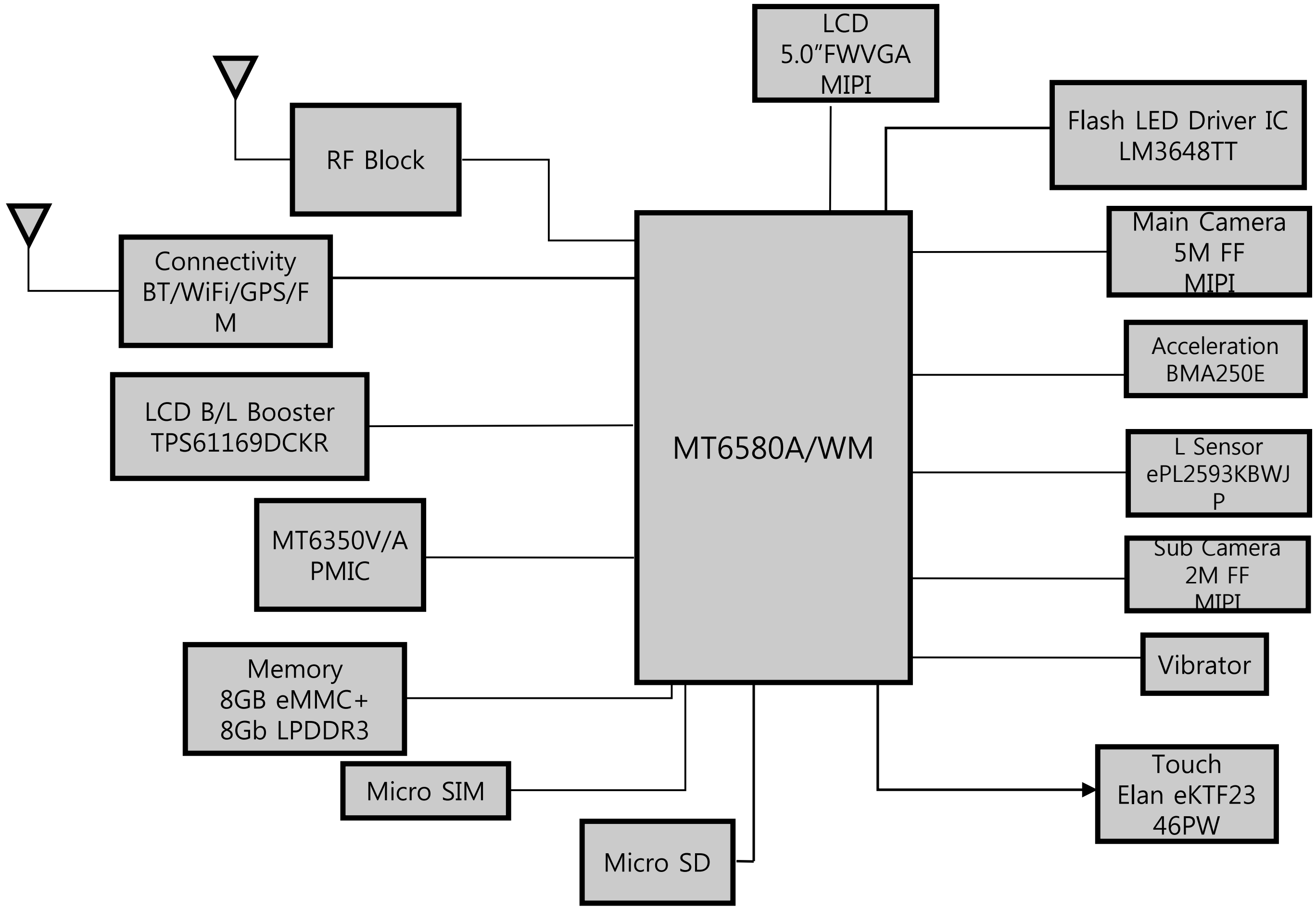


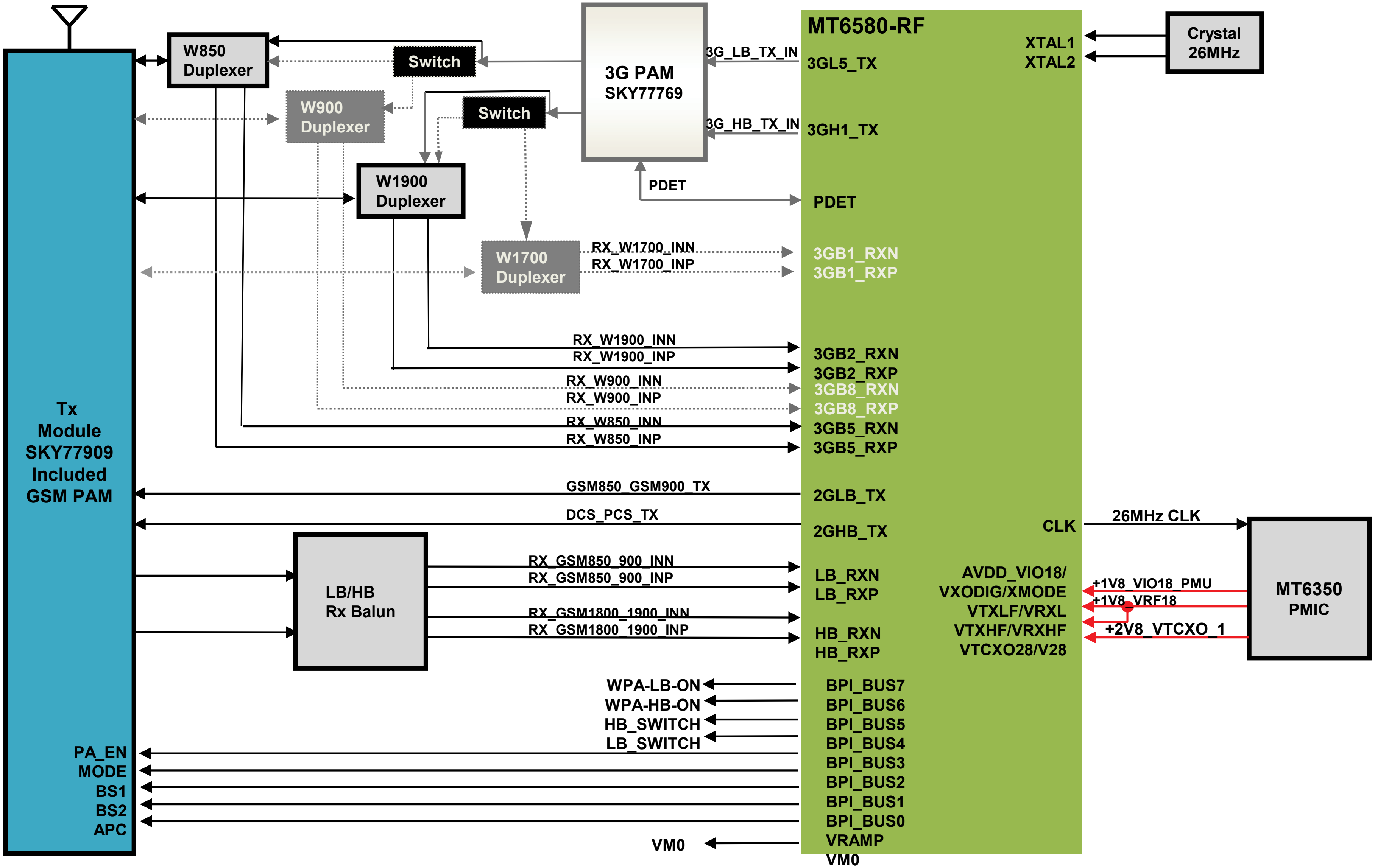


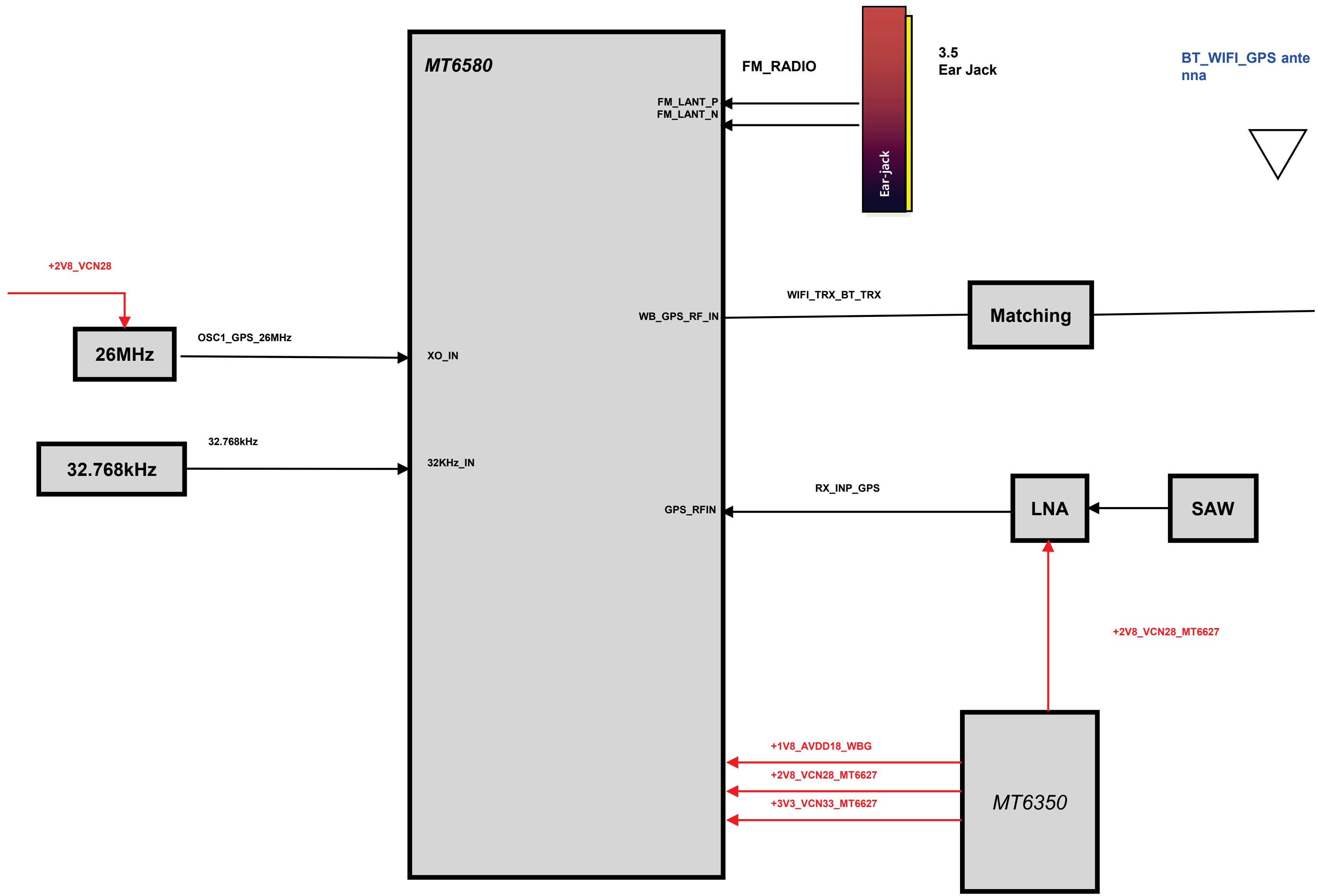
Circuit Diagram



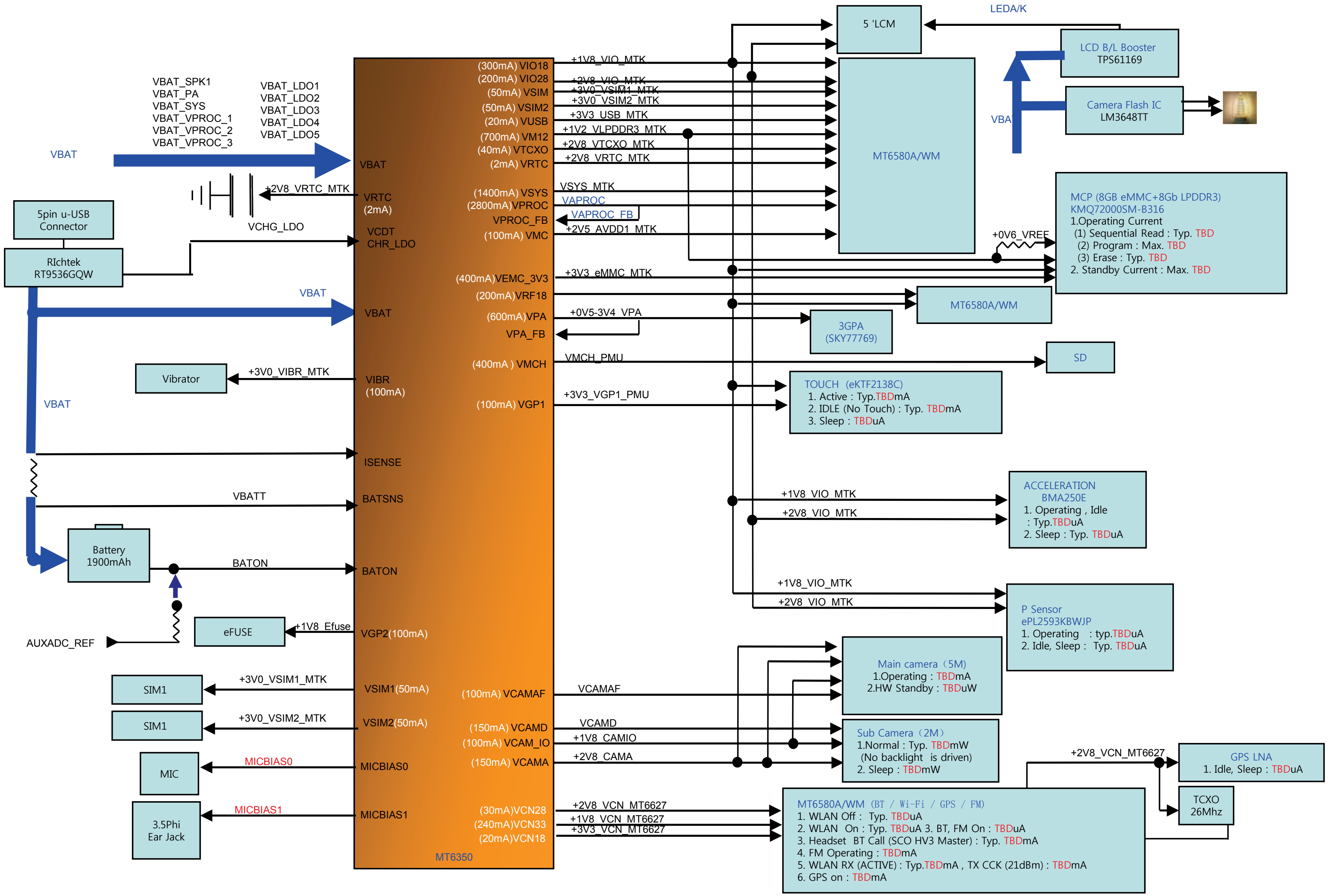
4. BLOCK DIAGRAM

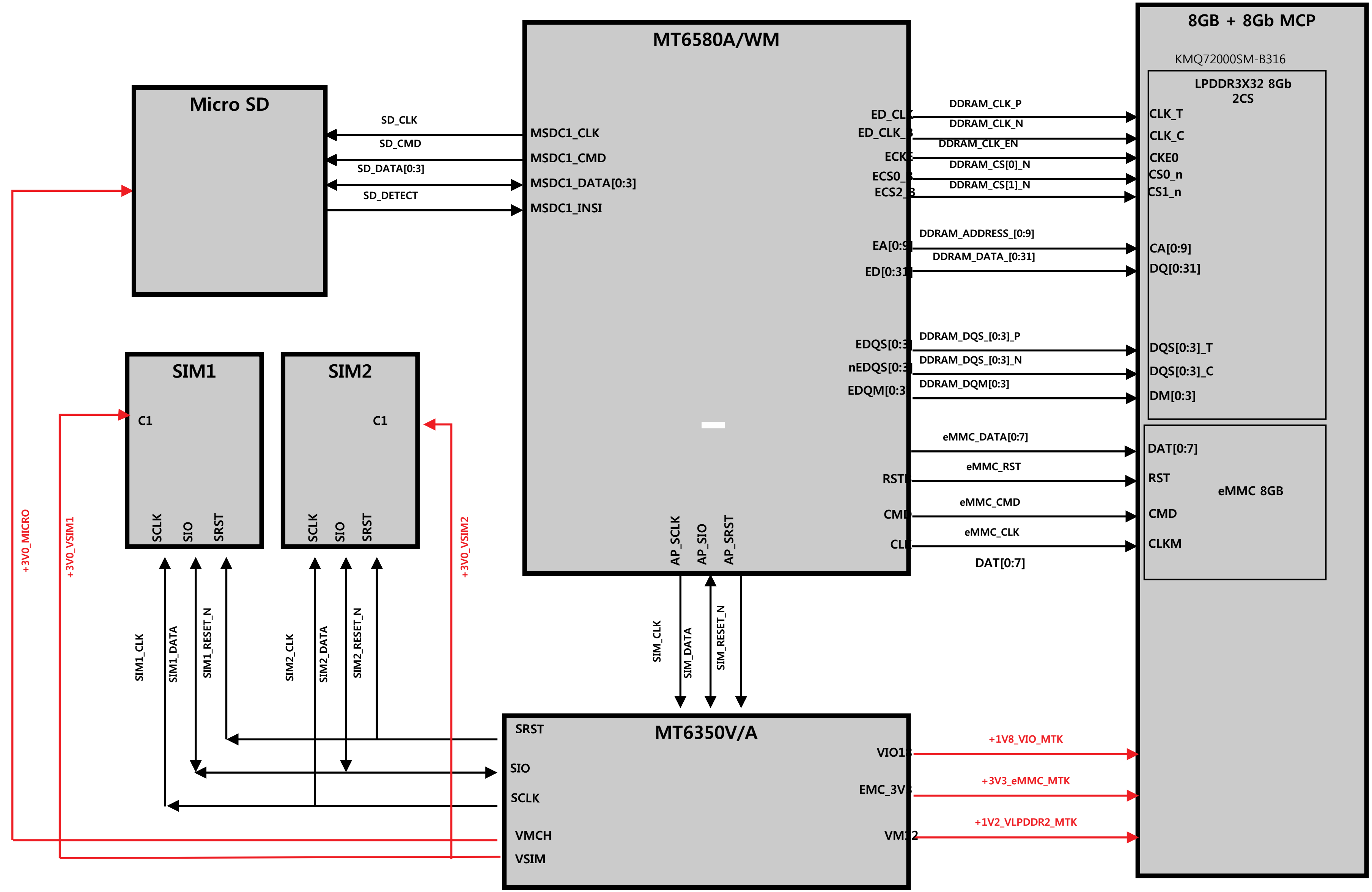


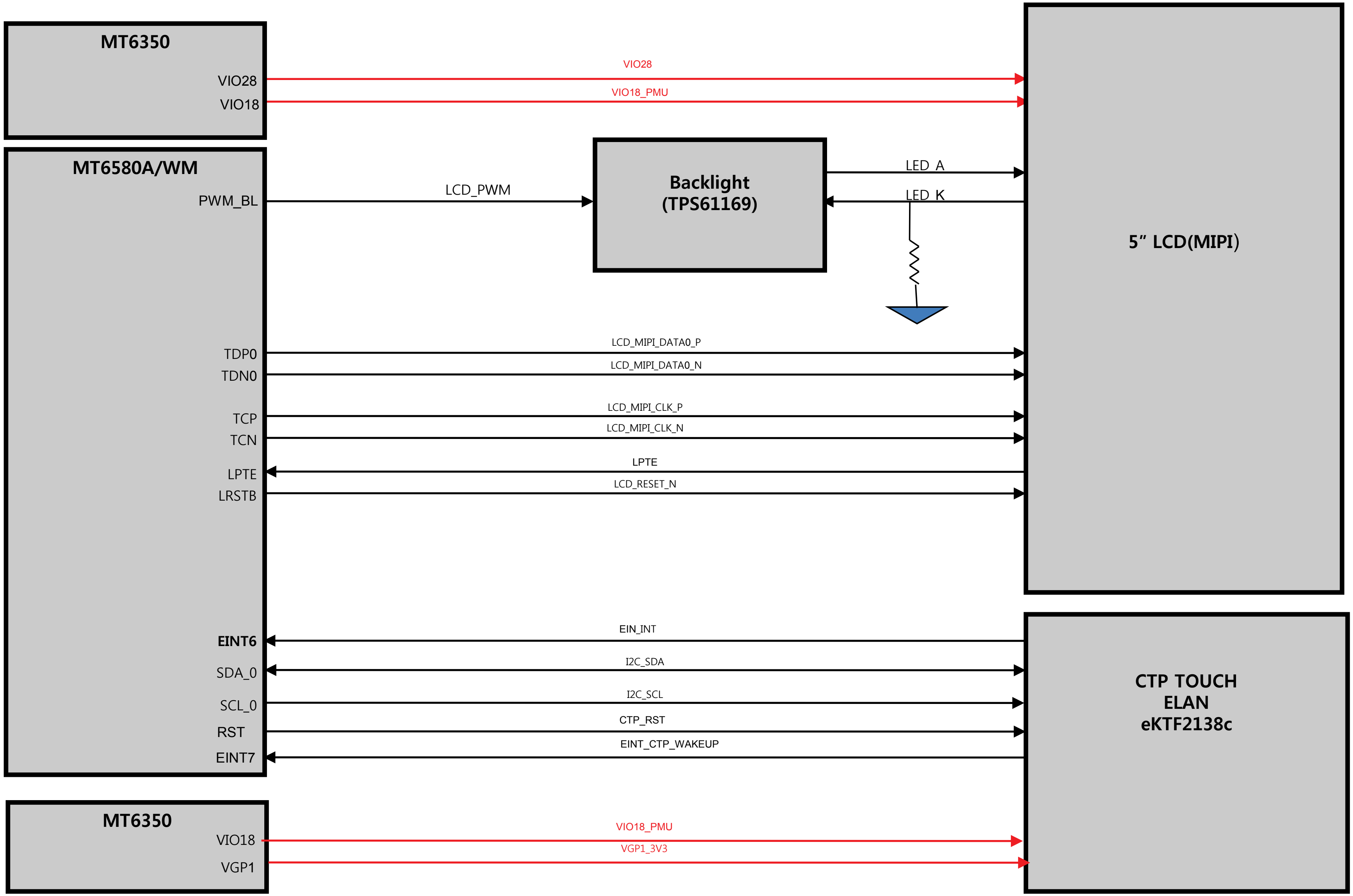

W850
Duplexer
Switch
3G PAM
SKY77769
MT6580-RF
Crystal
26MHz
3G_LB_TX_IN
3GL5_TX
XTAL1
XTAL2
W900
Duplexer
Switch
3G_HB_TX_IN
3GH1_TX
W1900
Duplexer
PDET
PDET
W1700
Duplexer
RX_W1700_INN
RX_W1700_INP
3GB1_RXN
3GB1_RXP
RX_W1900_INN
RX_W1900_INP
RX_W900_INN
RX_W900_INP
RX_W850_INN
RX_W850_INP
3GB2_RXN
3GB2_RXP
3GB8_RXN
3GB8_RXP
3GB5_RXN
3GB5_RXP
Tx
Module
SKY77909
Included
GSM PAM
GSM850_GSM900_TX
2GLB_TX
DCS_PCS_TX
2GHB_TX
CLK
26MHz CLK
MT6350
PMIC
LB/HB
Rx Balun
RX_GSM850_900_INN
RX_GSM850_900_INP
RX_GSM1800_1900_INN
RX_GSM1800_1900_INP
LB_RXN
LB_RXP
HB_RXN
HB_RXP
AVDD_VIO18/
VXODIG/XMODE
VTXLF/VRXL
VTXHF/VRXHF
VTCXO28/V28
+1V8_VIO18_PMU
+1V8_VRF18
+2V8_VTCXO_1
WPA-LB-ON
WPA-HB-ON
HB_SWITCH
LB_SWITCH
BPI_BUS7
BPI_BUS6
BPI_BUS5
BPI_BUS4
BPI_BUS3
BPI_BUS2
BPI_BUS1
BPI_BUS0
VRAMP
VM0
PA_EN
MODE
BS1
BS2
APC
VM0

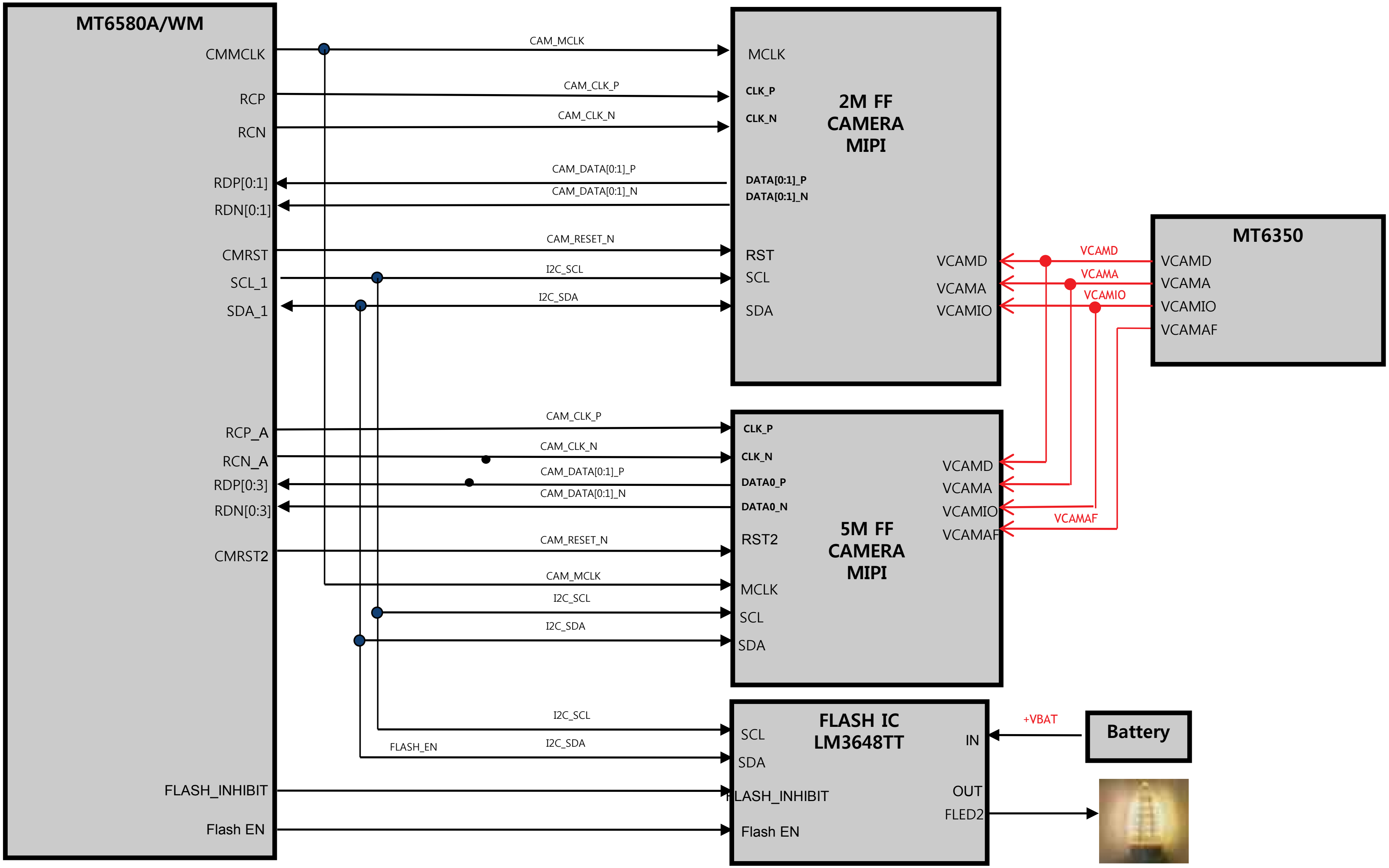

MT6580
FM_RADIO
FM_LANT_P
FM_LANT_N
Ear-jack
3.5
Ear Jack
BT_WIFI_GPS ante
nna
+2V8_VCN28
WIFI_TRX_BT_TRX
WB_GPS_RF_IN
Matching
26MHz
OSC1_GPS_26MHz
XO_IN
32.768kHz
32.768kHz
32KHz_IN
RX_INP_GPS
GPS_RFIN
LNA
SAW
+2V8_VCN28_MT6627
+1V8_AVDD18_WBG
+2V8_VCN28_MT6627
+3V3_VCN33_MT6627
MT6350

4. Power block diagram

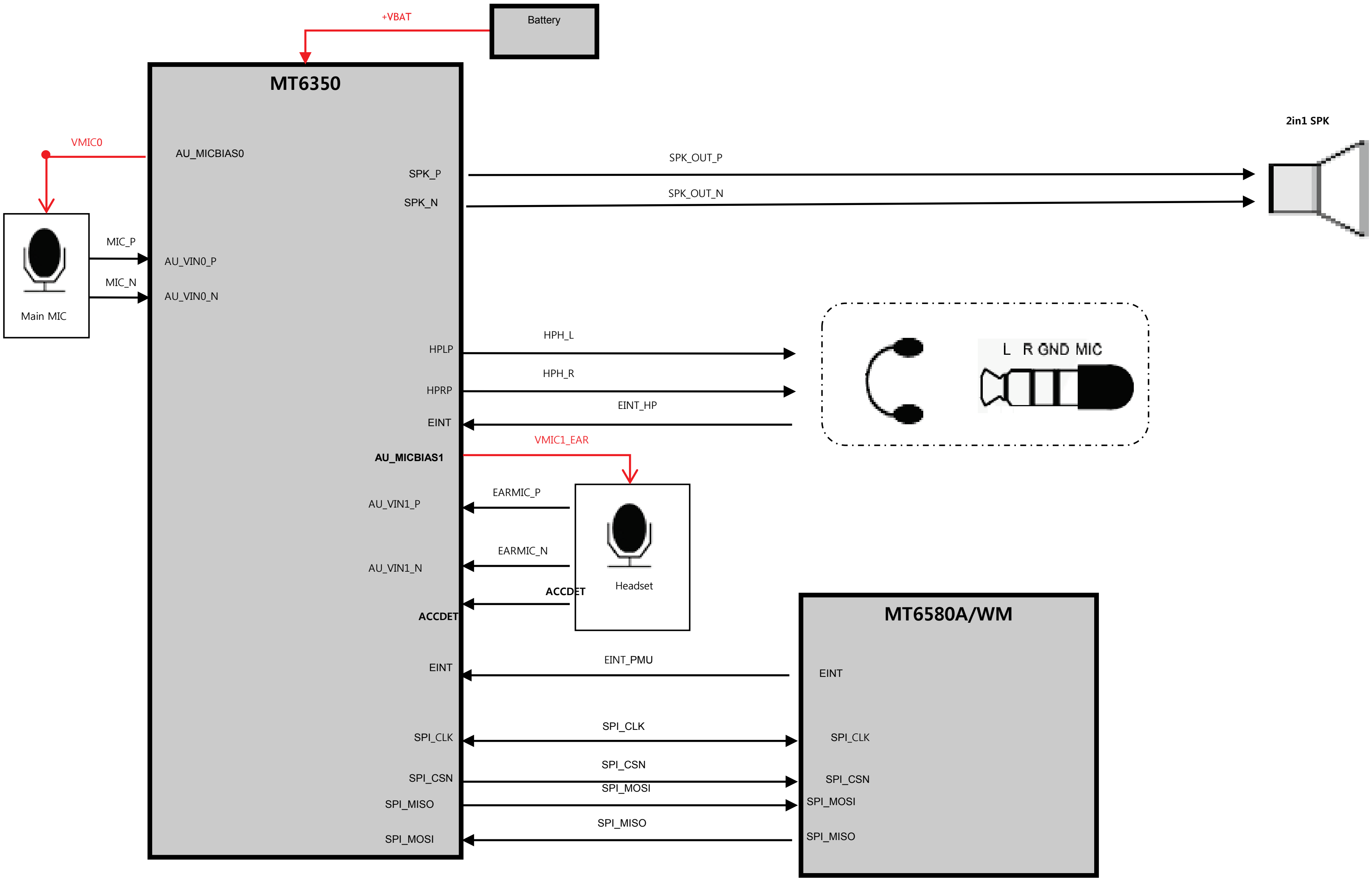


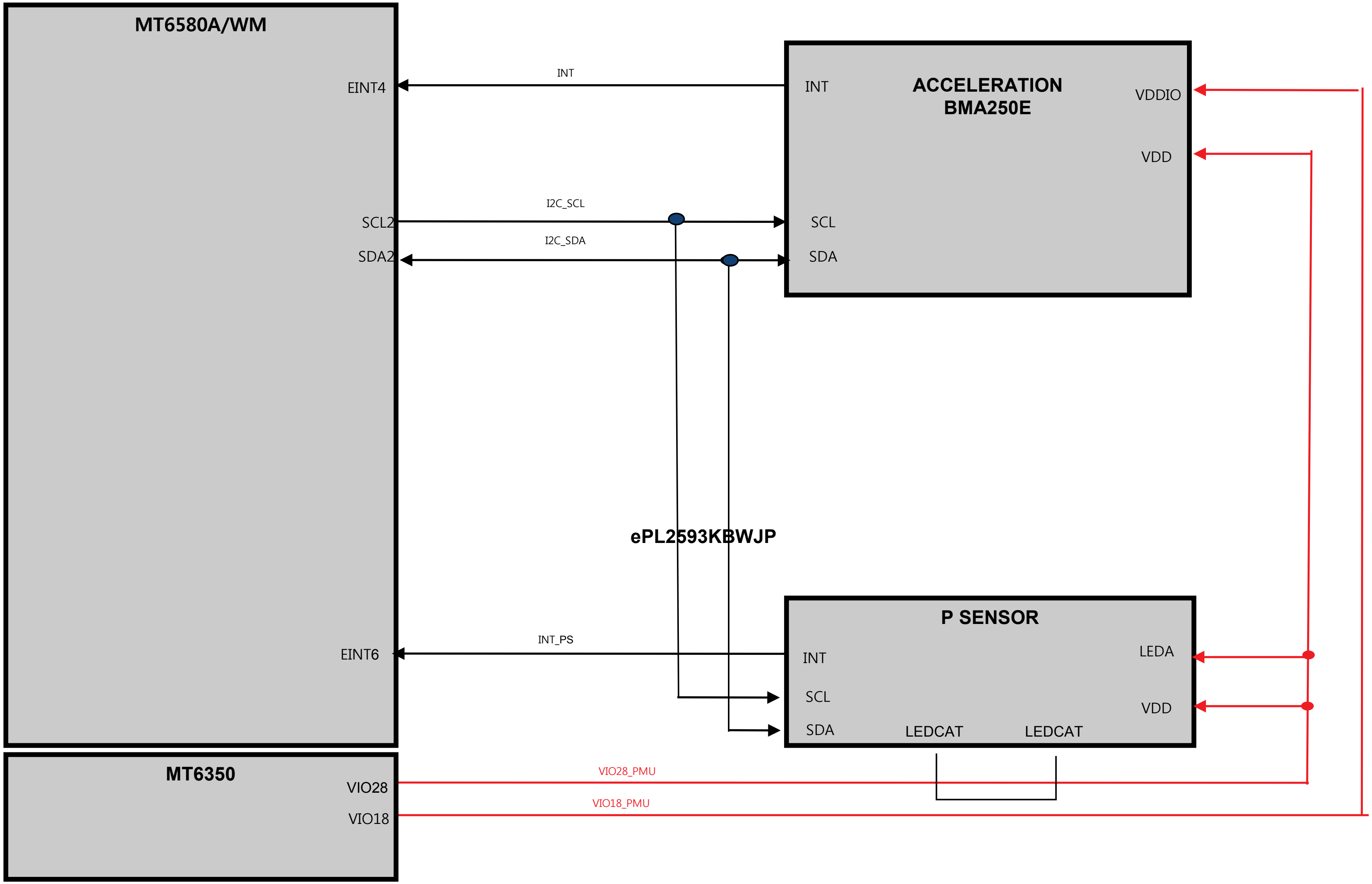

MT6580A/WM
Micro SD
SD_CLK
SD_CMD
SD_DATA[0:3]
SD_DETECT
MSDC1_CLK
MSDC1_CMD
MSDC1_DATA[0:3]
MSDC1_INSI
8GB + 8Gb MCP
KMQ72000SM-B316
LPDDR3X32 8Gb
2CS
ED_CLK
ED_CLK_B
ECKE
ECS0_B
ECS2_B
DDRAM_CLK_P
DDRAM_CLK_N
DDRAM_CLK_EN
DDRAM_CS[0]_N
DDRAM_CS[1]_N
CLK_T
CLK_C
CKE0
CS0_n
CS1_n
EA[0:9]
ED[0:31]
DDRAM_ADDRESS_[0:9]
DDRAM_DATA_[0:31]
CA[0:9]
DQ[0:31]
EDQS[0:3]
nEDQS[0:3]
EDQM[0:3]
DDRAM_DQS_[0:3]_P
DDRAM_DQS_[0:3]_N
DDRAM_DQM[0:3]
DQS[0:3]_T
DQS[0:3]_C
DM[0:3]
eMMC_DATA[0:7]
eMMC_RST
eMMC_CMD
eMMC_CLK
RST
CMD
CLK
DAT[0:7]
DAT[0:7]
RST
eMMC 8GB
CMD
CLKM
SIM1
C1
SCLK
SIO
SRST
SIM2
C1
SCLK
SIO
SRST
AP_SCLK
AP_SIO
AP_SRST
+3V0_MICRO
+3V0_VSIM1
+3V0_VSIM2
SIM1_CLK
SIM1_DATA
SIM1_RESET_N
SIM2_CLK
SIM2_DATA
SIM2_RESET_N
SIM_CLK
SIM_DATA
SIM_RESET_N
MT6350V/A
SRST
SIO
SCLK
VMCH
VSIM
VIO18
EMC_3V3
VM12
+1V8_VIO_MTK
+3V3_eMMC_MTK
+1V2_VLPDDR2_MTK


MT6350
VIO28
VIO18
VIO28
VIO18_PMU
MT6580A/WM
PWM_BL
LCD_PWM
Backlight
(TPS61169)
LED_A
LED_K
5" LCD(MIPI)
TDP0
TDN0
TCP
TCN
LPTE
LRSTB
LCD_MIPI_DATA0_P
LCD_MIPI_DATA0_N
LCD_MIPI_CLK_P
LCD_MIPI_CLK_N
LPTE
LCD_RESET_N
EINT6
SDA_0
SCL_0
RST
EINT7
EIN_INT
I2C_SDA
I2C_SCL
CTP_RST
EINT_CTP_WAKEUP
CTP TOUCH
ELAN
eKTF2138c
MT6350
VIO18
VGP1
VIO18_PMU
VGP1_3V3


MT6580A/WM
CMMCLK
RCP
RCN
RDP[0:1]
RDN[0:1]
CMRST
SCL_1
SDA_1
RCP_A
RCN_A
RDP[0:3]
RDN[0:3]
CMRST2
FLASH_INHIBIT
Flash EN
CAM_MCLK
CAM_CLK_P
CAM_CLK_N
CAM_DATA[0:1]_P
CAM_DATA[0:1]_N
CAM_RESET_N
I2C_SCL
I2C_SDA
FLASH_EN
2M FF CAMERA MIPI
MCLK
CLK_P
CLK_N
DATA[0:1]_P
DATA[0:1]_N
RST
SCL
SDA
VCAMD
VCAMA
VCAMIO
MT6350
VCAMD
VCAMA
VCAMIO
VCAMAF
5M FF CAMERA MIPI
CLK_P
CLK_N
DATA0_P
DATA0_N
RST2
MCLK
SCL
SDA
VCAMD
VCAMA
VCAMIO
VCAMAF
FLASH IC LM3648TT
SCL
SDA
FLASH_INHIBIT
Flash EN
IN
OUT
FLED2
+VBAT
Battery

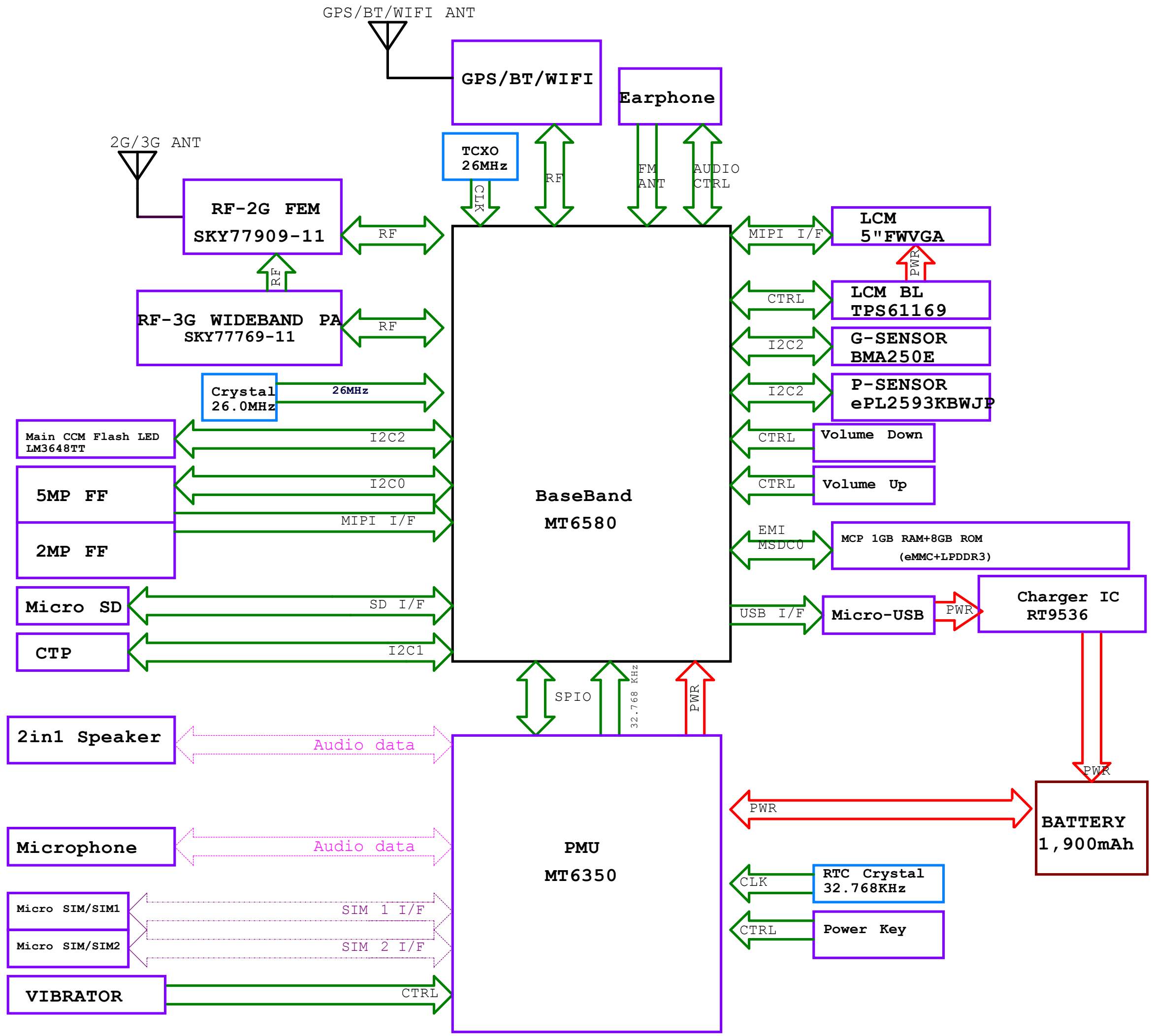
4. BLOCK DIAGRAM




MT6580A/WM
EINT4
SCL2
SDA2
EINT6
INT
I2C_SCL
I2C_SDA
INT_PS
ACCELERATION
BMA250E
INT
VDDIO
VDD
SCL
SDA
ePL2593KBWJP
P SENSOR
INT
SCL
SDA
LEDA
VDD
LEDCAT
LEDCAT
MT6350
VIO28
VIO18
VIO28_PMU
VIO18_PMU

CIRCUIT DIAGRAM

BLOCK DIAGRAM



Arima Communications Corp.

Title	D3	
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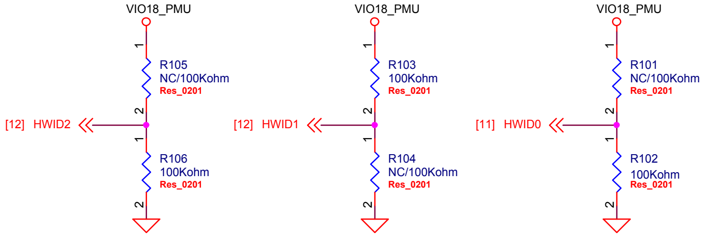
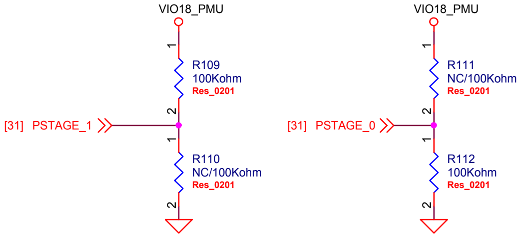
HW ID



Region	Country	LG Model	Operator	MP	Specification										Arima Model No.
					Multi SIM	Network	Rear Camera	Front Camera	Battery	DTV	Ear phone	HWID2 GPIO19	HWID1 GPIO16	HWID0 GPIO34	
LATAM	Mexico	LG-X220g	Telcel	'16.3.14	Single	B2+B5	5MP FF	2MP	1900mAh	N	Y	0	0	0	5542
	Peru		Claro	'16.4.11	Single	B2+B5	5MP FF	2MP	1900mAh	N	Y				
	Panama	LG-X220m	Open	'16.4.11	Single	B2+B5+B8	5MP FF	2MP	1900mAh	N	Y	0	1	1	5547
		LG-X220dsh	Open	'16.4.11	Dual	B1+B2+B5	5MP FF	2MP	1900mAh	N	Y	1	1	0	5545
	Chile (Base)	LG-X220mb	Open	'16.4.11	Single	B2+B4+B5+B8	5MP FF	2MP	1900mAh	N	Y	0	0	1	5548
CIS	CIS (Russia)	LG-X220ds	Open	16.3.25 (TBD)	Dual	B1+B8	5MP FF	2MP	1900mAh	N	Y	0	1	0	5546

Pilot Phase ID



PCB Stage	GPIO82 PSTAGE_1	GPIO84 PSTAGE_0	Remark
002	0	0	PP2
003	0	1	DV
004	**1**	**0**	**PV**
005	1	1	

Arima Communications Corp.

Title	D3
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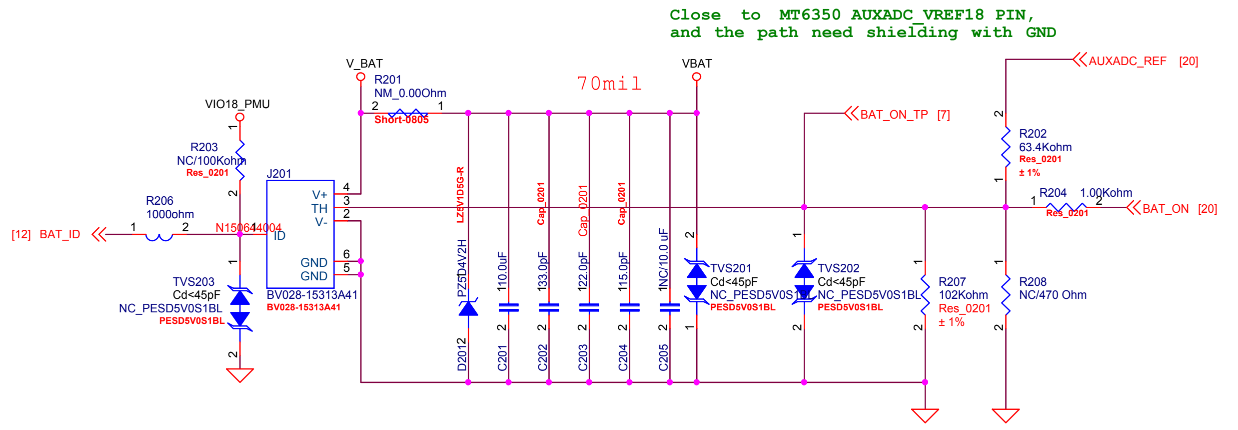
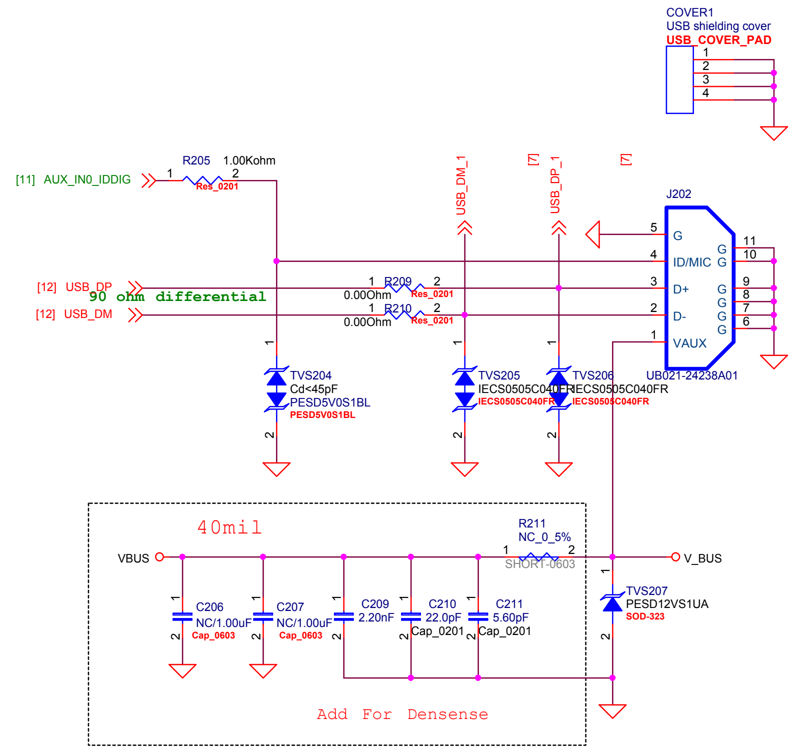
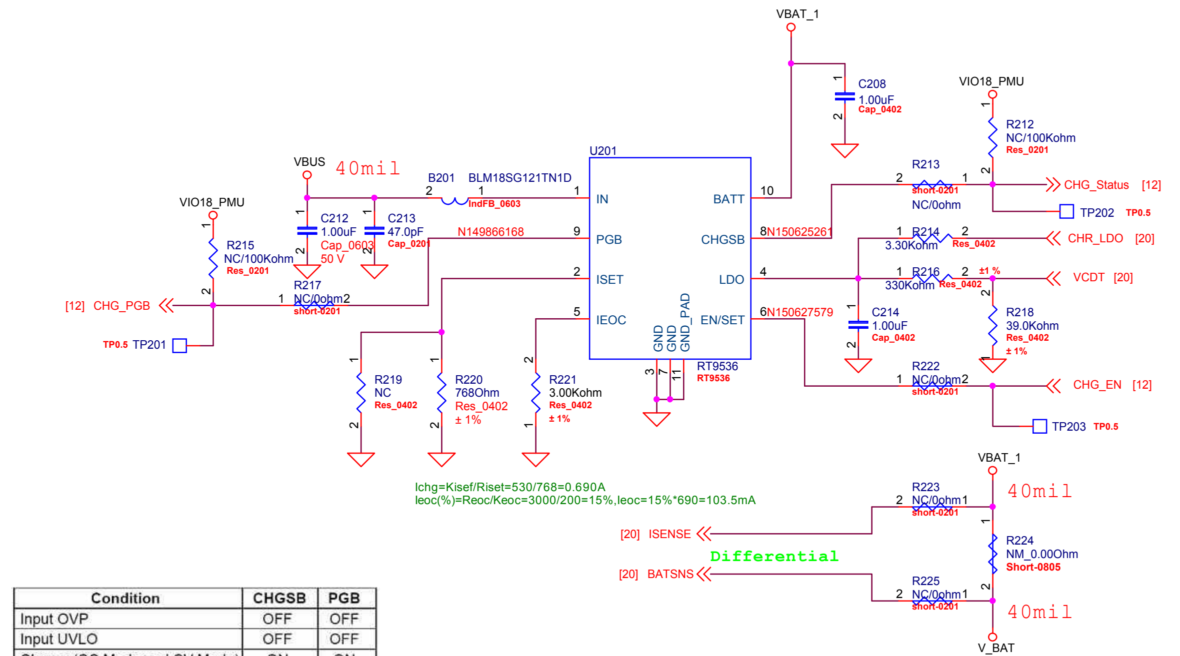
BATTERY CONNECTOR



USB HS IF



OVP & CHARGE



Condition	CHGSB	PGB
Input OVP	OFF	OFF
Input UVLO	OFF	OFF
Charge (CC Mode and CV Mode)	ON	ON
Charge Done (IFULL)	OFF	ON

Arima Communications Corp.

Title: D3

Size: A3 | Document ID: 02_USB_OVP_CHG_BAT | Rev: PA1

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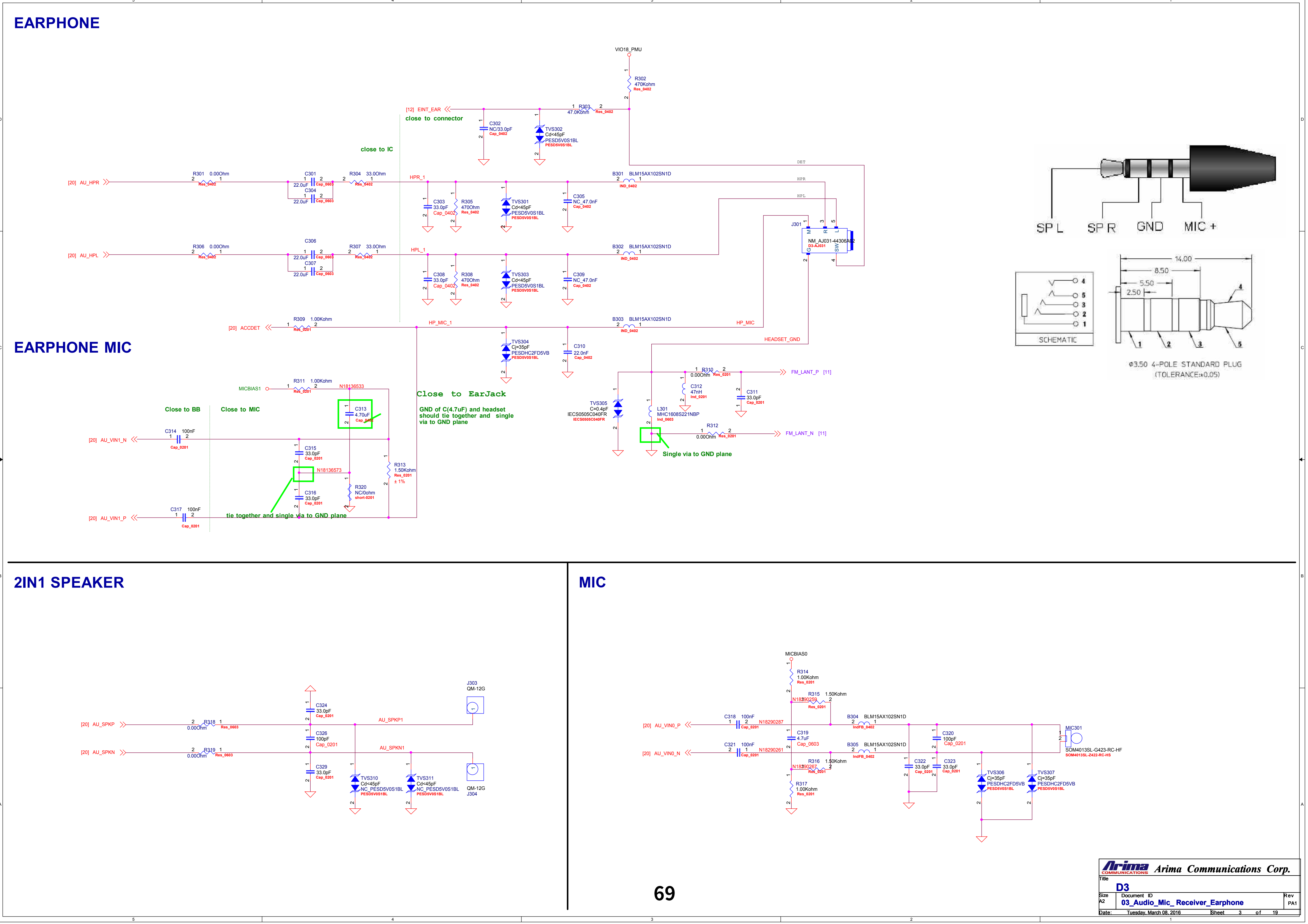
EARPHONE
VIO18_PMU
R302
470Kohm
Res_0402
[12] EINT_EAR
R303
47.0Kohm
Res_0402
close to connector
C302
NC/33.0pF
Cap_0402
TVS302
Cd<45pF
PESD5V0S1BL
close to IC
DET
HPR
HPL
[20] AU_HPR
R301 0.00Ohm
Res_0402
C301
22.0uF
Cap_0603
C304
22.0uF
Cap_0603
R304 33.0Ohm
Res_0402
HPR_1
C303
33.0pF
Cap_0402
R305
470Ohm
Res_0402
TVS301
Cd<45pF
PESD5V0S1BL
C305
NC_47.0nF
Cap_0402
B301 BLM15AX102SN1D
IND_0402
J301
NM_AJ031-44306A02
D3-AJ031
[20] AU_HPL
R306 0.00Ohm
Res_0402
C306
22.0uF
Cap_0603
C307
22.0uF
Cap_0603
R307 33.0Ohm
Res_0402
HPL_1
C308
33.0pF
Cap_0402
R308
470Ohm
Res_0402
TVS303
Cd<45pF
PESD5V0S1BL
C309
NC_47.0nF
Cap_0402
B302 BLM15AX102SN1D
IND_0402
[20] ACCDET
R309 1.00Kohm
Res_0201
HP_MIC_1
TVS304
Cj=35pF
PESDHC2FD5VB
PESD5V0S1BL
C310
22.0nF
Cap_0402
B303 BLM15AX102SN1D
IND_0402
HP_MIC
HEADSET_GND
SP L
SP R
GND
MIC +
14.00
8.50
5.50
2.50
SCHEMATIC
Ø3.50 4-POLE STANDARD PLUG
(TOLERANCE:±0.05)
EARPHONE MIC
R311 1.00Kohm
Res_0201
MICBIAS1
N18136533
C313
4.70uF
Cap_0402
Close to BB
Close to MIC
C314 100nF
Cap_0201
[20] AU_VIN1_N
C315
33.0pF
Cap_0201
N18136573
R313
1.50Kohm
Res_0201
± 1%
R320
NC/0ohm
short-0201
C316
33.0pF
Cap_0201
C317 100nF
Cap_0201
[20] AU_VIN1_P
tie together and single via to GND plane
Close to EarJack
GND of C(4.7uF) and headset should tie together and single via to GND plane
R310
0.00Ohm
Res_0201
FM_LANT_P [11]
C312
47nH
Ind_0201
C311
33.0pF
Cap_0201
TVS305
C=0.4pF
IECS0505C040FR
L301
MHC1608S221NBP
Ind_0603
R312
0.00Ohm
Res_0201
FM_LANT_N [11]
Single via to GND plane
2IN1 SPEAKER
[20] AU_SPKP
R318
0.00Ohm
Res_0603
[20] AU_SPKN
R319
0.00Ohm
Res_0603
C324
33.0pF
Cap_0201
C326
100pF
Cap_0201
C329
33.0pF
Cap_0201
AU_SPKP1
AU_SPKN1
TVS310
Cd<45pF
NC_PESD5V0S1BL
PESD5V0S1BL
TVS311
Cd<45pF
NC_PESD5V0S1BL
PESD5V0S1BL
J303
QM-12G
QM-12G
J304
MIC
MICBIAS0
R314
1.00Kohm
Res_0201
R315 1.50Kohm
Res_0201
N18290259
[20] AU_VIN0_P
C318 100nF
Cap_0201
N18290287
C319
4.7uF
Cap_0603
[20] AU_VIN0_N
C321 100nF
Cap_0201
N18290261
R316 1.50Kohm
Res_0201
N18290267
R317
1.00Kohm
Res_0201
B304 BLM15AX102SN1D
IndFB_0402
B305 BLM15AX102SN1D
IndFB_0402
C320
100pF
Cap_0201
C322
33.0pF
Cap_0201
C323
33.0pF
Cap_0201
TVS306
Cj=35pF
PESDHC2FD5VB
PESD5V0S1BL
TVS307
Cj=35pF
PESDHC2FD5VB
PESD5V0S1BL
MIC301
SOM4013SL-G423-RC-HF
SOM4013SL-Z422-RC-HS
Arima Communications Corp.
Title
D3
Size A2
Document ID
03_Audio_Mic_ Receiver_Earphone
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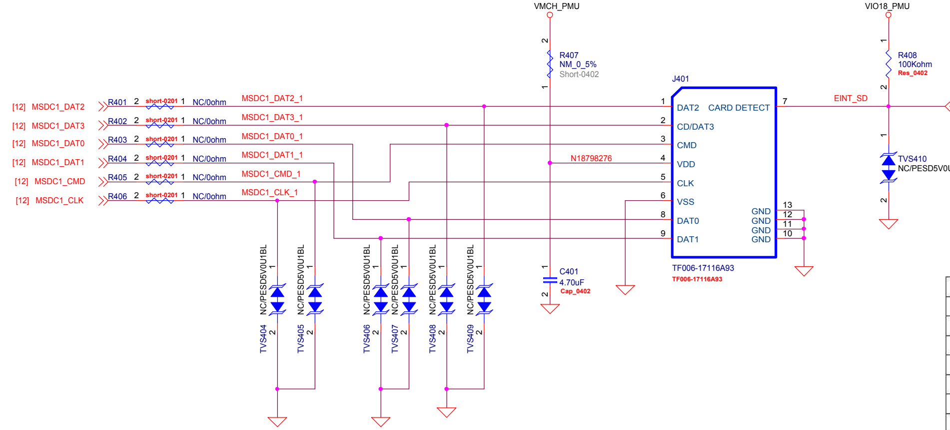
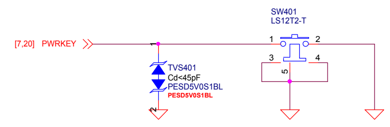
SD CARD

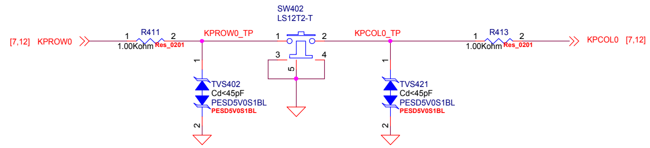


PIN NO.	PIN NAME
PIN 1	DAT 2
PIN 2	CD/DAT 3
PIN 3	CMD
PIN 4	VDD
PIN 5	CLK
PIN 6	VSS
PIN 7	DETECT
PIN 8	DAT 0
PIN 9	DAT 1

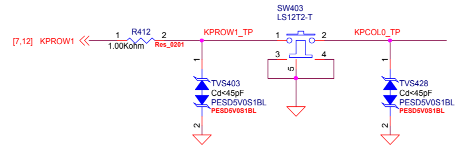
POWER KEY



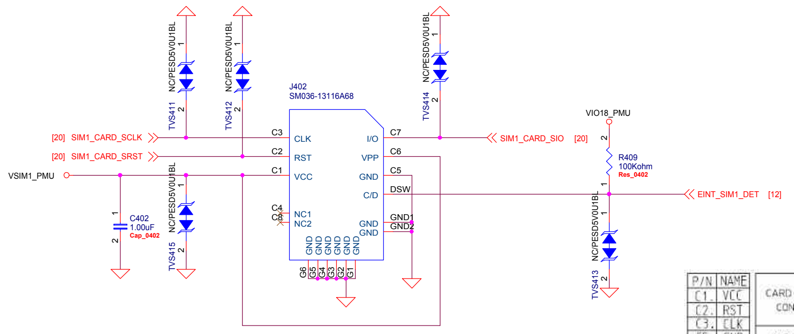
VOLUME UP



VOLUME DOWN



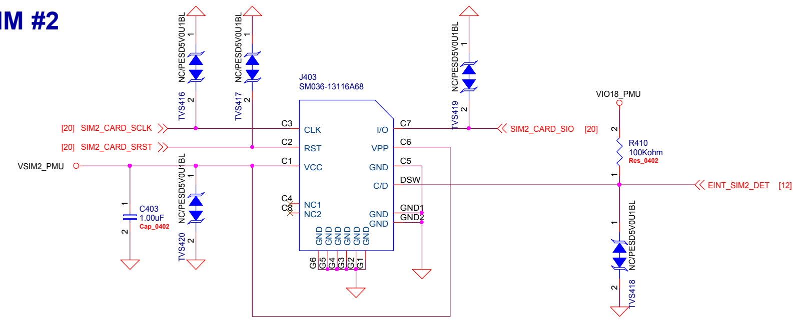
SIM #1



P/N	NAME
C1.	VCC
C2.	RST
C3.	CLK
C5.	GND
C6.	VPP
C7.	I/O

CARD INSERTION CONDITION	CARD DETECT SWITCH	CIRCUIT C/D GND
WITHOUT CARD	CLOSE	
CARD INSERTION	OPEN	

SIM #2



LG Model	Arima Model No.	Multi SIM	J403	C403	R410
LG-X220g	5542	Single	NC	NC	NC
LG-X220m	5547	Single			
LG-X220mb	5548	Single			
LG-X220dsh	5545	Dual	314-0000-00965	302-G000-02034	301-G000-00085
LG-X220ds	5546	Dual	SM036-13116A68	1.00 uF_± 10%_X5R_10	100 Kohm_± 5%_1/16

Arima Arima Communications Corp.

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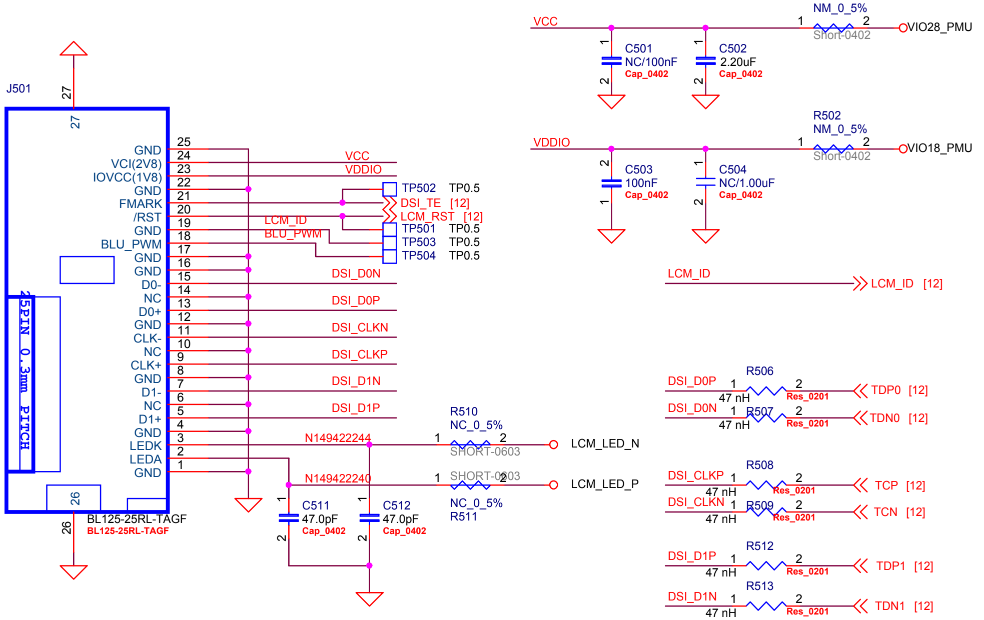
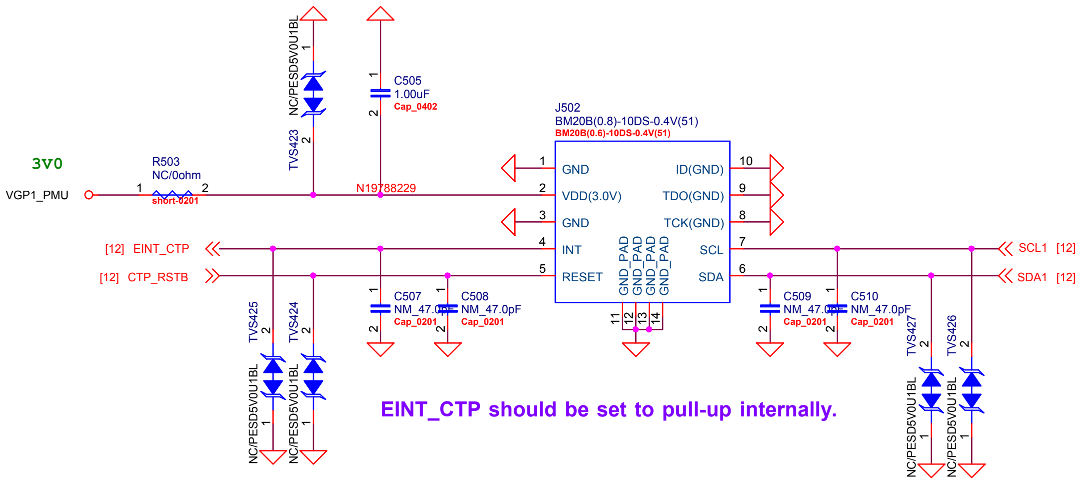
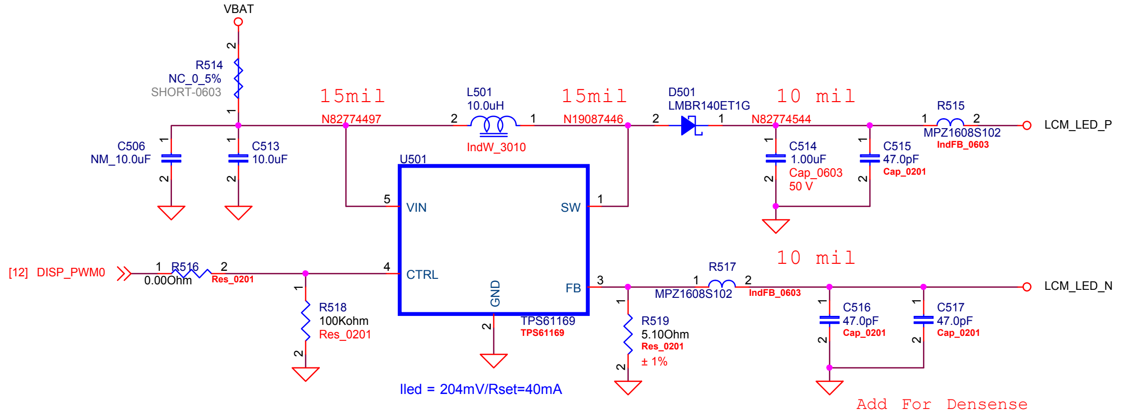
LCM



TOUCH PANEL



BACKLIGHT DRIVER



Arima Communications Corp.

Title	D3
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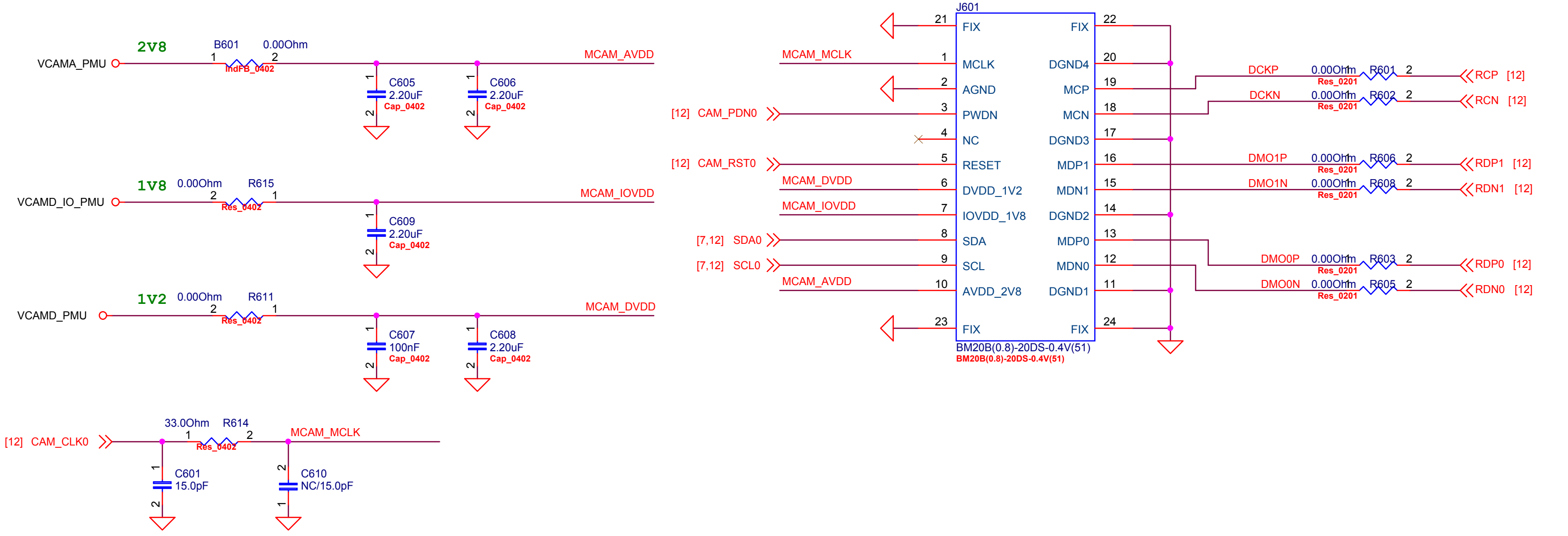
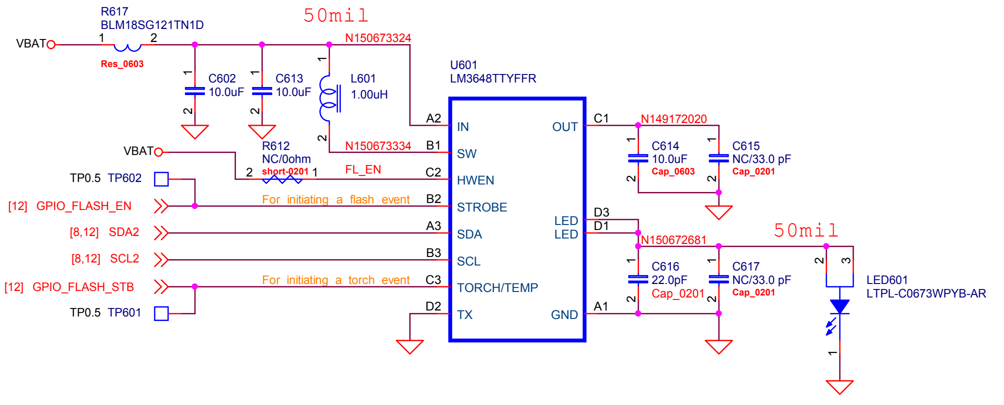
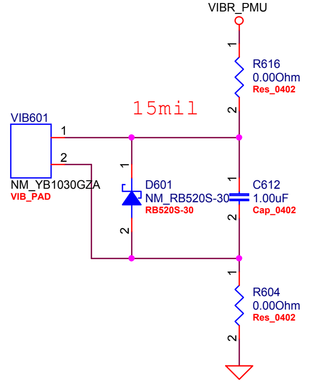
MAIN CAMERA 5MP FF



FLASH LED DRIVER



VIBRATOR



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Arima Communications Corp.

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Size A3	Document ID 06_Main Camera_FlashLED_Vib	Rev PA1
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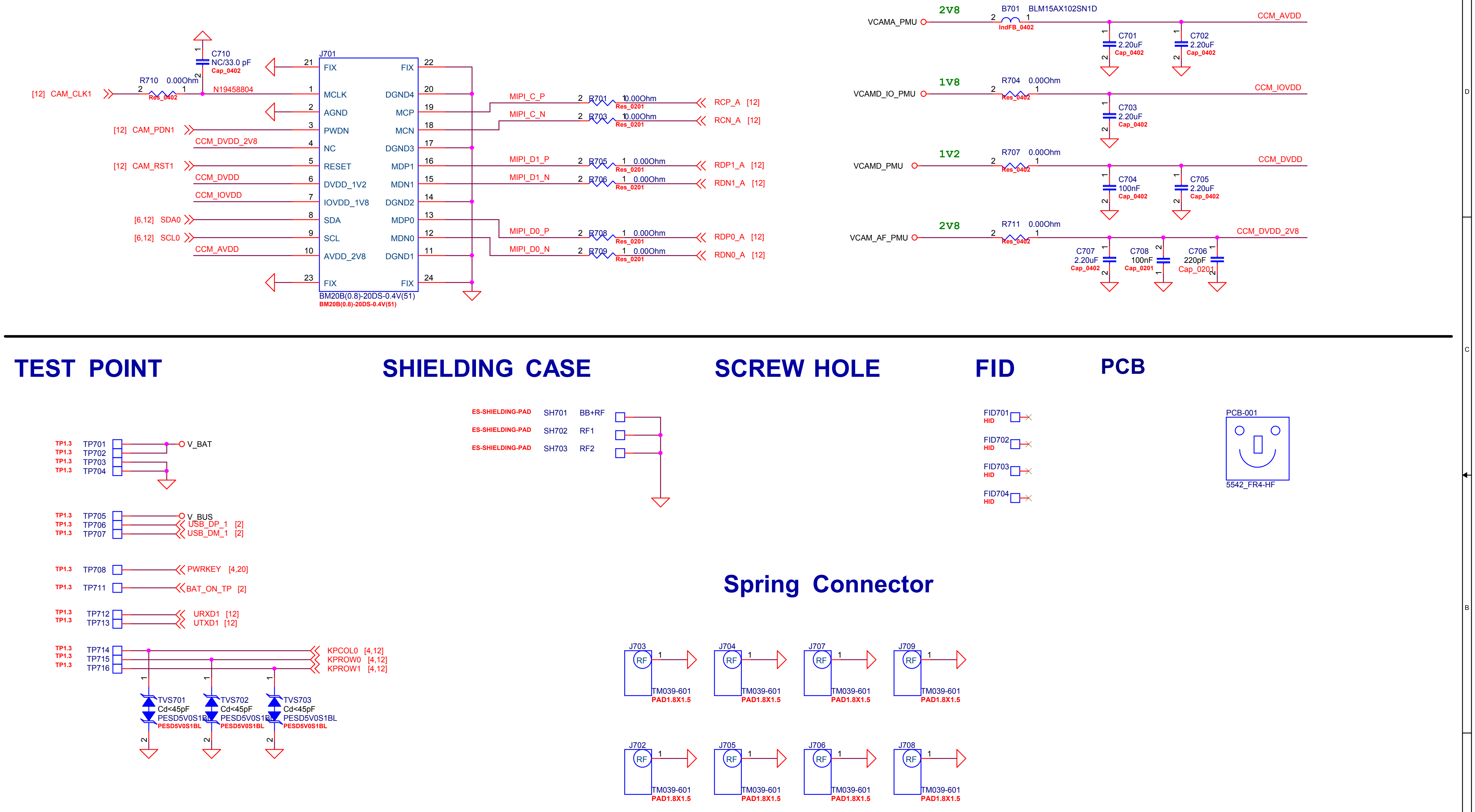
SUB CAMERA 2MP FF



TEST POINT

SHIELDING CASE

SCREW HOLE

FID

PCB

Spring Connector

Arima Communications Corp.

Title	D3
Size A3	Document ID: 07_Sub Camera_Test point — Rev PA1
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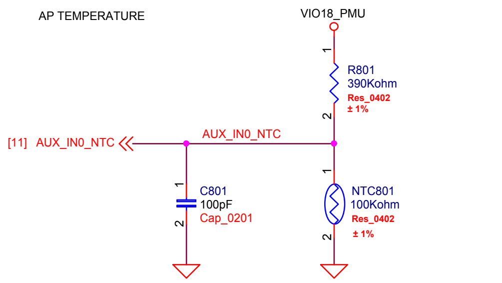
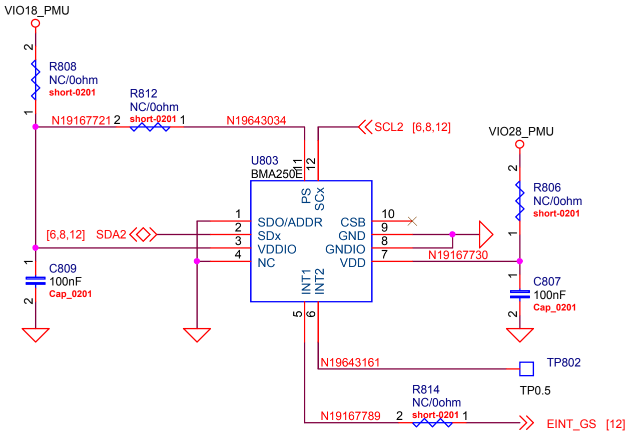
THERMISTOR

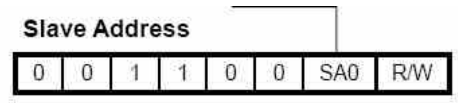


C801 close to BB

NTC801 close to BB ,and located in the same layer
(can refer to thermal design notice)

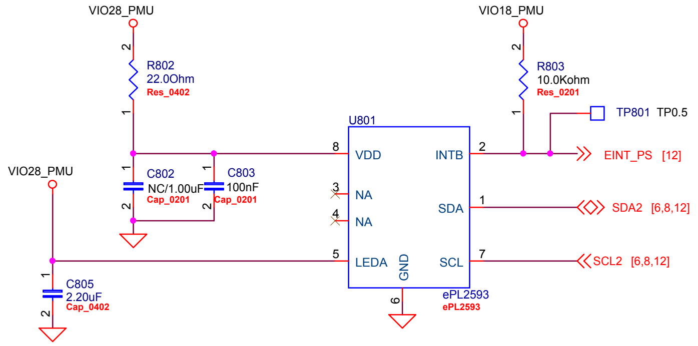
G SENSOR





7-bit Slave Address	
SA0	Address
GND	0x18
VDDIO	0x19

P SENSOR



Ordering Chip Information	Write to Slave	Read from Slave
ePL25903	10010010	10010011

Arima COMMUNICATIONS Arima Communications Corp.
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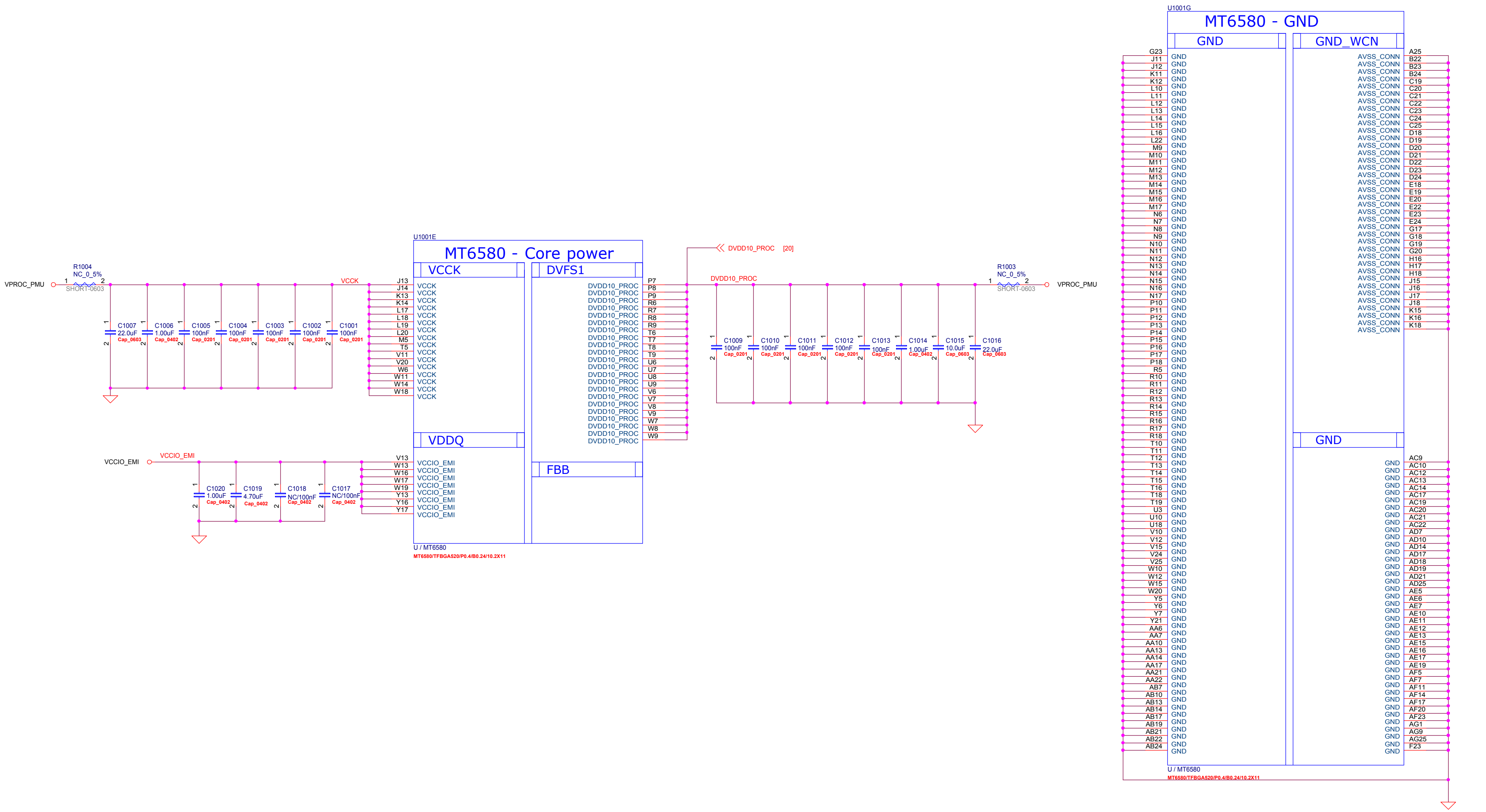

Arima Communications Corp.

Title	D3
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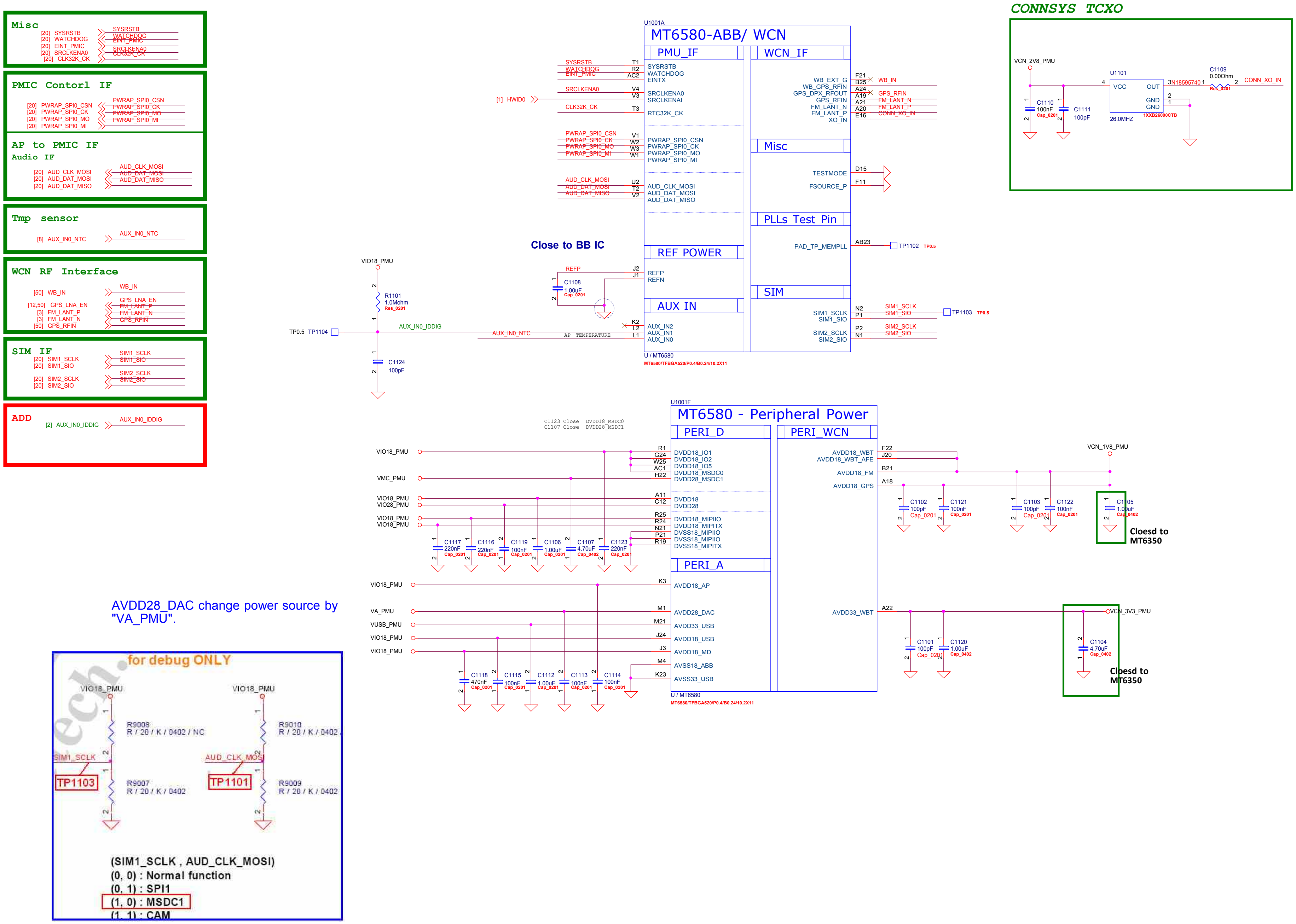
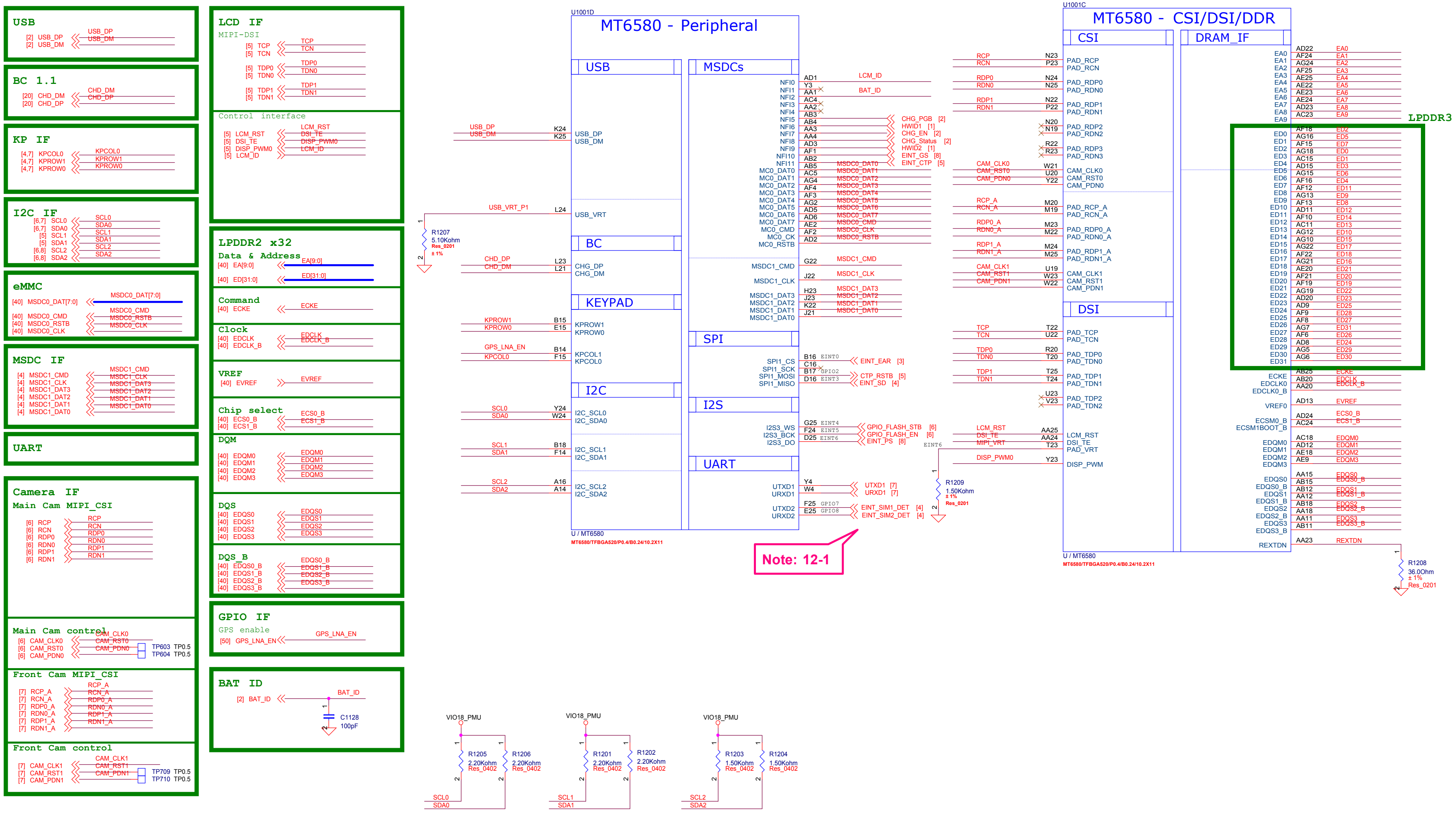


Arima Arima Communications Corp.

Title	D3	
Size A2	Document ID 11_BB_1	Rev PA1
Date: Tuesday, March 08, 2016	Sheet 11 of 19	



Schematic design notice of "12_BB_2" page.

Note 12-1: Do not use UART1 as Power solution enable.
Do not use UART2 as power solution enable, if select UART1 as D/L or calibration path,

Arima COMMUNICATIONS Arima Communications Corp.

Title	D3	
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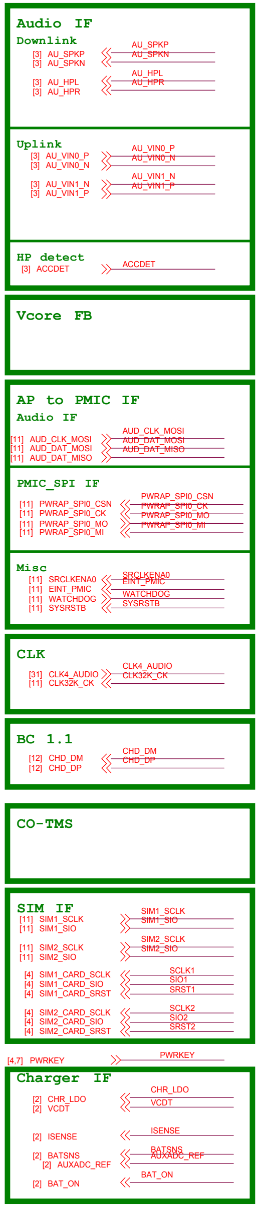
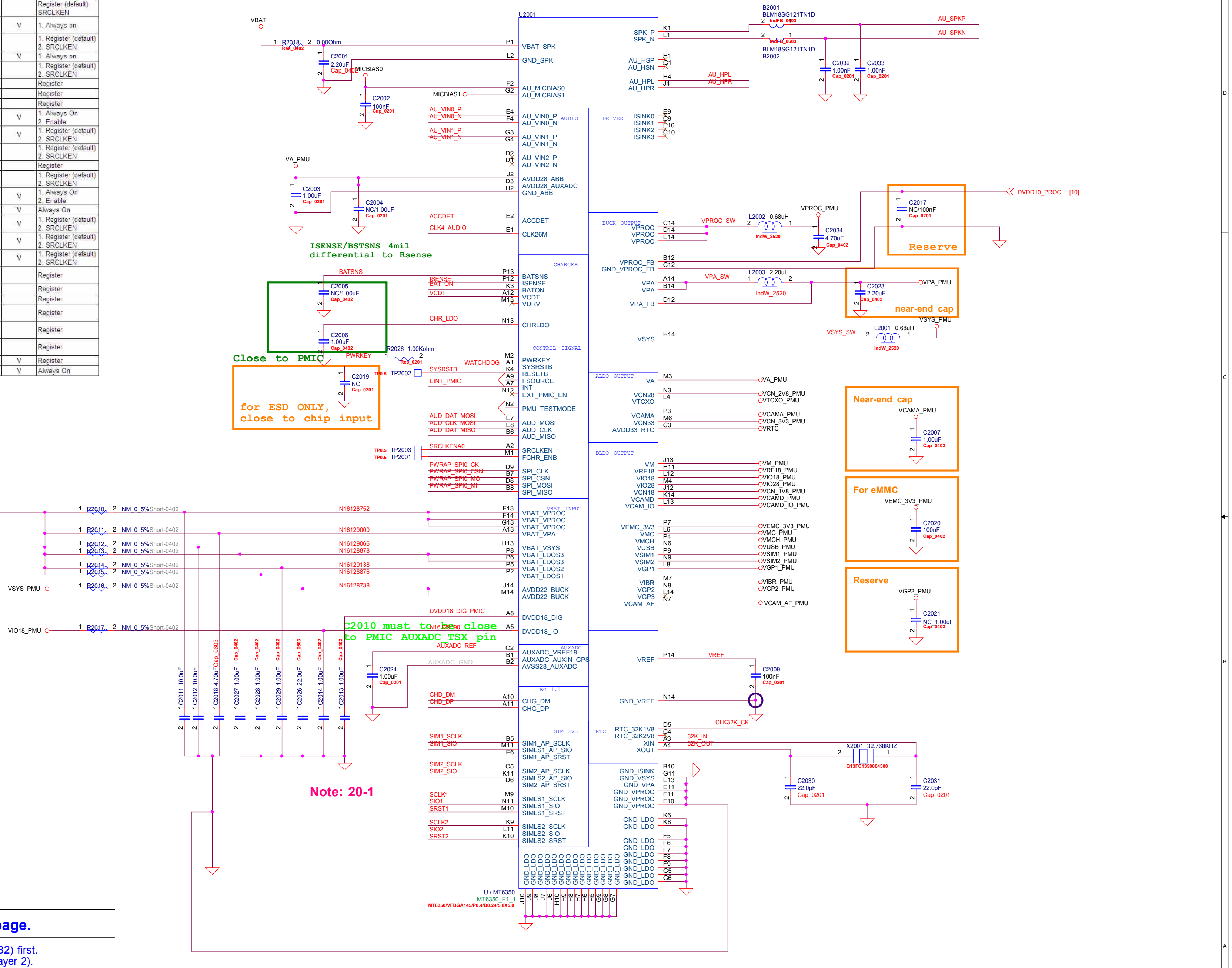


Block	Name	Vout	Current	Default On	On/Off control
DC/DC	VCORE	0.7~1.4 (DC/DC+DVS)	2800	V	1. Always On
	VSYS	1.825/2.2	1400	V	1. Always on
	VPA	0.5~3.4V	600		Register (default) SRCLKEN
VSYS LDO	VM	1.2/1.35/1.5/1.8	300 (1.8V) 700 (1.2V)	V	1. Always on
	VRF18	1.825	200		1. Register (default) 2. SRCLKEN
	VIO18	1.8	300	V	1. Always on
	VCN_1V8	1.8	120		1. Register (default) 2. SRCLKEN
	VCAMD	1.2/1.3/1.5/1.8	150		Register
	VCAM_IO	1.8	100		Register
	VGP3	1.2/1.3/1.5/1.8	200		Register
Analog LDO	VA	2.8	150	V	1. Always On 2. Enable
	VTCXO	2.8	40	V	1. Register (default) 2. SRCLKEN
	VCN28	2.8	30		1. Register (default) 2. SRCLKEN
	VCAMA	1.5/1.8/2.5/2.8	150		Register
Digital LDO	VCN33	3.3/3.4/3.5/3.6	240		1. Register (default) 2. SRCLKEN
	VIO28	2.8	200	V	1. Always On 2. Enable
	VUSB	3.3	20	V	Always On
	VMC	1.8/3.3	100	V	1. Register (default) 2. SRCLKEN
	VMCH	3.0/3.3	400 (support SD 3.0)	V	1. Register (default) 2. SRCLKEN
	VEMC_3V3	3.0/3.3	400	V	1. Register (default) 2. SRCLKEN
	VCAM_AF	1.2/1.3/1.5/1.8/2.0 2.8/3.0/3.3	100		Register
	VSIM1	1.8/3.0	50		Register
	VSIM2	1.8/3.0	50		Register
	VGP1	1.2/1.3/1.5/1.8/2.0 2.8/3.0/3.3	100		Register
	VGP2	1.2/1.3/1.5/1.8/2.0 /2.5/2.8/3.0	100		Register
	VIBR	1.2/1.3/1.5/1.8/2.0 2.8/3.0/3.3	100		Register
	VDIG18	1.8	20	V	Register
VRTC	VRTC	2.8	2	V	Always On



Schematic design notice of "20_Power_MT6350" page.

Note 20-1: Connect TMS GND (Pin2) to PMIC AUXADC_GND (Pin B2) first.
Then connect to main GND by inner layer short pad (i.e., layer 2).

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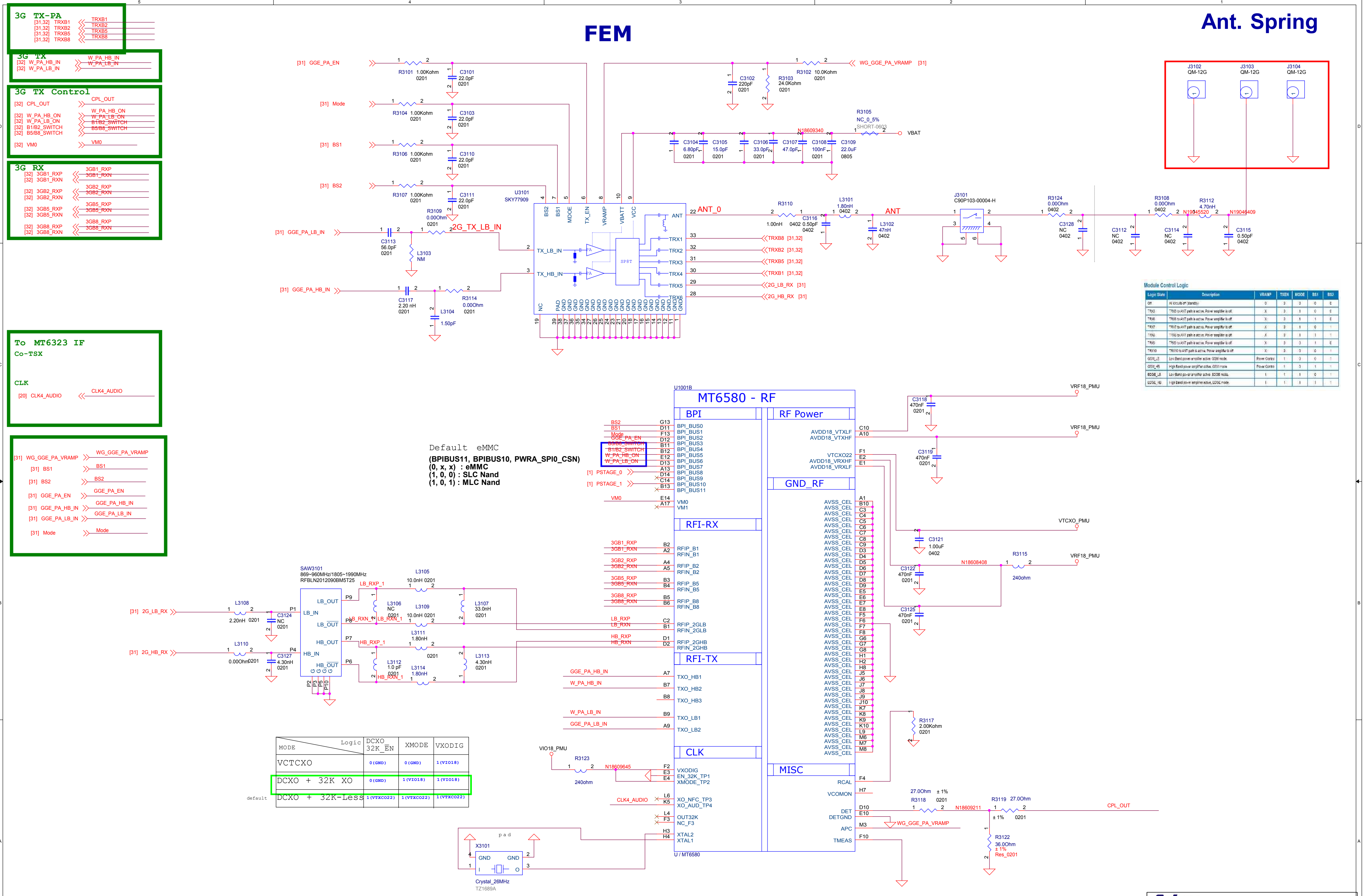

MODE / Logic	DCXO 32K_EN	XMODE	VXODIG
VCTCXO	0(GND)	0(GND)	1(VIO18)
DCXO + 32K XO	0(GND)	1(VIO18)	1(VIO18)
default DCXO + 32K-Less	1(VTCXO22)	1(VTCXO22)	1(VTCXO22)

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Title: D3

Size	Document ID	Rev
A2	31_RF_TXM_2G	PA1

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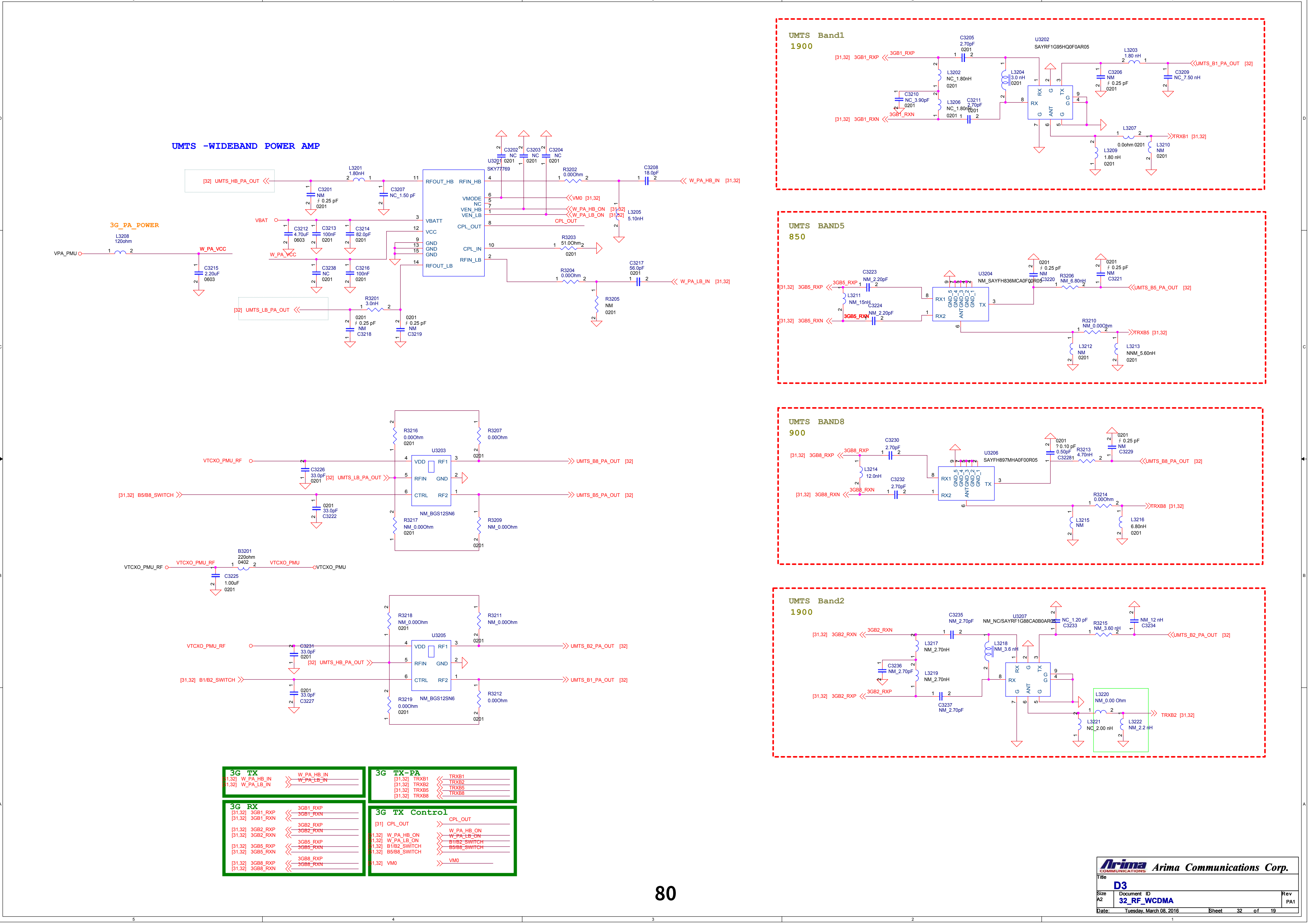
UMTS -WIDEBAND POWER AMP
3G_PA_POWER
UMTS Band1
1900
UMTS BAND5
850
UMTS BAND8
900
UMTS Band2
1900
3G TX
3G TX-PA
3G RX
3G TX Control
U3201
SKY77769
U3202
SAYRF1G95HQ0F0AR05
U3204
NM_SAYFH836MCA0F00R05
U3206
SAYFH897MHA0F00R05
U3207
NM_NC/SAYRF1G88CA0B0AR05
U3203
NM_BGS12SN6
U3205
NM_BGS12SN6
Arima Communications Corp.
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D3
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A2
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32_RF_WCDMA
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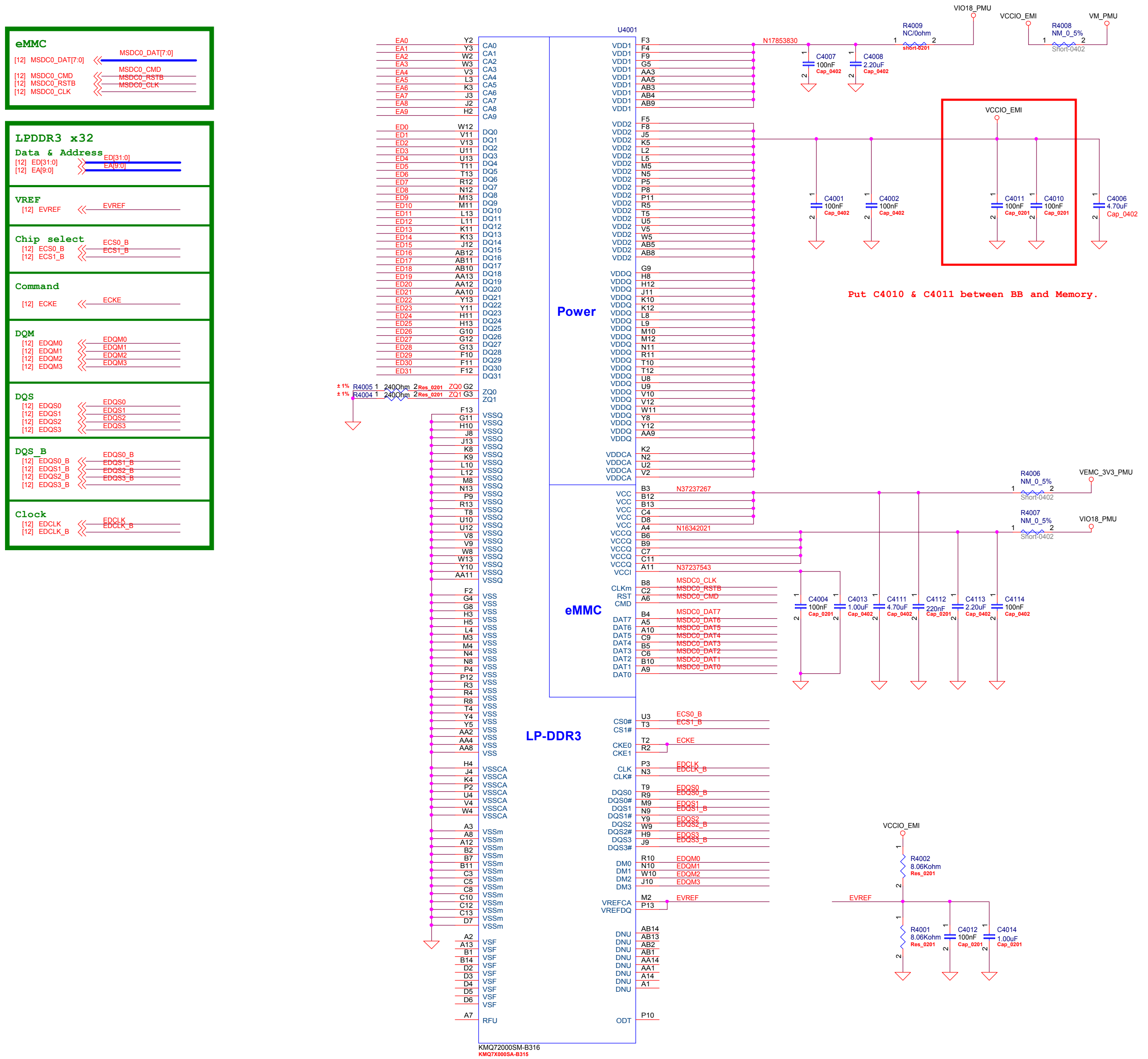
eMMC+LPDDR3

8GB+1GB: KMQ72000SM-B316 (V10)



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Title	D3
Size A2	Document ID: 40_Memory_eMMC_NAND_LPDDR2
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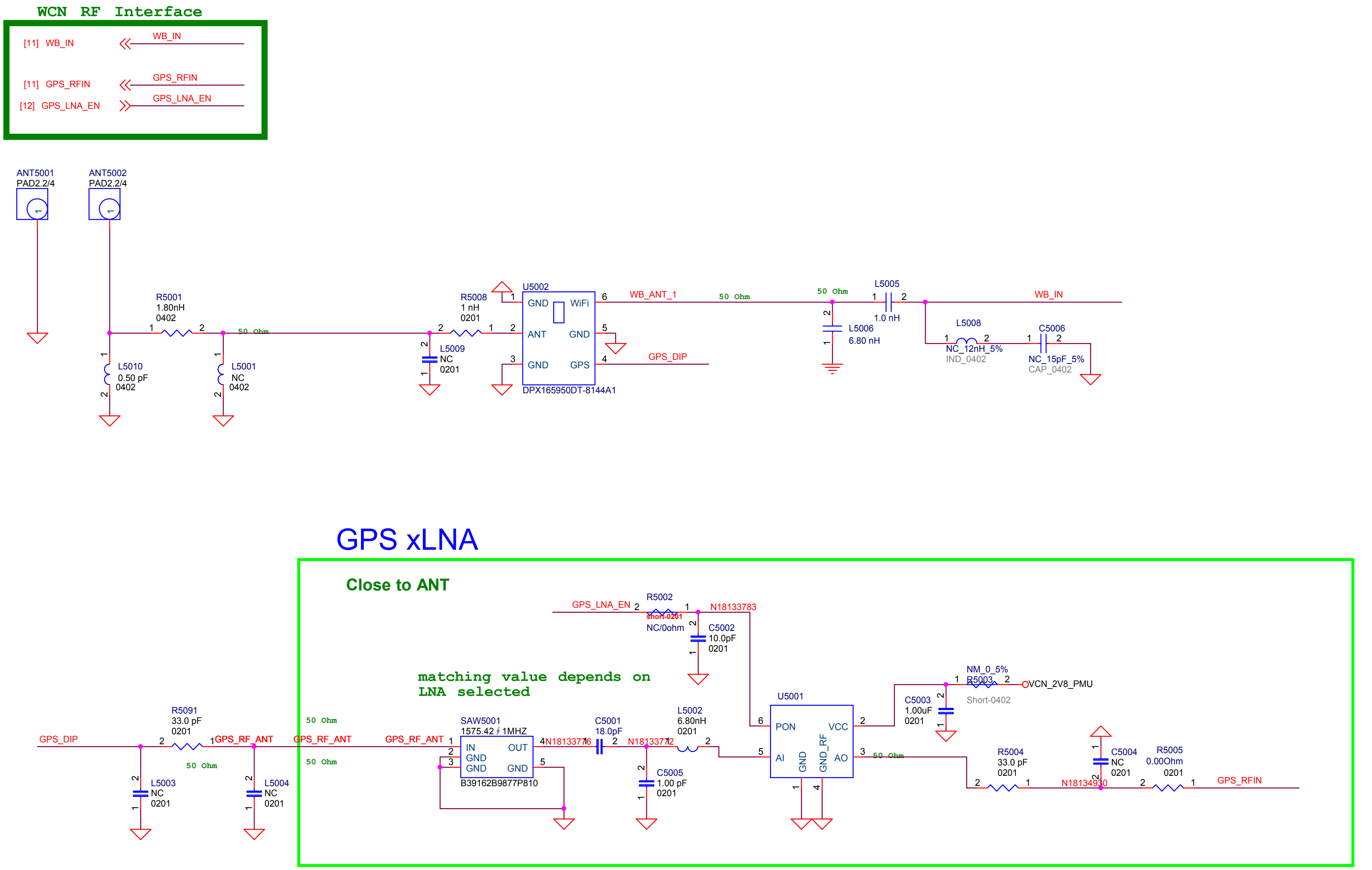

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Title	D3	
Size A3	Document ID 50_BT_FM_GPS_WiFi	Rev PA1
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Change Note

PP1

8/10 BB Part：
1) Del U804
2) Change Main Camera to 5M FF

8/11 EM/ME Part：
1) Change J202 to 314-NEW, UB021-24236A01
2) Change SW401,402,403 to 315-0000-00047
3) Change J402,403 to 314-0000-00965
4) Change J301 to 314-NEW, footprint follow V20
5) Del J702-708
6) Change J303,304 to 314-0000-00924
7) Change U801 to 311-0000-01773
8) Change J502 to 314-0000-00565
9) Change LED601 to 309-0000-00270

8/12 EM/ME Part：
1) Change J402,403 footprint to SM036-13116A66A Follow V10
2) Del Receiver, change to 2in1

8/14 RF Part：
1) Del DTV
2) Add U5002 diplexer for GPS/WIFI/BT single antenna

8/17 EM/ME Part：
1) Change U801 to 311-0000-01970, follow D5
2) Change J303,304 to 314-0000-00937

8/18：
1) Change Project code to D3

8/19 RF Part：
1) Change J3101 to MM8130-2600RA2, follow ME
2) Add R3123 for VIO18_PMU DESENSE
3) Add R3115 for VRF18_PMU
4) Change R5002 from 0402 to 0201
5) Add R3124 & C3128 for matching
5) Add R3216 R3217 R3218 R3219 for matching

8/19 BB Part：
1) Reserve TVS for TP
2) Change BAT_ID to AA1
3) Add TVS for Key Test Point

8/20：
1) Change HWID to D3
2) Change C209 C210 C211 to 0201 size
3) Change J3101 to C90P106-00004-H

8/20 EM/ME Part：
1) Change J301 follow D5

8/21 BB Part：
1) Change CHG_EN and HWID1 Netlist
2) Change C615,C617 to 0201 Size(NC);
3) Change C616 to 22pF_0201
4) Del C2010
5) Add PCB-001
6) Update J301 symbol

8/25：
1) Update SKU

8/26 BB Part：
1) Change C201,C205 to 302-G000-02113
2) Change C209 to 302-G000-00381

8/27 BB Part：
1) Change C613,C2011,C2012 to 302-G000-02113
2) Change L601 to 303-1000-00916, following S1
3) Change C513 to 302-G000-02232
4) Change C2001 to 302-1000-01476
5) Change X2001 to 305-0000-00221
6) Del TVS428-430,R2007
7) Change R514 to 301-G000-01049
8) Update J202 to 314-0000-00990
9) Del R618,R619, add TP601,TP602
10) Add TP603,TP604,TP709,TP710,TP801,TP1104
11) NM C507,C508,C509,C510
12) Modify VBAT net, marked V_BAT

8/27 EM/ME Part：
1) Change J3102-4 to 314-0000-00937

8/28 BB Part：
1) Add TP2003
2) Add C2034
3) Change TP_1V8 to VGP2_PMU
4) Update TP Conn. J502 Symbol
5) Change C506 to 302-G000-02034
6) Change C209 to 302-G000-00368

8/28 BB Part：
1) Connect J501 Pin17/22 to GND
2) Update J501 Symbol

8/30 BB Part：
1) Add V_BUS net

8/31 BB Part：
1) Change R201 connection
2) Change VDDIO of U803 to VGP2

8/31 BB Part：
1) Change LED601 to 309-0000-00285
2) Del TP716, change Volume Key connection
3) Connect J502 Pin3 to GND, update J502 Symbol
4) Del R505,TVS422,C506

8/31 BB Part：
1) Del Screw Holes
2) Change VDDIO of U803 to VIO18_PMU

8/31：
1) Change MIC301 to 312-0000-00073

9/1 BB Part：
1) Change Key connection back to D5
2) Reserve TVS428

9/1 EM/ME Part：
1) Add USB Cover, COVER1
2) Change SW401,402,403 to 315-0000-00086

9/2 BB Part：
1) Change C209 to 302-0225-53085
2) Update PCB to 321-0000-00640
3) Update Audio Jack to 314-0000-00991

9/2 Footprint：
1) Change J303,J304,J3102,J3103,J3104 to QM-12G-2
2) Change J601,J701 to AXQ1201

9/3 BB/RF Part：
1) Change R201 from 0603 to 0805 for large current.
2) Change RF power from V_BAT to VBAT for easy routing

9/6 Audio Part：
1) Change C319 to 302-0231-62133

9/6 BB Part：
1) Change R1203,R1204 to 301-G000-00045
2) Mount TVS204,TVS401~3,TVS421,TVS428
3) Add TVS701,TVS702,TVS703
4) Change Key Test Point net connection

9/7 Camera Part：
1) Add DVDD_2V8 for Sub Camera compatible design
2) Add R711,C706,C707,C708

9/7 Symbol/Footprint：
1) Update J202 symbol, add Pin10,11 to GND
2) Update LED601 symbol, add Pin3
3) Update LED601 footprint to LTPL-C0673WPYB-AR
4) Update J502,J601,J701 symbol, add FIX pin to GND

9/8 RF Part:
1) Change J3101 from C90P106-00004-H to C90P103-00004-H

9/8 EM/ME Part:
1) Change J601,J701 to 314-0000-00807

9/8 BB Part:
1) Add TP501,TP502

9/9 BB/RF Part:
1) Add R520,R521,R522 for TP compatible design
2) Change U3201 to SKY77769
3) Change R520,R521,R522 to 0201 size
4) Change 3G PA VBATT power to VBAT
5) Change U3101 to SKY77909

9/10 BB Part:
1) Mount TVS301,302,303 for ESD
2) Connect Pin6,10,14 of J501 to GND

9/18 BB Part:
1) Change R1201,R1202,R1205,R1206 to 301-G000-00236

9/21 BB Part:
1) Change C2009 to 302-G000-00182

PP2

9/24 EM/ME Part:
1) Change J401 to 314-0000-00985
2) Change J502 to 314-0000-00784
3) Del R520,R521,R522
4) Change U803 Symbol and Footprint name
5) Change U601 to 311-0000-01997
6) Add R612,R613 to 301-G000-01047

9/28 BB Part:
1) Change U601 control signals setting, add TP60
2) Update L601 to 303-1000-00981

10/8 BB Part:
1) Change U803 to 311-0000-01293
2) NM R814, mount R812,R813

10/9 BB Part:
1) Change C516 to 302-G000-02113, reserve C506
2) Add C602, change R617 to 303-1000-00885
3) Change L501 to 303-1000-01035
4) NM R813, mount R814

10/12:
1) Update Phase ID table
2) Change D501 to 309-0000-00292

10/21:
1) Add R604

10/21_1:
1) Del TP1101
2) Del C3123, C3126

10/22:
1) Add J702,J703(TM039-601) as antenna required

10/28:
1) Change B601 to 301-G000-01048

DV

11/10:
1) Change HW-ID table
2) Change Pilot Phase table to DV

11/18:
1) Change HW-ID table

11/24:
1) Add J704~709, 314-0000-01021

11/24_1:
1) Remove J702,J703
1) Change J704~709 to TM039-601, 314-0000-NEW

11/25:
1) Change U1001,U2001 footprint to B0.24

12/2:
1) Add J702, J703

12/7:
1) Update SKU

PV

12/15:
1) Add Pin19 of J501 for LCM ID, connect to PinAD1 of U1001
2) Add TP503, TP504
3) Update Pilot Phase ID to PV

12/23:
1) Change R213,R217,R222,R401~R407,R612,R806,R808,R812, R814,R2010,R2012,R2017 to shortpad
2) Remove R613,TP605,R813,R1102,R1103,C1125
3) Add TP802

12/24:
1) Change R3105, R5002, R5003 to shortpad

12/24_1:
1) Change R503,R4009 to shortpad-0201
2) Change R2011,R2013~R2016,R501,R502,R4006,R4007,R4008 to shortpad-0402
3) Change R211,R510,R511,R514,R1003,R1004 to shortpad-0603
4) Change R201,R224 to shortpad-0805
5) Change R207 to 301-1000-00955 for PIF

12/28:
1) NC C2021

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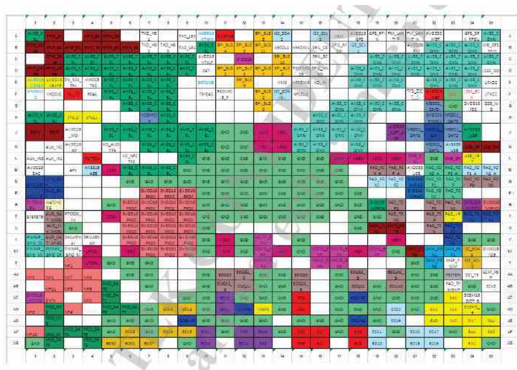
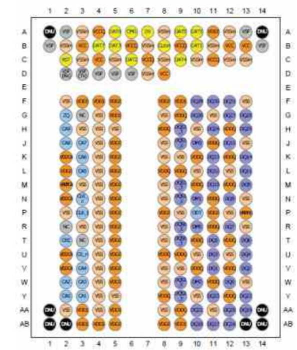
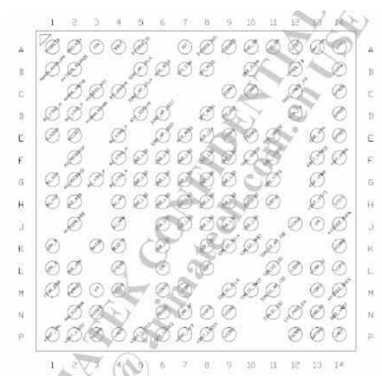
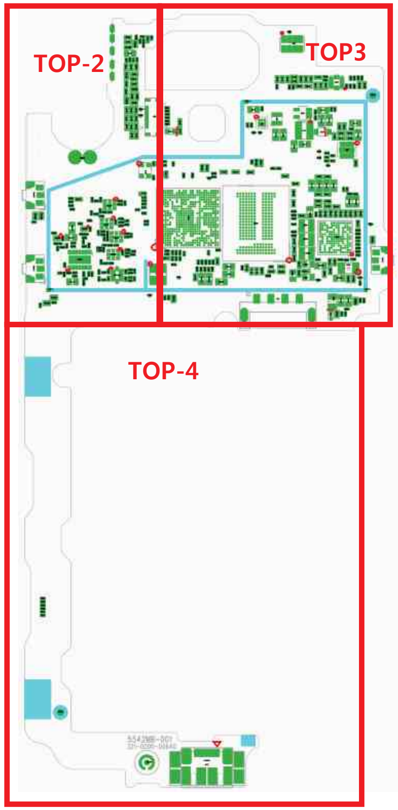
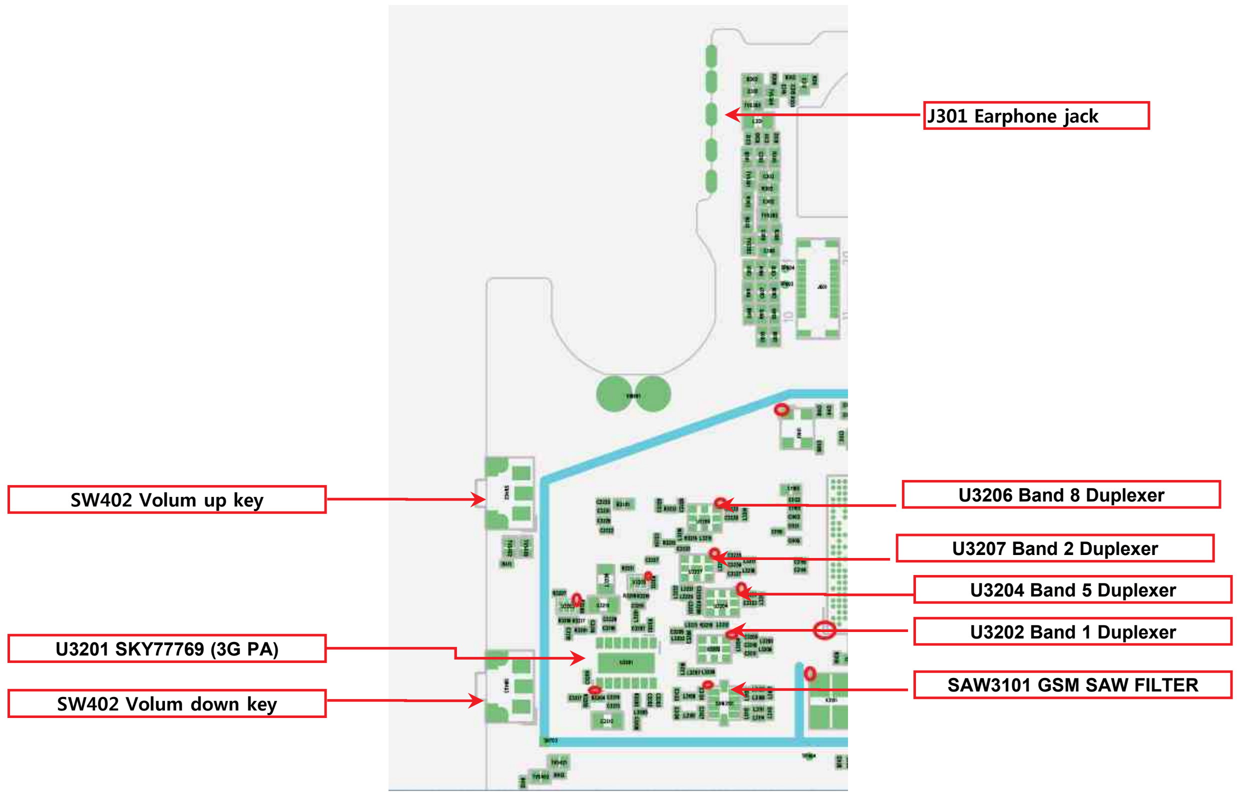
U1001_MT6580_BB chip IC, Digital Baseband Processor(Top)	U4001_H9TQ17A8GTMCUR-KUM,IC,MCP. eMMC+LPDDR3(Top)
 O Not used	 O Not used
U2001_MT6350_IC,PMIC (Top)	
 O Not used	O Not used

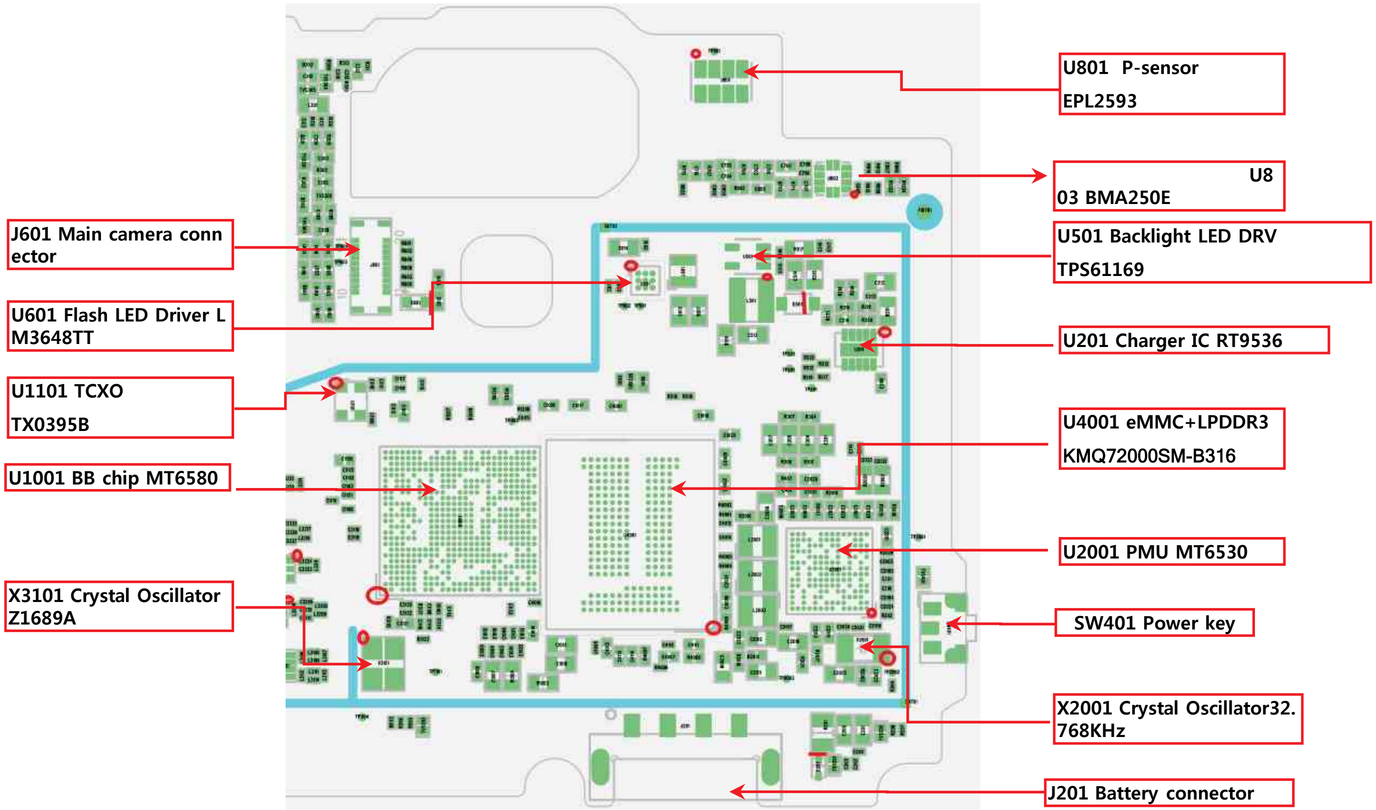


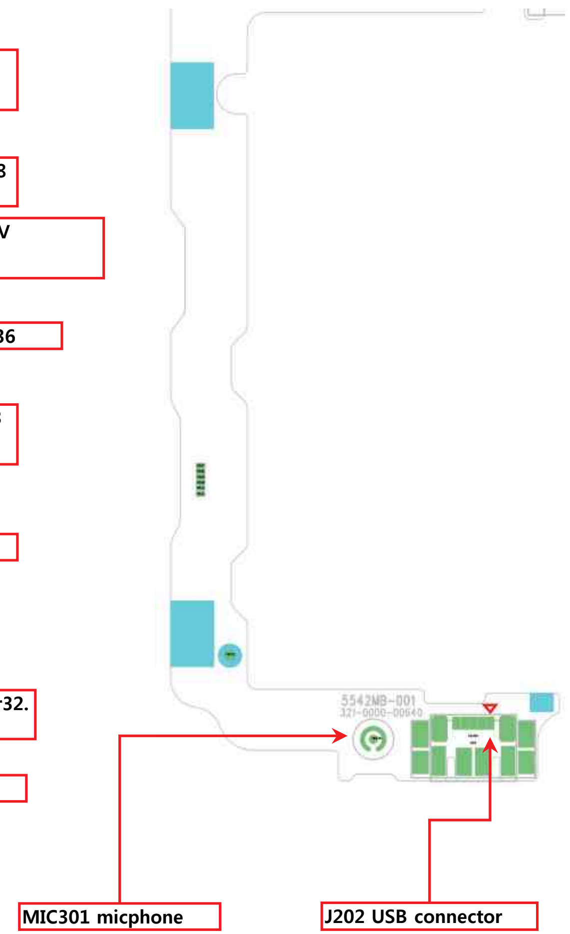
LG-X220ds-MAIN_TOP-2

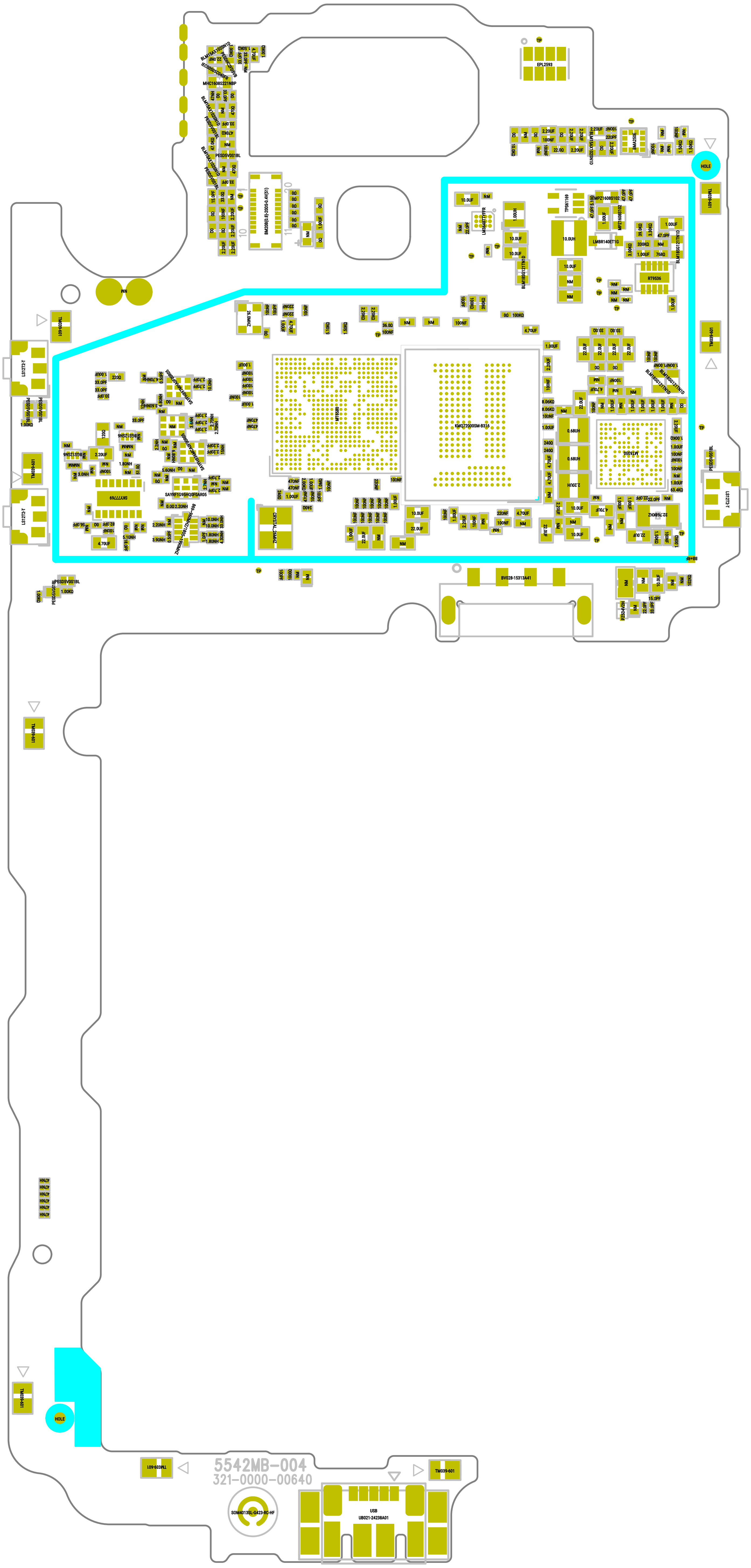


LG-X220ds-MAIN_TOP-3

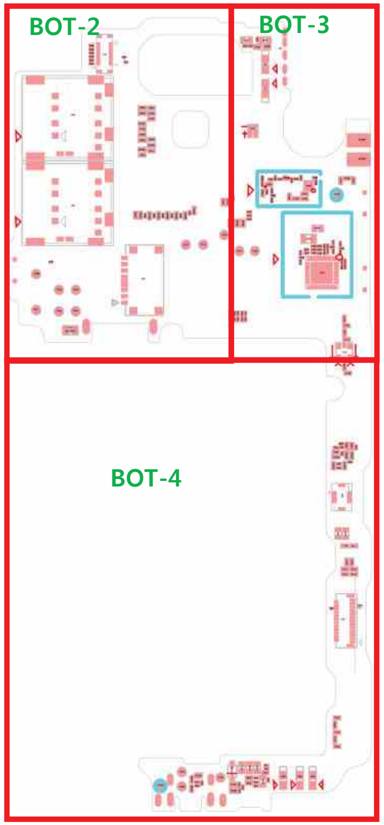
LG-X220ds-MAIN_TOP-4



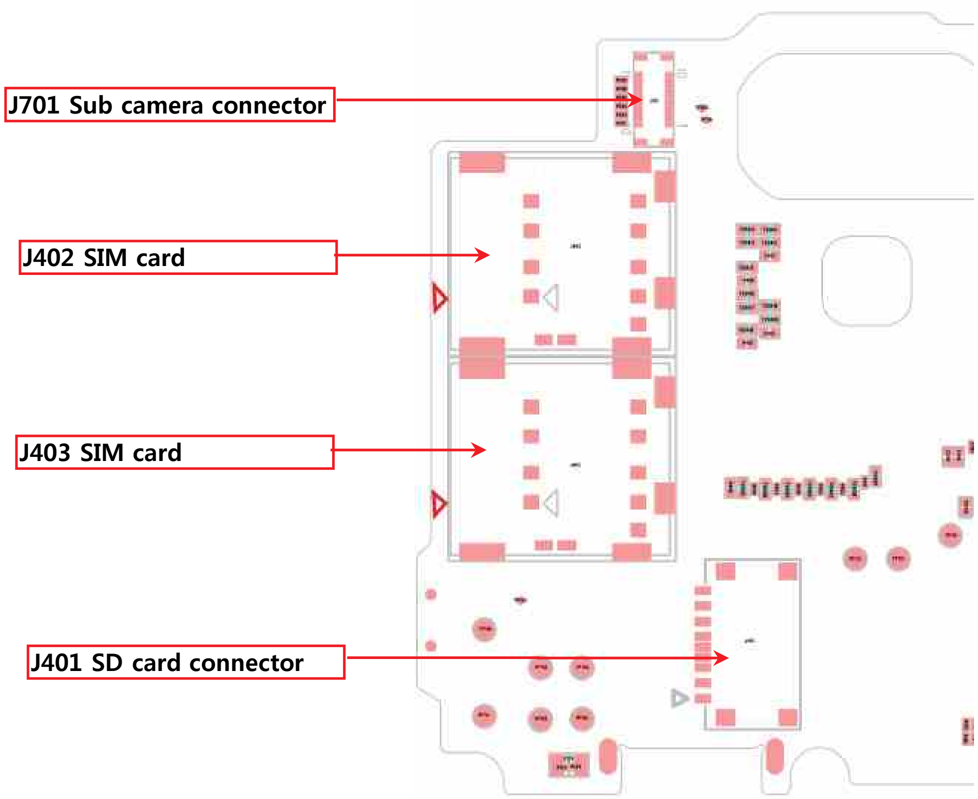



5542MB-004
321-0000-00640
USB

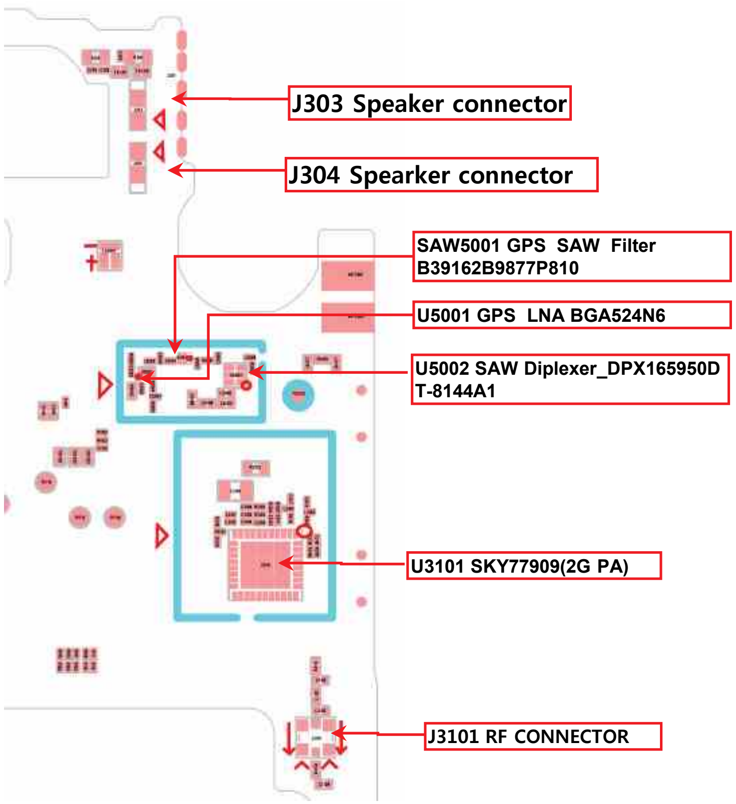
LG-X220ds-MAIN_BOT-1



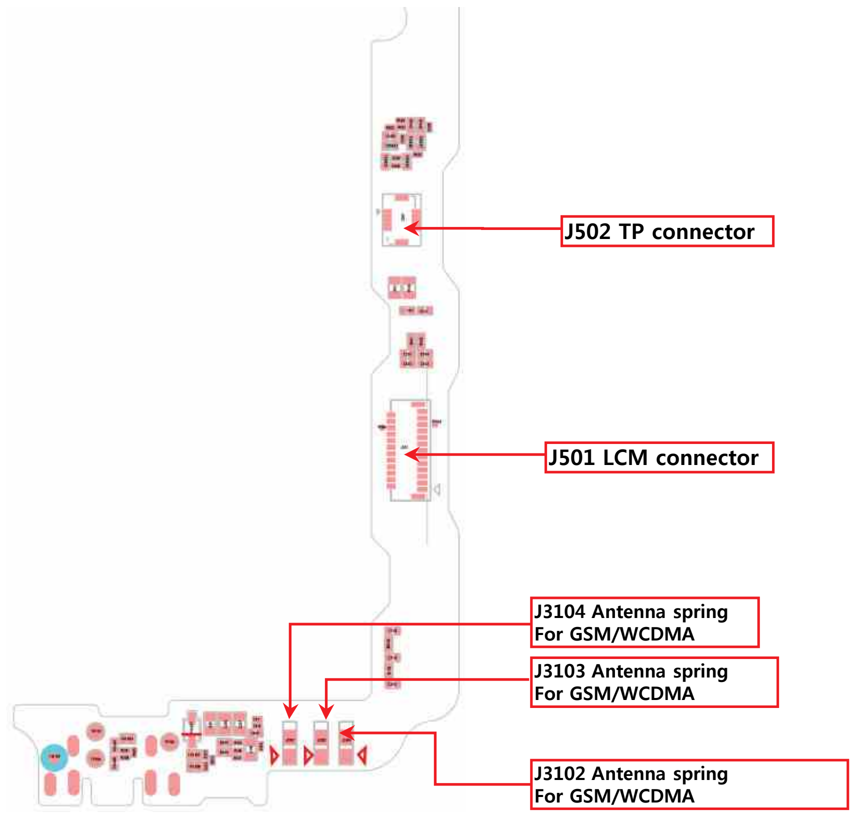
LG-X220ds-MAIN_BOT-2

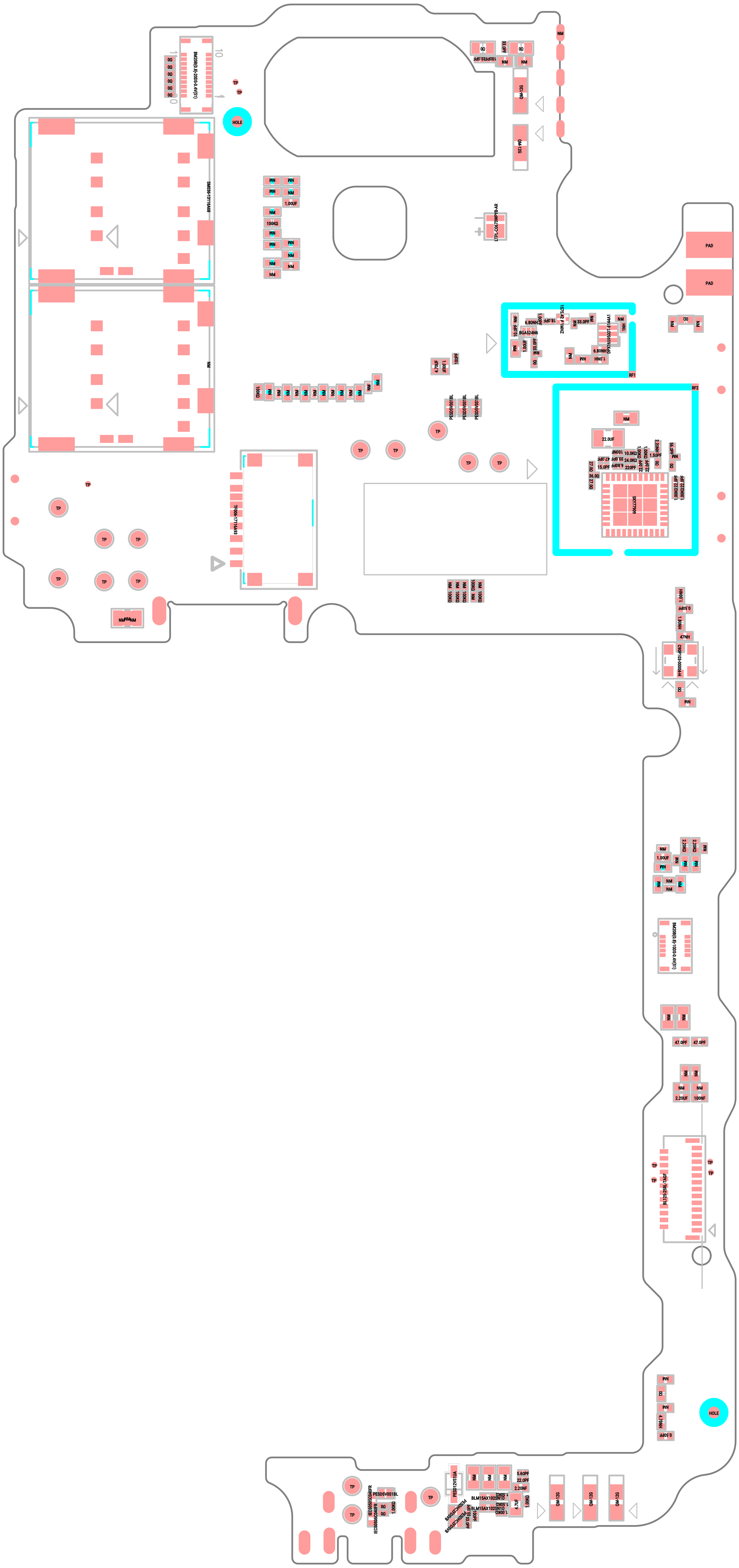


LG-X220ds-MAIN_BOT-3

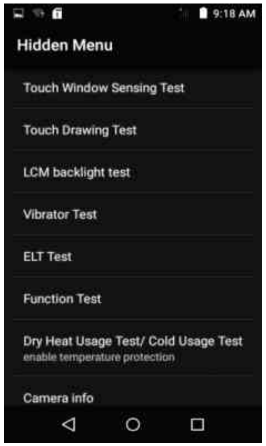


LG-X220ds-MAIN_BOT-4



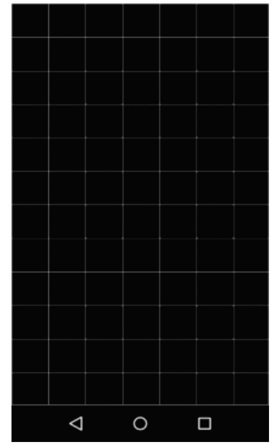

HOLE
PAD
PAD
HOLE

1. Hidden Menu Start



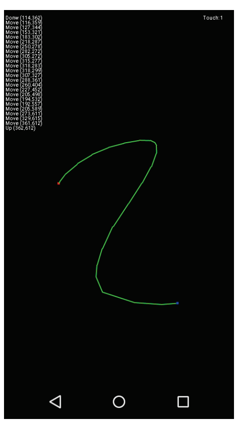
- Start shortcut key: *#546368#*220#
- Hidden Menu List
 : Start the desired menu, click

2. Touch window sensing test



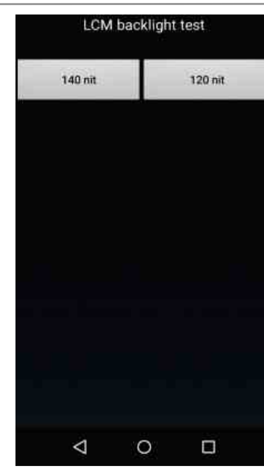
- Recording touch count
- Recording touch point
- Recording touch key count
- press volume up to back

3. Touch draw test



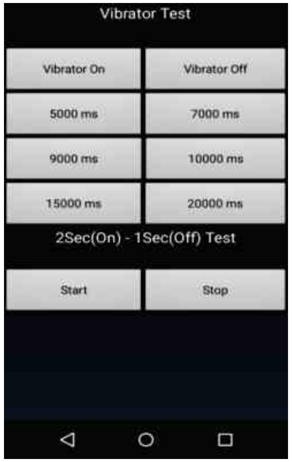
- Recording touch trace when slide on secreen
- Recording touch count when slide on secreen

4. LCM backlight test



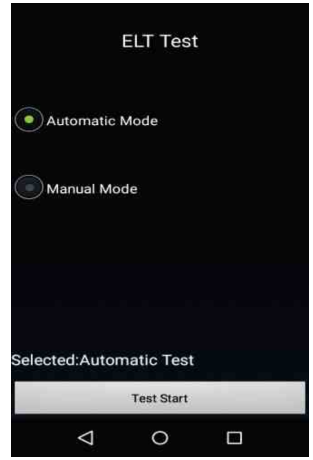
-Test back light
(two levels: 140 nit ,120 nit)

5. Vibrator Test



- Select one item from menu to start the vobrator test

6. ELT Test

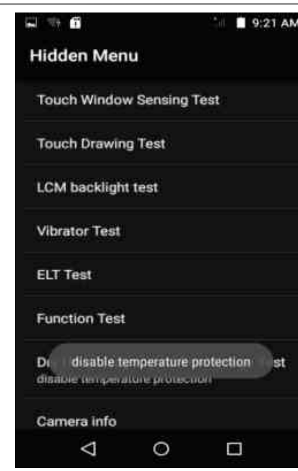


- ELT auto mode test
- ELT manual mode test
- press test start button to start ELT test

7. Function Test

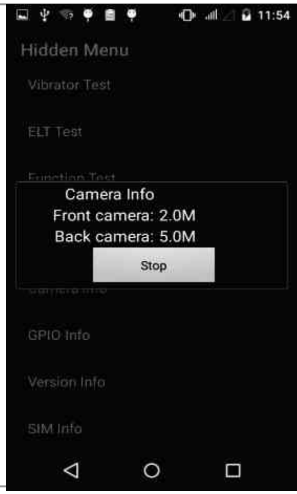
- test function one by one

8. Dry Heat Usage Test / Cold Usage Test

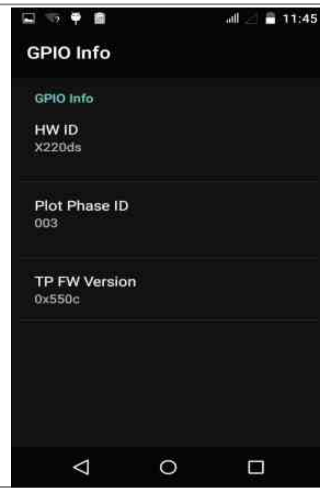


- Over temperature protection switch

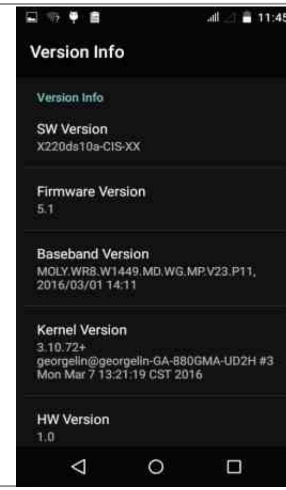
9. Camera info



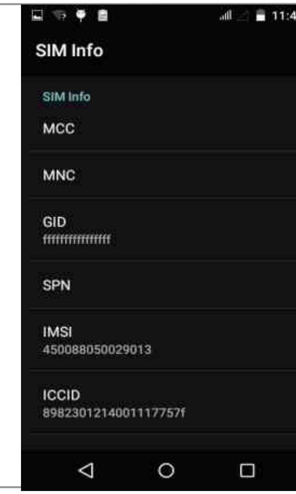
10. GPIO info



11. Version info

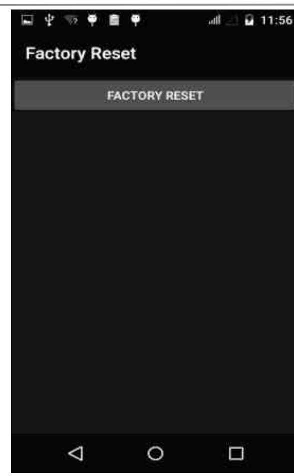


12. SIM Info

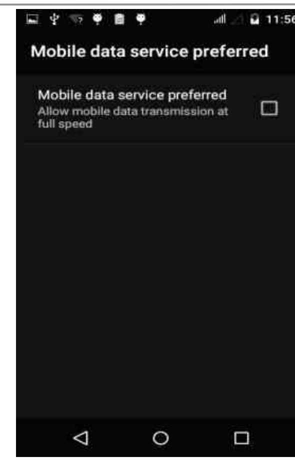


13. VSIM TEST

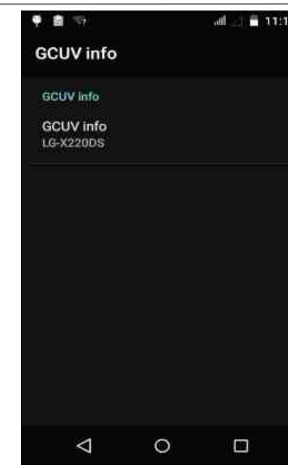
14. Factory Reset



15. Mobile data service preferred

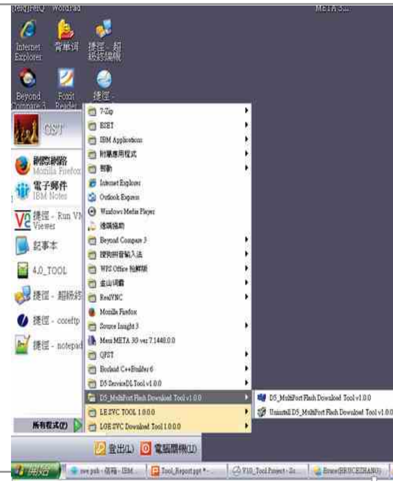


16. GCUV Info

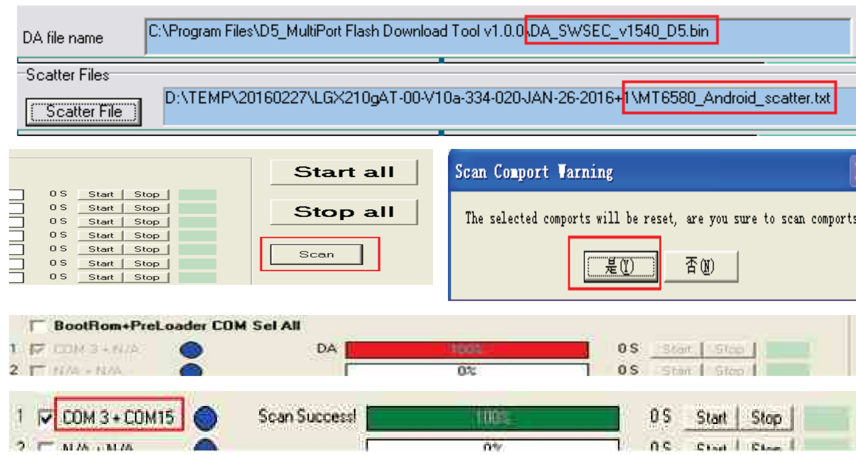


1. Summary

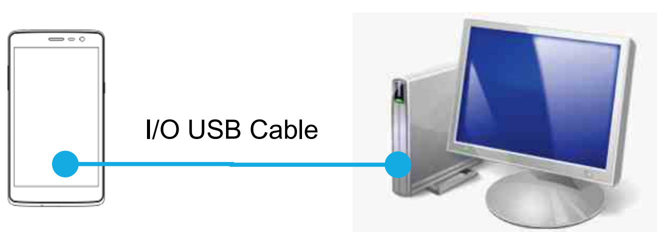
Tool Version	DLL name	USB Driver	
MultiPort Flash Download Setup v1.0.0	FlashToolLib.dll (version 1540)	LGUnitedMobileDriver_S51MAN312AP22_ML_WHQL_Ver_3.14.1	
H/W			
	Name	Part No.	SW
D/L Cable	USB Cable 2.0	N/A	N/A



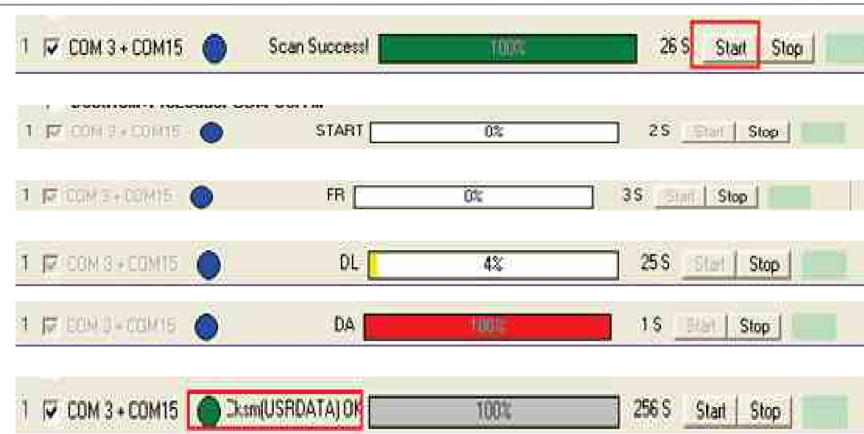
2. USB COM port Setting



3. USB D/L Cable setting



4. Flash tool D/L setting

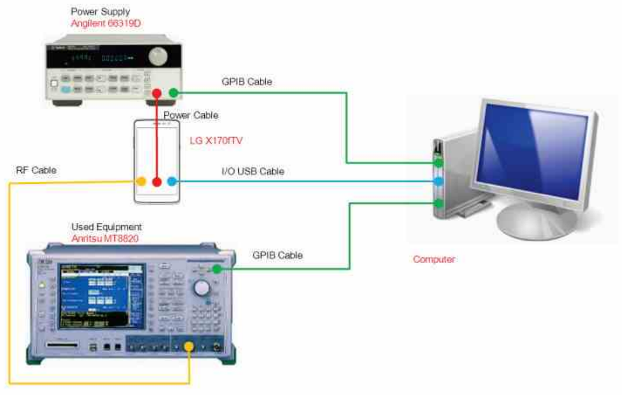


※ If you want more information, please refer to "Download User Guide".

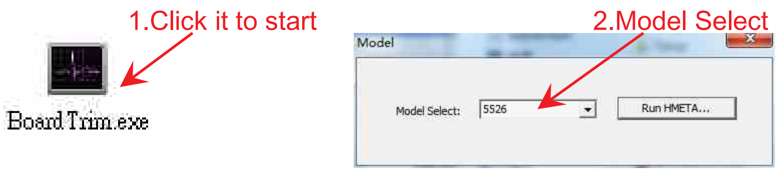
1. Summary

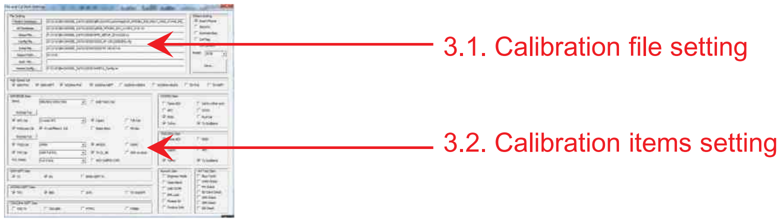
CAL INFORMATION		
S/W VERSION		
H/W		
	Name	Part No.
PIF	N/A	N/A
USB Cable	USB Cable	
Power Cable	DC Power Cable	
I/O Cable	N/A	
RF Cable_Main	MXHQ87WJ3000	RAD32847889
Power Supply_PIF	N/A	
Power Supply_Phone	Power supply + dummy Battery	
PF Test Equipment	Agilent	
Notice		

2. Calibration Cable setting

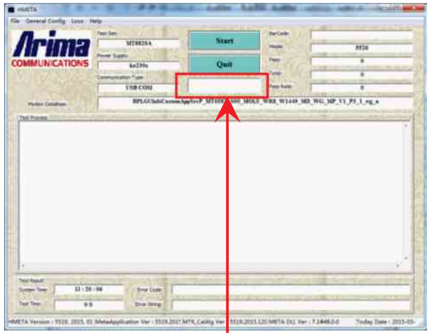


3. Calibration setting 1

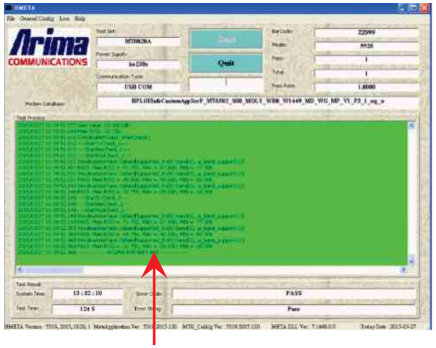




4. Calibration setting 2



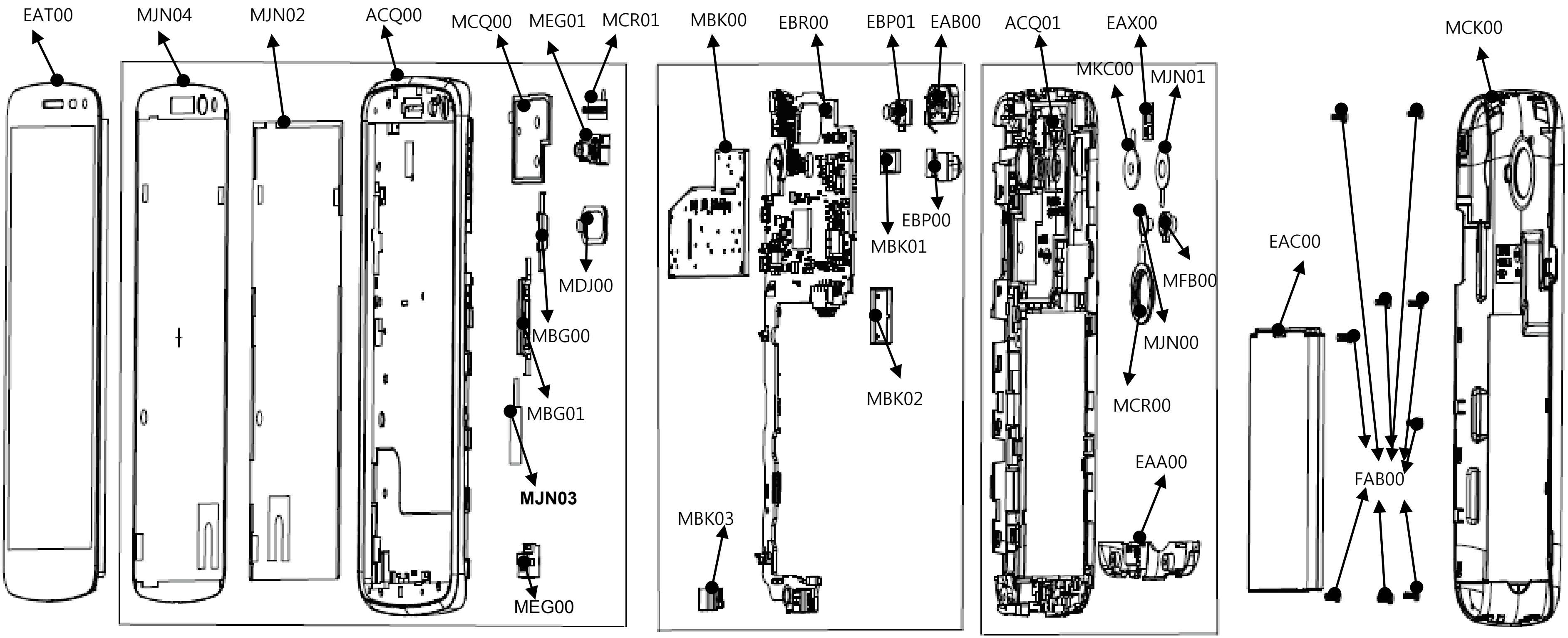
Input Barcode in the red box,
and enter “Enter” key.
The tool will start to calibration.



If success , test Process will
turn green , else will turn red.

1. Remove screws (9point)	2. Disassemble Rear cover	3. Disassemble Connector	4. Disassemble Main PCB
			
5. Disassemble HW parts(3ea)	6. Disassemble HW parts(1ea)	7. Disassemble main camera	
			

8. Complete Disassembling



Location no	Description	Location no	Description	Location no	Description	Location no	Description
EAT00	Hybrid Touch LCD Module,OLE Dy	MJN04	Tape,Window	MBK03	Can,Shield	EAX00	PCB,Flexible
ACQ00	Cover Assembly,Bar	MDJ00	Filter	EAB00	Speaker Module	MFB00	Lens,Flash
MEG00	Holder	MJN02	Tape	EBP00	Camera Module	MJN00	Tape
MEG01	Holder	MJN03	Tape	EBP01	Camera Module	MJN01	Tape,Camera
MCQ00	Damper,Speaker	EBR00	PCB Assembly,Main	ACQ01	Cover Assembly,Rear	EAC00	Rechargeable Battery,Lithium Ion
MBG00	Button,Side	MBK00	Can,Shield	MKC00	Window,Camera	FAB00	Screw,Machine
MBG01	Button,Side	MBK01	Can,Shield	MCR00	Decor	MCK00	Cover,Battery
MCR01	Decor	MBK02	Can,Shield	EAA00	Antenna,Multiple		

P/N	Item	Description	Quantity	Location no
AAD87875807	- AAD87875807	Addition Assembly	1	AAD000000
EAC62378301	EAC62378301	Rechargeable Battery,Lithium Ion	1	EAC00
SSAD0038307	SSAD0038307	Adapters	1	EAY060001
MBM65442101	MBM65442101	Card,Quick Reference	1	MBM062600
MBM64995926	MBM64995926	Card,Warranty	1	MBM087201
MFL69450601	MFL69450601	Manual,Operation	0	MFL053800
EAD62377903	EAD62377903	Cable,Assembly	1	EAD010001
MCK69294303	MCK69294303	Cover,Battery	1	MCK00
AGF78382609	- AGF78382609	Package Assembly	1	AGF000000
AGJ73458330	- AGJ73458330	Pallet Assembly	1	AGJ000000
MBEC0004405	MBEC0004405	Box,Carton	0.00208	MAY010800
MCCL0002604	MCCL0002604	Cap,Box	0.00416	MBL007000
MCQ66486911	MCQ66486911	Damper	0.00208	MCQ000000
MLAJ0004402	MLAJ0004402	Label,Master Box	0.00833	MEZ047200
MLAZ0050901	MLAZ0050901	Label	0.00208	MEZ000000
MPCY0012406	MPCY0012406	Pallet	0.00208	MGA000000
MAY67459304	MAY67459304	Box,Unit	1	MAY084000
MBEE0061004	MBEE0061004	Box,Master	0.1	MAY047100
MEZ64571201	MEZ64571201	Label,Unit Box	1	MEZ084100
MEZ64686921	MEZ64686921	Label,Unit Box	1	MEZ084101
MLAC0004541	MLAC0004541	Label,Barcode	1	MEZ003500
MLAJ0004402	MLAJ0004402	Label,Master Box	0.1	MEZ047200
MAF63267402	MAF63267402	Bag,Vinyl	1	MAF086500
AGQ88865507	- AGQ88865507	Phone Assembly	1	AGQ000000
ACQ88828407	- ACQ88828407	Cover Assembly,EMS	1	ACQ100402
EBR82420303	- EBR82420303	PCB Assembly,Main	1	EBR00
EBR82420403	- EBR82420403	PCB Assembly,Main,SMT	1	EBR071801
EAF62430001	EAF62430001	Varistor	3	EAF050000,EAF050001,EAF050002

P/N	Item	Description	Quantity	Location no
EAG64109901	EAG64109901	Connector,RF	1	EAG080000
EAX67092801	EAX67092801	PCB,Sub	1	EAX010100
EAG64952201	EAG64952201	Card Socket	1	EAG120500
EAG64890901	EAG64890901	Connector,BtoB	2	EAG020000,EAG020001
EAG64670801	EAG64670801	Connector,FFC/FPC/PIC	1	EAG060000
EAG64671001	EAG64671001	Connector,Terminal Block	1	EAG110200
EAG64873401	EAG64873401	Socket,DIMM/SIMM	2	EAG120600,EAG120601
EAN62677301	EAN62677301	IC,Charger	1	EAN050301
EAN64308901	EAN64308901	IC,Proximity	1	EAN051100
EAN63865901	EAN63865901	IC,PMIC	1	EAN050300
EAH63353001	EAH63353001	Diode,TVS	1	EAH050003
EAN64106501	EAN64106501	IC,Digital Baseband Processor,3G	1	EAN090200
EAM64072801	EAM64072801	Filter,Saw	1	EAM080001
MCE63192801	MCE63192801	Contact	5	MCE000000,MCE000001,MCE000002,MCE000003,MCE000004
EAM63890501	EAM63890501	Filter,Bead	2	EAM020016,EAM020017
EDTY0010701	EDTY0010701	Diode,TVS	12	EAH050004,EAH050005,EAH050006,EAH050007,EAH050008,EAH050009,EAH050010,EAH050011,EAH050012,EAH050013,EAH050014,EAH050015
EAH63192701	EAH63192701	Diode,TVS	3	EAH050000,EAH050001,EAH050002
EAM63930201	EAM63930201	Filter,Bead	1	EAM020005
EAM63870701	EAM63870701	Filter,Saw	1	EAM080002
EBR82420501	EBR82420501	PCB Assembly,Main,SMT Bottom	1	EBR071600
EAN64067101	EAN64067101	IC,Buffer	1	EAN050101
EAM63370101	EAM63370101	Filter,LCR	1	EAM010000
EAG64952101	EAG64952101	Connector,BtoB	1	EAG020002
EAN64248501	EAN64248501	IC,D/A Converter	1	EAN050500
EAN63366602	EAN63366602	IC,MCP,eMMC	1	EAN011300
EAH62692701	EAH62692701	Diode,Zener	1	EAH070000
EAV63591901	EAV63591901	Numeric Displays	1	EAV100000
EAM64151801	EAM64151801	Filter,Duplexer	1	EAM090000
MBK64794801	MBK64794801	Can,Shield	1	MBK00

P/N	Item	Description	Quantity	Location no
EAM63910501	EAM63910501	Filter,Bead	6	EAM020010,EAM020011,EAM020012,EAM020013,EAM020014,EAM020015
EAM63271201	EAM63271201	Filter,Bead	5	EAM020000,EAM020001,EAM020002,EAM020003,EAM020004
MBK64794804	MBK64794804	Can,Shield	1	MBK01
EAN64268601	EAN64268601	IC,Power Amplifier	1	EAN050102
EAG64951301	EAG64951301	Connector,BtoB	1	EAG020003
EAH63433001	EAH63433001	Diode,Schottky	1	EAH030000
EAN64289201	EAN64289201	IC,LED Driver	1	EAN050600
EAN63466001	EAN63466001	IC,RF Amplifier	1	EAN050100
EAW62543401	EAW62543401	Oscillator,Crystal	1	EAW010100
EAM63293001	EAM63293001	Filter,Saw	1	EAM080000
EAM64151701	EAM64151701	Filter,Bead	2	EAM020008,EAM020009
MBK64794803	MBK64794803	Can,Shield	1	MBK02
EBR82420601	EBR82420601	PCB Assembly,Main,SMT Top	1	EBR071700
SAD35901501	SAD35901501	Software,Mobile	1	SAD010000
EAM64111501	EAM64111501	Filter,Bead	1	EAM020007
EAW62563601	EAW62563601	Oscillator,TCXO	1	EAW010400
EAW63083801	EAW63083801	Crystal	1	EAW030100
MCE63212801	MCE63212801	Contact	8	MCE000005,MCE000006,MCE000007,MCE000008,MCE000009,MCE000010,MCE000011,MCE000012
EBF63134701	EBF63134701	Switch,Tact	3	EBF120000,EBF120001,EBF120002
EAM63293401	EAM63293401	Filter,Bead	1	EAM020006
MBK64794802	MBK64794802	Can,Shield	1	MBK03
EAB64368501	EAB64368501	Microphone,Condenser	1	EAB020200
EAN63466401	EAN63466401	IC,Acceleration Sensor	1	EAN051101
EAT63401301	EAT63401301	Hybrid Touch LCD Module,OLED	1	EAT00
EAG64911901	EAG64911901	Connector,FFC/FPC/PIC	1	EAG060000
EAU63563201	EAU63563201	Motor,DC	1	EAU010000
EAB64348701	EAB64348701	Speaker Module	1	EAB00
EBP62922201	EBP62922201	Camera Module	1	EBP00
EBP62962101	EBP62962101	Camera Module	1	EBP01
GMZZ0017701	GMZZ0017701	Screw,Machine	9	FAB00
MEZ66451301	MEZ66451301	Label,Approval	1	MEZ002101

P/N	Item	Description	Quantity	Location no
MEZ66653101	MEZ66653101	Label,Approval	1	MEZ002100
MJN70031902	MJN70031902	Tape,USP Film	1	MJN107400
ACQ89028303	- ACQ89028303	Cover Assembly,Rear	1	ACQ01
MKC66119901	MKC66119901	Window,Camera	1	MKC00
MCR66610203	MCR66610203	Decor	1	MCR00
EAA64549001	EAA64549001	PIFA Antenna,Multiple	1	EAA00
EAX67103601	EAX67103601	PCB,Flexible	1	EAX00
MFB64333301	MFB64333301	Lens,Flash	1	MFB00
MJN70054801	MJN70054801	Tape	1	MJN00
MJN70033901	MJN70033901	Tape,Camera	1	MJN01
ACQ89016503	- ACQ89016503	Cover Assembly,Bar	1	ACQ00
MCK69278503	MCK69278503	Cover,Front	1	MCK032700
MEG64900601	MEG64900601	Holder	1	MEG00
MEG64900701	MEG64900701	Holder	1	MEG01
MCQ68909401	MCQ68909401	Damper,Speaker	1	MCQ00
MBG66165103	MBG66165103	Button,Side	1	MBG00
MBG66225003	MBG66225003	Button,Side	1	MBG01
MCR66630001	MCR66630001	Decor	1	MCR01
MJN70033801	MJN70033801	Tape,Window	1	MJN04
MDJ65005301	MDJ65005301	Filter	1	MDJ00
MJN70015501	MJN70015501	Tape	1	MJN03
MJN70076001	MJN70076001	Tape	1	MJN02

P/N : FLXXXXXXXX(X.X)